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Quality-Driven Next-Best-View Planning for Target-Conditioned Egocentric 3D Reconstruction

Qualitätsgetriebene Next-Best-View-Planung für zielkonditionierte egozentrische 3D-Rekonstruktion

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Abstract

Active perception couples sensing and reconstruction: what can be reconstructed depends on what a sensing action makes visible. Under a limited acquisition budget, next-best-view planning must therefore choose which feasible observation should come next. The literature reviewed in Chapter 2 treats scene coverage, one-step reconstruction-quality ranking, target-aware information criteria, and sequential coverage separately, but leaves their conjunction for target-specific egocentric reconstruction unresolved. This thesis studies that conjunction as finite candidate selection under hard validity constraints in Aria Synthetic Environments (ASE). It adapts Relative Reconstruction Improvement (RRI) to a target-cropped point–mesh objective, separates privileged task construction and oracle evaluation from the deployable actor path, and marks GT-derived target or selected-depth inputs as non-deployable controls. The primary experiment first tests whether bounded oracle lookahead provides endpoint-quality headroom over one-step oracle greedy; a learned policy is evaluated only if that headroom passes a prespecified meaningful-effect and uncertainty rule. The current implementation supports target-specific oracle scoring and selected-action replay, with rendering memory limiting scale. Actor-visible target matching, metric repeatability, a validated held-out population, headroom, and paired policy outcomes remain unestablished. The present contribution is therefore an auditable method and experimental design, not evidence of policy superiority or deployment readiness.

Zusammenfassung

Aktive Wahrnehmung beruht auf einer geometrischen Prämisse: Was rekonstruiert werden kann, hängt davon ab, was eine Sensorhandlung sichtbar macht. Bei begrenztem Aufnahmebudget muss die Next-Best-View-Planung daher entscheiden, welche zulässige Beobachtung als Nächstes erfolgen soll. Die in Kapitel 2 ausgewertete Literatur behandelt Szenenabdeckung, einstufige Rekonstruktionsqualität, zielbezogene Informationsmaße und sequenzielle Abdeckung getrennt, lässt deren Verbindung für die zielspezifische egozentrische Rekonstruktion jedoch offen. Diese Arbeit untersucht diese Verbindung als Auswahl aus einer endlichen Kandidatenmenge unter harten Zulässigkeitsbedingungen in ASE. Sie überträgt die RRI auf ein zielbeschnittenes Punkt–Mesh-Maß, trennt privilegierte Aufgabenerzeugung und Orakelevaluation vom einsatzfähigen Akteurpfad und kennzeichnet GT-basierte Ziel- oder Tiefeneingaben als nicht einsetzbare Kontrollen. Das primäre Experiment prüft zunächst, ob eine beschränkte Orakelvorausschau gegenüber einer einstufigen gierigen Orakelstrategie einen Vorteil bei der Endpunktqualität bietet; eine gelernte Strategie wird nur bewertet, wenn dieser Vorteil ein vorab festgelegtes Relevanz- und Unsicherheitskriterium erfüllt. Die aktuelle Implementierung unterstützt zielspezifische Orakelbewertung und Replay ausgewählter Aktionen, wobei der Speicherbedarf des Renderers die Skalierung begrenzt. Nicht belegt sind bislang eine beobachtungsbasierte Zielzuordnung, die Wiederholbarkeit des Qualitätsmaßes, eine validierte zurückgehaltene Studienpopulation, ein Vorteil der Orakelvorausschau und gepaarte Strategieergebnisse. Der gegenwärtige Beitrag ist daher ein prüfbarer Methoden- und Versuchsaufbau, kein Nachweis für Strategieüberlegenheit oder Einsatzreife.

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Glossary and Abbreviations

Dataset

***ADT* – Aria Digital Twin**: Project Aria dataset with real-world captures and digital-twin scene annotations. ADT is adjacent to ARIA-NBV's sim-to-real context; the current experiments focus on ASE and EFM3D/ATEK exports, while ADT remains a relevant transfer surface.

***AEO* – Aria Everyday Objects**: Small-scale real-world Project Aria object dataset used by EFM3D for egocentric 3D perception evaluation. AEO is relevant as a possible sim-to-real check for ARIA-NBV ideas that are first developed on ASE mesh-supervised snippets.

***ASE* – Aria Synthetic Environments**: Large-scale synthetic indoor dataset with simulated Project Aria sensor characteristics, egocentric trajectories, and scene annotations. ARIA-NBV uses ASE snippets and the public mesh-supervised subset to generate oracle RRI labels from ground-truth meshes and semi-dense point clouds.

***CPF* – Central Pupil Frame**: Coordinate frame placed at the midpoint between the left and right eye boxes of Project Aria glasses. CPF is used by Project Aria for gaze-related quantities and should remain distinct from rig, camera, world, and PyTorch3D frames in ARIA-NBV docs.

***MPS* – Machine Perception Services**: Project Aria processing services that derive pose, mapping, gaze, hand, and related perception outputs from sensor recordings. ARIA-NBV mainly depends on MPS-style pose and semi-dense mapping products as the current reconstruction state for RRI computation.

***MTD* – Motion Trajectory Data**: Device poses over time, usually represented as a sequence of 6-DoF transformations. MTD supplies the logged egocentric trajectory used to define snippet state, current reconstruction context, and candidate-view reference poses.

***MFCD* – Multi-Frame Camera Data**: Synchronized camera streams from multiple Project Aria cameras over a temporal window. MFCD provides the egocentric image evidence consumed by EVL and aligned with pose, calibration, and semi-dense point streams.

***MSDPD* – Multi-Semi-Dense Point Data**: Semi-dense 3D point observations generated by SLAM-style processing across a snippet or trajectory window. MSDPD is the sparse observed geometry that ARIA-NBV fuses with candidate point clouds and compares against ground-truth meshes for RRI labels.

***Project Aria* – Project Aria**: Meta's egocentric research-device and tooling ecosystem for calibrated, time-aligned multimodal sensing. ARIA-NBV treats Project Aria and its MPS-style products as the actor-visible sensing contract, while ASE meshes remain offline supervision and evaluation assets.

***SSL* – SceneScript Language**: Structured language representation for indoor scene layout using primitives such as walls, doors, windows, and objects. SceneScript is relevant to ARIA-NBV as a possible semantic/global planning layer and as one source of ASE scene-structure context.

***snippet* – Snippet**: Short synchronized temporal window of Aria sensor data used as one EVL/VIN input sample. A snippet typically contains RGB or grayscale streams, poses, calibration, semi-dense points, and scene metadata that EVL lifts into a voxel grid.

***VRS* – Virtual Reality Standard**: File format used by Project Aria tooling to store multi-modal sensor recordings efficiently. VRS is part of the upstream Project Aria ecosystem, while ARIA-NBV primarily consumes ASE/ATEK-derived tensorized snippets for current experiments.

Entity

***target* – Target of Interest**: Selected entity, object crop, point, region, or surface-deficit hypothesis whose reconstruction quality should be improved. Target-conditioned ARIA-NBV variants use this target as an explicit input to candidate scoring and planning instead of optimizing only scene-level RRI.

Geometry

***DoF* – Degrees of Freedom**: Number of independent pose parameters available to a camera, object, or action representation. ARIA-NBV distinguishes full 6-DoF pose estimation from reduced 5-DoF candidate or action spaces used for practical view planning.

***frustum* – Frustum**: Truncated pyramidal camera-visible volume bounded by near and far clipping planes plus lateral field-of-view planes. Candidate-view frusta define which scene surfaces can project into a camera image and are therefore central to visibility, rendering, and RRI diagnostics.

***LUF* – Left-Up-Forward**: Camera coordinate convention whose x axis points left, y axis points up, and z axis points forward. LUF is one of the coordinate-frame conventions that must stay explicit when moving between Project Aria cameras, PyTorch3D cameras, and ARIA-NBV candidate poses.

***OBB* – Oriented Bounding Box**: 3D bounding box with arbitrary orientation, used to represent object extent more tightly than an axis-aligned box. OBBs are a natural target encoding for entity-aware ARIA-NBV because they provide center, extent, orientation, semantic class, and confidence signals.

***6DoF* – Six Degrees of Freedom**: Pose parameterization with three translational and three rotational degrees of freedom. Project Aria poses, candidate cameras, and world-to-device transforms are usually represented as 6-DoF rigid-body transforms.

Metrics

***AUC* – Area Under Curve**: Aggregate score computed by integrating a metric curve over an acquisition, threshold, or ranking axis. AUC can summarize how quickly reconstruction quality, coverage, or ranking quality improves as views are acquired or candidate thresholds change.

***CD* – Chamfer Distance**: Historical bidirectional distance family used to compare reconstructed points against reference geometry. Thesis-facing ARIA-

NBV notation uses point-mesh error D with directional components $D_{P \to M}$ and $D_{M \to P}$; older seminar material may still call this CD.

***CR* – Coverage Ratio**: Fraction of a target surface, scene, or region treated as observed under a chosen visibility or distance threshold. Coverage ratio is a useful diagnostic baseline for NBV, but ARIA-NBV treats reconstruction-quality improvement through RRI as the preferred optimization target.

***GT* – Ground Truth**: Reference data or annotations treated as the trusted target for training, validation, or evaluation. In ARIA-NBV, ground truth usually refers to ASE meshes, object annotations, poses, or labels used to compute oracle supervision and diagnostic metrics.

***RRI* – Relative Reconstruction Improvement**: Metric quantifying the relative reconstruction-quality improvement obtained by adding a candidate observation to the current reconstruction. ARIA-NBV computes RRI by comparing reconstruction error before and after fusing candidate-view geometry, usually against an ASE ground-truth mesh for oracle supervision and evaluation.

***reward* – Target Root-Gain Reward**: Quality-only immediate reward for thesis-core target-aware rollouts. It is the target-specific point-mesh error reduction normalized by the rollout-root target error; scalar motion, rule, diversity, and validity penalties are later ablations.

***target RRI* – Target-Specific RRI**: RRI computed only on the ground-truth and reconstructed geometry associated with a selected target of interest. Target-specific RRI lets the thesis compare views by how much they improve a selected object or region, even when scene-level RRI would prefer large background surfaces.

Model

***Q-CORAL* – CORAL Q-Value Decoder**: Ordinal Q-decoder ablation that discretizes continuous fitted-Q targets into ordered classes and decodes cumulative threshold outputs to a scalar conditional value through fixed return representatives.

EFM3D **– Egocentric Foundation Model 3D**: Egocentric 3D foundation-model stack used as the frozen spatial backbone for ARIA-NBV candidate scoring. The current project uses EFM3D and its EVL architecture to expose voxel occupancy, centerness, semantic, and OBB evidence for VIN-style RRI prediction.

EVL **– Egocentric Voxel Lifting**: EFM3D architecture that lifts synchronized egocentric observations into a gravity-aligned 3D voxel feature volume. ARIA-NBV uses EVL head outputs, pre-head features, and OBB predictions as local actor-visible evidence and target support, not as the complete frozen scene representation.

Q_H **– Finite-Horizon Q Function**: Target-conditioned finite-candidate value family Q_theta(s,e,a,h) that scores each valid candidate for an explicit requested residual horizon h up to a configured maximum H. The thesis-core architecture is one shared horizon-conditioned scorer; dense Q_1 and exact Q_2 targets establish the recursion, horizon-indexed fitted Q supplies longer-horizon targets, and Double Q remains an optional correction for a noisy learned successor maximum.

target-conditioned scorer **– Target-Conditioned Scorer**: VIN-style candidate scorer that receives scene state, a candidate view, and an encoding of the target of interest. The scorer predicts target-specific utility so view ranking can prioritize a selected entity or region instead of only optimizing scene-level RRI.

VIN **– View Introspection Network**: Learned candidate-view scorer that predicts RRI or an ordinal RRI-derived utility without capturing the candidate view. ARIA-NBV adapts VIN-NBV by placing a lightweight RRI prediction head on top of frozen EVL features and candidate-pose evidence from ASE snippets.

Planning

***cost* – Acquisition Cost**: Budget consumed to acquire observations, measured by view count, path length, elapsed time, invalid-action rate, or a weighted combination. The thesis should first report cumulative root-normalized target gain, diagnostic target RRI, and acquisition cost separately, then use scalarized objectives only when the tradeoff is explicit.

***candidate* – Candidate View**: Proposed camera pose whose expected reconstruction utility is evaluated before selecting the next observation. ARIA-NBV samples candidate views around a reference pose or target, renders candidate depths from the mesh for oracle labels, and scores candidates with RRI or learned VIN predictions.

***transition* – Counterfactual Transition**: Deterministic or replayable rollout update that renders or retrieves only the selected candidate observation, backprojects it to candidate geometry, and fuses it into the accumulated reconstruction proxy.

***action set* – Finite Candidate Action Set**: Valid discrete action set for ARIA-NBV planning, obtained by masking a sampled candidate view table. The action is the candidate-table index, not a continuous pose; the selected pose is $q_t = q_{t,a_t}$. Invalid candidates are constraints, not low-quality actions.

***return* – Finite-Horizon Return**: Symbolic H-step return used by ARIA-NBV rollout evaluation and Q_H targets. The formula keeps γ explicit while the thesis-core comparison reports cumulative root-normalized target gain under equal acquisition budget.

***5DoF* – Five Degrees of Freedom**: Reduced camera-action parameterization commonly used when roll is fixed or otherwise constrained. ARIA-NBV uses 5-DoF language for candidate and planning abstractions where position and viewing direction matter while roll is not an independently optimized control dimension.

***CF+ state* – Geometry-Rich Counterfactual State**: Ablation actor state that extends the minimal counterfactual state with selected prior synthetic observations only. The executable CF-GT carrier stores rendered depth, valid mask, calibration, selected-camera pose, source identity, and causal support; implemented S1 derives an identity-start, fixed-width current-camera surface-point residual from it.

historic state **– Historic Snippet State**: Raw actor-visible state available along the logged Project Aria/ASE trajectory before counterfactual rollout synthesis. It includes captured camera streams, calibration, timestamps, trajectory/gravity, semidense points with support or uncertainty, EVL/EFM evidence, and detected or predicted OBBs; GT mesh and GT OBBs are excluded as actor inputs.

CF0 state **– Minimal Counterfactual Actor State**: Main thesis-core actor-visible counterfactual rollout state. It keeps the fused point proxy $\mathcal{P}_t$ as broad scene state, optional lifted image-foundation features attached to points, local root EVL evidence for target support, selected-view history, target descriptor, budget, candidate table, validity masks and reason codes, and current candidate-query features, without exposing all-candidate GT renders.

NBV **– Next-Best-View**: Problem of selecting the next sensor viewpoint to improve an active reconstruction or inspection objective under a limited acquisition budget. In ARIA-NBV, the preferred objective is reconstruction quality improvement rather than coverage alone, with candidate views ranked by oracle or learned RRI scores.

oracle state **– Oracle Rollout State**: Privileged state used only for ASE mesh-supervised label generation, upper-bound planning, and evaluation. It includes GT mesh and target crops, all-candidate renders, points, normals, mesh-face visibility, RRI and point-mesh error metrics, oracle scores, and GT OBBs.

offline state **– Persisted Offline Sample State**: Compact VIN offline-store state persisted for training and diagnostics. It contains the VinSnippetView payload, candidate table and cameras, candidate count, validity or count metadata, oracle metrics as labels, optional rendered depths, compact OBBs, trajectory metadata, and selected EVL numeric fields.

***state* – Rollout State**: Family of rollout state variants used by ARIA-NBV. The actor-visible variants contain logged or replayed observations, reconstruction proxies, candidates, validity masks, target descriptors, history, and budget; the oracle variant adds privileged GT geometry and all-candidate labels for supervision and evaluation only.

***NBV MDP* – Target-Conditioned NBV MDP**: Finite-horizon Markov decision process used to define ARIA-NBV's target-conditioned rollout and fitted Q_H training contract. The actor observes only current reconstruction state, sampled candidate views, validity masks, target descriptors, history, and budget state; oracle meshes and GT crops are supervision and evaluation signals.

***mask* – Validity Mask**: Hard candidate-level feasibility mask used during candidate ranking, rollout generation, and Q_H training. The scalar mask $m_{t,i}$ gates selectable rows, while the reason code $\rho_{t,i}$ records why a row was rejected. Invalidity is represented separately from reconstruction quality and should not be encoded as the lowest RRI class.

Protocol

***GT-EVAL* – Ground-Truth Target Evaluation**: Main thesis protocol component restricting ground-truth OBBs and target mesh crops to oracle labels, target-RRI computation, matching checks, and evaluation. GT target crops are not actor-visible inputs in the main result.

***OBS-SEL* – Observed Target Selection**: Main thesis protocol component requiring target selection to use only actor-visible evidence such as predicted or tracked OBBs, class probabilities, confidence, projected area, and semidense or EVL support. Ground-truth target annotations are not visible to the selector in this protocol.

***PRED-Q* – Predicted-Target Q**: Main thesis protocol component requiring the scorer or fitted Q_H model to condition on predicted or observed target descriptors rather than ground-truth target inputs. It covers target-conditioned one-step scoring and finite-candidate Q_H selection.

Reconstruction

***MVS* – Multi-view Stereo**: Reconstruction family that estimates dense or semi-dense scene geometry from multiple calibrated views. MVS is part of the broader reconstruction context for NBV, although ARIA-NBV's current oracle labels are built from ASE meshes and Aria-style semi-dense point observations.

***PC* – Point Cloud**: Set of 3D points representing observed scene geometry. ARIA-NBV compares current and candidate-fused point clouds against ground-truth meshes to compute oracle RRI and surface-distance diagnostics.

***SLAM* – Simultaneous Localization and Mapping**: Method for estimating sensor motion while building a map of the surrounding scene. In ARIA-NBV, logged SLAM poses and semi-dense points form the current reconstruction state against which candidate-view RRI is evaluated.

***track* – Track**: Temporal sequence of corresponding image-feature detections across frames, usually carrying per-frame image coordinates, timestamps, and camera IDs. Tracks can be triangulated or optimized into 3D points, making them a bridge between image evidence and the semi-dense point clouds used by ARIA-NBV.

***VIO* – Visual-Inertial Odometry**: Pose-estimation method that combines visual measurements from cameras with inertial measurements from IMUs. Project Aria pose and mapping products build on visual-inertial estimation, and ARIA-NBV depends on those pose streams for snippet state and candidate frame contracts.

Representation

***Occupancy Grid* – Occupancy Grid**: Spatial grid whose cells encode whether space is occupied, free, unknown, or represented by a related occupancy probability. Occupancy-style voxel evidence appears in EVL outputs and VIN feature construction, where it helps summarize local 3D scene state for candidate scoring.

***3DGS* – 3D Gaussian Splatting**: Explicit radiance-field representation using optimized 3D Gaussian primitives for real-time novel-view synthesis. ARIA-NBV treats active 3DGS work as proposal, uncertainty, and simulator-bridge literature, not as a replacement for ASE mesh-supervised RRI.

Supervision

***oracle RRI* – Oracle RRI**: RRI label computed with privileged ground-truth geometry, used for supervised training and evaluation. The current oracle renders candidate depth maps from ASE ground-truth meshes, backprojects candidate point clouds, fuses them with the current semi-dense reconstruction, and scores the resulting surface-distance improvement.

List of Symbols

Symbol	Meaning	Key
	Oracle and Reconstruction	
RRI	Relative Reconstruction Improvement score for a candidate view.	`oracle.rri`
$\mathcal{P}$	Actor-visible point set accumulated from observed or replayed views.	`oracle.points`
$\mathcal{P}_q$	Candidate-view point contribution used in oracle or counterfactual updates.	`oracle.points_q`
$\mathcal{Q}$	Finite candidate-view set considered by the planner.	`oracle.candidates`
$\mathcal{Q}_t$	Candidate-view set available at rollout step t.	`oracle.candidates_t`
$q_{t,i}$	Candidate pose i at rollout step t.	`oracle.candidate_qti`
$\boldsymbol{c}$	World-space camera-center vector; subscripts identify a rollout root or candidate pose.	`oracle.center`
$\boldsymbol{D}_q$	Oracle-rendered or candidate-specific depth map for a proposed view.	`oracle.depth_q`
$D_{P\to M}$	Point-to-mesh directional error component.	`oracle.acc`
$D_{M\to P}$	Mesh-to-point directional error component.	`oracle.comp`
D	Aggregate point-mesh reconstruction error used by RRI definitions.	`oracle.err`
	ASE Assets	
$\mathcal{M}^{\mathrm{GT}}$	Ground-truth scene mesh used for oracle labels and evaluation.	`ase.mesh`
$\mathcal{M}_e^{\mathrm{GT}}$	Ground-truth mesh crop for target e.	`ase.mesh_target`
$\mathcal{F}^{\mathrm{GT}}$	Ground-truth mesh faces used in point-mesh distance calculations.	`ase.faces`
$\boldsymbol{T}^{\mathrm{w}}_{\mathrm{rig}}(t)$	Project Aria rig pose in the world frame at time t.	`ase.traj`
$\boldsymbol{T}^{\mathrm{w}}_{\mathrm{rig}}(T)$	Final rig pose used as an anchor for gravity-aligned local fields.	`ase.traj_final`
	VIN Scorer	
r	Target or scene RRI value used as a model target.	`vin.rri`
$\hat{r}$	Predicted RRI value emitted by the learned scorer.	`vin.rri_hat`
$\mathcal{L}$	Training loss for scorer or value-model objectives.	`vin.loss`
m	Candidate validity indicator used by scorer diagnostics.	`vin.cand_valid`

Symbol	Meaning	Key
	Entity and Target	
RRI_e	Target-specific Relative Reconstruction Improvement for entity e.	`entity.rri_e`
$\mathrm{RRI}^e_{t,i}$	State-relative marginal target RRI for candidate i at rollout step t.	`entity.target_rri_marginal`
$C_t^{\mathrm{RRI},e}$	Running sum of selected state-relative target RRIs.	`entity.target_rri_cumulative`
J_t^e	Running cumulative target gain normalized by rollout-root error.	`entity.target_root_gain_cumulative`
Δ_t^e	Target-specific reconstruction error at rollout step t.	`entity.target_error`
$\boldsymbol{\varphi}_e$	Actor-visible target/entity descriptor used to condition target-specific value prediction.	`entity.target_desc`
$\boldsymbol{p}_e^w$	World-space center of the selected target entity e.	`entity.center`
	Planning and Rollout	
s	Abstract planning state.	`rl.s`
a	Action index selecting a candidate row.	`rl.a`
r	Scalar reward or immediate gain.	`rl.r`
G	Finite-horizon return accumulated across rollout steps.	`rl.G`
j	Factual rollout-chain index used to preserve common trajectory heritage across selected steps.	`rl.rollout_index`
$\mathcal{M}_{\mathrm{NBV}}$	Target-conditioned finite-candidate NBV decision process.	`rl.mdp_nbv`
$\mathcal{A}(s_t)$	Valid action set available in state s_t.	`rl.action_set`
T	State-transition operator for selected candidate actions.	`rl.transition`
s_t^{hist}	Logged historic snippet state at rollout step t.	`rl.s_hist`
s_t^{off}	Persisted offline sample state used by training readers.	`rl.s_off`
s_t^{cf0}	Minimal actor-visible counterfactual state.	`rl.s_cf0`
$s_t^{\mathrm{S0\text{-}pose}}$	Implemented fixed-root pose-history state used by qh_cf0_v1.	`rl.s_pose`
$s_t^{\mathrm{cf+}}$	Geometry-enriched counterfactual successor state.	`rl.s_cf_geom`
$s_t^{\mathrm{CF\text{-}GT\text{-}carrier}}$	Implemented privileged selected-depth carrier used by qh_cfplus_gt_depth_v1.	`rl.s_cf_gt_carrier`
s_t^{oracle}	Oracle-only state used for labels, upper bounds, and evaluation.	`rl.s_oracle`
r_t^e	Target-specific reward at rollout step t.	`rl.reward_target`

Symbol	Meaning	Key
	Entity and Target	
r_t^e	Canonical target-specific reward at rollout step t.	`entity.target_reward`
	Planning and Rollout	
$G_t^{(H)}$	H-step finite-horizon return from rollout step t.	`rl.return_h`
Q_H	Finite-horizon candidate-value function.	`rl.qh`
$Q_{h,\theta,e,i}^{\mathrm{cond}}$	Action-mask-independent conditional candidate value emitted by the scorer.	`rl.conditional_q`
$\ell_{t,i}^{\mathrm{feas}}$	Physical or observed feasibility logit emitted by the scorer's feasibility head.	`rl.feasibility_logits`
e_k^Q	Fixed continuous fitted-Q boundary between adjacent CORAL classes.	`rl.coral_q_edge`
u_k^Q	Fixed continuous-Q representative used to decode one CORAL class.	`rl.coral_q_value`
c_n^Q	Ordinal class assigned to one continuous fitted-Q target.	`rl.coral_q_label`
γ	Return discount factor.	`rl.gamma`
H	Planning or rollout horizon length.	`rl.H`
$H_{\max}$	Maximum residual horizon admitted by a scorer/data contract.	`rl.H_max`
h	Scalar residual horizon requested for one scorer query; it must satisfy 1 <= h <= b_t <= H_max.	`rl.requested_horizon`
$m_{t,i}$	Hard validity mask for candidate i at rollout step t.	`rl.validity_mask`
$m_{t,i}^{\mathrm{cand}}$	Materialized candidate row versus structural padding.	`rl.candidate_row_mask`
$m_{t,i}^{\mathrm{act}}$	Authoritative physically valid action support.	`rl.action_mask`
$m_{t,i}^{Q,h}$	Availability of a finite factual value target for candidate i at requested horizon h.	`rl.q_label_mask`
$m_{t,i}^{F}$	Availability of a trustworthy feasibility label for candidate i.	`rl.feasibility_label_mask`
m_t^{succ}	Availability of a factual successor backup for transition t.	`rl.successor_mask`
$\ell_{t,i}^{\mathrm{src}}$	Actor-oracle source-provenance role for candidate evidence.	`rl.source_role`
$\rho_{t,i}$	Invalid-action reason code for candidate i at rollout step t.	`rl.invalid_reason`
e_t	Selected target entity at rollout step t.	`rl.target`
b_t	Remaining acquisition budget at rollout step t.	`rl.budget`

Symbol	Meaning	Key
$C(\tau)$	Acquisition cost of a selected trajectory.	`rl.acquisition_cost`
	Shape and Size	
N_q	Number of candidate views in a candidate table.	`shape.Nq`
K	Number of ordinal bins or discrete levels when used in scorer losses.	`shape.K`
	scene	
$\boldsymbol{M}_t^{\mathrm{ray}}$	Sparse actor-visible ray-aware memory separating occupied, free, and unknown support at step t.	`scene.ray_memory_t`
	Observation and Actor State	
$\boldsymbol{F}_t^{\mathrm{DINO@pt}}$	Visibility-gated logged DINO feature bank attached to semidense or fused world points.	`obs.dino_point_bank_t`
$\boldsymbol{X}_t^{\mathrm{pt}}$	Point-token set combining geometry, support, uncertainty, history, and optional compressed logged descriptors.	`obs.point_tokens_t`
	scene	
$\boldsymbol{\Phi}_t^{\mathrm{scene}}$	Composite actor-visible scene-memory descriptor queried by the finite-candidate value model.	`scene.scene_memory_t`
$\boldsymbol{E}_0^{\mathrm{EVL\text{-}local}}$	Root local EVL evidence field used for target/candidate support reads.	`scene.evl_local`
$\omega_{t,i}^{\mathrm{EVL}}$	Candidate or target-query fraction supported by the root EVL voxel extent.	`scene.evl_support_frac`
$\boldsymbol{g}_{t,i}^{\mathrm{EVL}}$	Pooled local EVL evidence token for a target, candidate frustum, or target-candidate intersection query.	`scene.evl_support_token`
$\boldsymbol{g}_e^{\mathrm{tgt}}$	Pooled target-support descriptor for the selected target hypothesis.	`scene.target_support_pool`
$\boldsymbol{g}_{t,i}^{\mathrm{fr}}$	Candidate-frustum support descriptor pooled from actor-visible scene memory.	`scene.frustum_support_pool`
$\boldsymbol{g}_{t,e,i}^{\mathrm{cap}}$	Pooled descriptor over the intersection between target support and candidate frustum support.	`scene.target_frustum_pool`
$\boldsymbol{g}_{t,i}^{\mathrm{ray}}$	Candidate-conditioned ray query over occupied, free, unknown, hit, target-support, and uncertainty summaries.	`scene.ray_query_ti`
RenderQuery	Abstract query operator that summarizes scene memory in a candidate camera/frustum without introducing fresh counterfactual RGB features.	`scene.render_query`
	spatial	

Symbol	Meaning	Key
r_t	Reference pose for candidate-relative descriptor construction at decision step t.	`spatial.ref_pose`
$\boldsymbol{T}^{\mathrm{rel}}_{r_t,i}$	Relative transform from the reference pose r_t to candidate camera i.	`spatial.ref_candidate_transform`
$\boldsymbol{h}^{\mathrm{pose}}_{t,i}$	Relative/local candidate pose descriptor derived from a reference pose rather than raw world coordinates.	`spatial.candidate_pose_feat`
$\boldsymbol{h}^{\mathrm{rel}}_{t,e\|i}$	Candidate-target relation descriptor in the candidate/query local frame.	`spatial.candidate_target_rel_feat`
$\boldsymbol{e}^{\mathrm{rel}}_{a\|i}$	Query-local relative positional embedding for target, history, support, or candidate relations.	`spatial.relation_rpe`
r_e	Geometric-mean target-proxy scale derived from the target OBB semi-axis lengths.	`spatial.target_obb_scale`
$\hat{\boldsymbol{\delta}}^{e}_{j,t}$	Unit factual selected-camera displacement expressed in target-object coordinates for rollout j and step t.	`spatial.target_frame_motion_direction`
$\hat{\boldsymbol{v}}^{e}_{j,t}$	Unit selected-camera forward optical axis expressed in target-object coordinates.	`spatial.target_frame_view_direction`
$\mathcal{F}^{e}_{j,t}$	Front-facing target-proxy surface directions that project inside the calibrated selected-camera image.	`spatial.target_frame_frustum`
$\kappa^{e}_{j,t}$	Fraction of the target-centred proxy sphere geometrically supported by one selected calibrated frustum.	`spatial.target_frame_frustum_fraction`
	model	
$\boldsymbol{h}^{\mathrm{tgt}}_{e}$	Learned selected-target token built from root-relative target pose and metric extents.	`model.target_token`
	spatial	
$\Omega^{\mathrm{FOV}}_{j,t}$	Intrinsic solid angle of one calibrated selected-camera pinhole field of view.	`spatial.frustum_solid_angle`
	model	
$\boldsymbol{h}^{\mathrm{phys}}_{t,i}$	Candidate-local physical token from root/current-relative candidate poses and compact root evidence.	`model.candidate_physical_token`
$\boldsymbol{x}_{t,i}$	Candidate-local conditional-value query formed by adding candidate-from-target pose encoding to the physical token.	`model.candidate_row`
$\boldsymbol{p}^{\mathrm{hist}}_{t,j}$	Previously selected pose j encoded from the current camera at decision state t.	`model.history_pose_feature`
$a^{\mathrm{hist}}_{t,j}$	Normalized relative age of selected pose j at decision state t; the immediate predecessor has age zero.	`model.history_relative_age`

Symbol	Meaning	Key
$\boldsymbol{h}_t^{\mathrm{hist}}$	Fixed-width causal selected-pose history token supplied to scorer state fusion.	`model.history_token`
Entity and Target		
$J_e^{(H)}$	Endpoint reconstruction gain for target e over horizon H.	`entity.endpoint_gain`
$J_{e,\log}^{(H)}$	Log-scale endpoint reconstruction gain for target e.	`entity.log_gain`
Δ_{look}	Gain available to an oracle lookahead policy beyond one-step selection.	`entity.lookahead_headroom`
η_Q	Fraction of the learned-myopic-to-oracle-lookahead endpoint gap closed by the Q policy.	`entity.q_recovery`
$G_t^{(H)}$	Target-conditioned finite-horizon return.	`entity.return_h`
Observation and Actor State		
$\boldsymbol{D}$	Depth observation sequence.	`obs.depth`
$\boldsymbol{n}$	Per-point or per-face normal observation.	`obs.face_normal`
$\boldsymbol{I}^{\mathrm{rgb}}$	RGB image observation.	`obs.img_rgb`
$\mathcal{P}^{\mathrm{cf}}$	Counterfactual point observation.	`obs.points_cf`
$\mathcal{P}^{\mathrm{semi}}$	Semi-dense observed point set.	`obs.points_semi`
$\mathcal{P}_t$	Point set available at rollout step t.	`obs.points_t`
Planning and Rollout		
$\mathcal{A}_t$	Finite feasible action set at rollout step t.	`rl.action_set_t`
$\mathcal{Q}_t$	Finite candidate-view table at rollout step t.	`rl.candidate_table`
$y_t^{(2,\mathrm{exact})}$	Exact two-step fitted-Q control using factual dense successor one-step rewards.	`rl.exact_q2_target`
$\varepsilon_t^{(2)}$	Absolute learned-recursion target error against the exact two-step control.	`rl.q2_recursion_error`
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1 Introduction

Active perception begins from a geometric premise: what can be reconstructed depends on what a sensing action makes visible [AWB88, Baj88]. A new view changes both the available surface evidence and the state from which later views are chosen. Next-best-view (NBV) planning turns this coupling between action, observation, and reconstruction into the repeated choice of where to look next [Jia+25, SRR03].

With a limited acquisition budget, a view is "best" only relative to an objective. Coverage rewards newly observed surface, information criteria reward reduced uncertainty, and quality-driven methods evaluate the reconstruction itself [Fra+25, GML22, JLD24]. These objectives can disagree. A view may reveal more of a room without exposing the occluded surface of a requested object, while a small side-step may improve that object but add little scene-wide coverage. Target conditioning therefore specifies **whose geometry** should improve; it is not generic semantic awareness.

Existing methods supply the necessary ingredients under different assumptions. VIN-NBV ranks sampled candidates by direct reconstruction improvement, object-aware 3D Gaussian Splatting conditions an information criterion on a requested object, and GenNBV and Hestia learn sequential coverage policies [Che+24, Fra+25, Jeo+26, Lu+26]. What remains unresolved is their conjunction: whether target-specific reconstruction quality contains useful multi-step structure and whether that structure can be recovered from a bounded egocentric information state.

The actor-state question includes a representation-transfer hypothesis. Prior NBV pipelines already use generic 2D pretraining or foundation-model components for view prediction, depth estimation, or scene construction [AKC20, Gué+23, Str+24a]. These uses do not establish whether an egocentric 3D representation that combines a frozen 2D foundation feature extractor with learned upsampling and 3D processing improves target-conditioned endpoint-value learning. ARIA-NBV therefore treats EFM3D/EVL evidence as a hypothesis to be tested against matched actor-visible geometric controls, not as an assumed source of gain [Str+24b].

Project Aria supplies calibrated egocentric observations, EFM3D lifts them into local 3D evidence, and ARIA Synthetic Environments (ASE) provides the privileged geometry needed for controlled target tasks and counterfactual evaluation [Eng+23, Met25, Str+24b]. This combination makes the information boundary testable. Task construction, action-set construction, hard admission, supervision, and evaluation may use privileged geometry, while the scorer consumes only the fields admitted by its declared actor-facing protocol. Scientific interpretation must account for both paths. Chapter 3 therefore treats actor visibility as a property of the complete decision process, not merely of the scorer tensor.

1.1 Aim and bounded problem

The evaluation proceeds in two stages. First, it measures whether bounded oracle lookahead yields better fixed-budget endpoint reconstruction of a requested object than one-step oracle-greedy selection. Conditional on such headroom, it then measures how much of the endpoint gap from an actor-visible learned-myopic control to oracle lookahead an offline finite-horizon model closes. This order separates the existence of a planning opportunity from the ability of a learned model to exploit it; the gap-closure ratio and oracle headroom have different baselines.

At every step, each policy selects from the same finite generated candidates; hard-invalid actions are excluded, and the target, horizon, candidate support, and endpoint evaluation are matched. The current oracle tasks use geometry-valid ground-truth boxes, while actor-visible target discovery and matching remain separate research requirements. The objective adapts VIN-NBV rather than copying it: VIN-NBV defines RRI from point-cloud Chamfer distance [Fra+25], whereas ARIA-NBV evaluates a target-cropped point–mesh error after an equal acquisition budget. The resulting claims concern this bounded finite-candidate setting, not a deployable target-selection system or an objective invariant to other reconstruction protocols.

1.2 Contributions and current evidence

The thesis operationalizes this evaluation through four scientific functions:

1. It separates one-step RRI, finite-horizon return, and fixed-budget endpoint gain within target-specific reconstruction quality.
2. It defines an end-to-end information boundary across task and action construction, state updates, model inputs, supervision, and evaluation.
3. It defines candidate value relationally over target, causal state, admissible action, budget, horizon, and continuation rule.
4. It evaluates measurement, support, oracle headroom, immediate-value learning, recursion, and learned-control endpoint gap closure in that order.

Before model capacity becomes the main question, the evidence needed to define the learning problem must itself be identifiable. The outcome must be repeatable, the study population and candidate support must contain the relevant opportunity, bounded lookahead must exhibit headroom, and factual replay must identify the required targets. Only after those conditions hold can a failure to recover non-myopic value be attributed primarily to representation or learning. The order makes a negative result informative: it distinguishes a measurement or support failure from a failure of immediate-value learning or finite-horizon recursion.

The present evidence supports a methodological contribution rather than a policy result. The implementation executes target-specific oracle scoring and selected-action replay, while rendering memory limits scale. Actor-visible target matching, metric repeatability, a validated held-out population, oracle-lookahead headroom, and paired policy outcomes remain unestablished. Accordingly, the thesis does not yet claim policy superiority or deployment readiness.

1.3 Research Questions

1.3.1 Research aim and evaluation logic

The study evaluates a sequence of linked questions. RQ1 defines the endpoint outcome. RQ2 then asks whether bounded oracle lookahead improves that outcome relative to one-step oracle-greedy selection and, conditionally, how much of the separate actor-visible myopic-to-oracle-lookahead gap an offline model closes. RQ3 defines the admissible information, and RQ4 establishes the target, action, and replay support over which both comparisons are interpretable. RQ5 and RQ6 extend the study only if the offline finite-candidate evidence warrants online or continuous-control claims.

1.3.2 RQ1 — Target-specific objective and endpoint outcome

Can target-conditioned finite-candidate NBV be evaluated through a stable, target-specific reconstruction objective under a fixed acquisition budget?

For a target e, the oracle measures reconstruction error on a frozen target crop. The primary estimand is the quality change from the root state to the endpoint after H acquisitions:

$$J_e^{(H)} = \frac{\Delta_0^e - \Delta_H^e}{\Delta_0^e + \varepsilon}$$

State-relative RRI is a one-step diagnostic, root-normalized gain supplies the training reward, and endpoint gain remains the policy outcome. A positive answer requires a frozen, repeatable metric and equal acquisition horizons; runtime, invalid actions, and oracle calls are reported separately.

1.3.3 RQ2 — Bounded lookahead and learned gap closure

Does bounded oracle lookahead improve fixed-budget endpoint target gain over one-step oracle-greedy selection and, if so, how much of the actor-visible myopic-to-oracle-lookahead endpoint gap does an offline finite-horizon value model close?

The paired endpoint difference under the same scene, target, candidates, validity rules, horizon, and budget defines oracle-lookahead headroom:

$$\Delta_{\text{look}} = J_e^{(H)}(\pi_{\text{oracle-look}}) - J_e^{(H)}(\pi_{\text{oracle-1}})$$

The learned comparison proceeds only if this headroom passes a predeclared meaningful-effect and uncertainty rule. Its ratio uses a different baseline:

$$\eta_Q = \frac{J_e^{(H)}(\pi_Q) - J_e^{(H)}(\pi_{\text{learned-1}})}{J_e^{(H)}(\pi_{\text{oracle-look}}) - J_e^{(H)}(\pi_{\text{learned-1}}) + \varepsilon}$$

Here the numerator is the gain of the finite-horizon learned policy over the actor-visible learned-myopic control, and the denominator is the full gap from that control to oracle lookahead. The ratio is therefore conditional learned gap closure, not a fraction of the oracle-lookahead headroom above oracle-greedy. It is reported only when the paired denominator $J_e^H(\pi_{\text{oracle-look}}) - J_e^H(\pi_{\text{learned-1}})$ exceeds a predeclared positive tolerance $\tau_{\text{gap}} > 0$; otherwise the analysis reports the raw paired policy effects and classifies gap closure as not estimable. Because the headroom rule and required gap-closure fraction are not yet frozen, RQ2 remains prospective. If headroom is absent, the evaluated support does not expose a non-myopic advantage; this does not establish its universal absence.

1.3.4 Conditions for interpretation

1.3.4.1 RQ3 — Actor-visible target and information state

Which end-to-end target, action-support, and history protocol supports one-step and finite-horizon candidate scoring without privileged target geometry, oracle labels, or unselected candidate renders at decision time?

Ground-truth target tasks may define supervision, evaluation, and a separately reported privileged `V0/GT` task-construction sanity control; they do not satisfy the core RQ3 gate. The core learned comparison requires an observation-derived target instruction and matching path (`V1/observed`) in addition to actor-visible candidate support, hard validity, selected-state updates, and scorer inputs. Until that route exists, the core actor-visible comparison remains unavailable. The scorer then receives only the declared target descriptor, causal egocentric evidence, remaining budget, and requested horizon; evaluation separately audits matching failures, ranking and calibration, and actor/oracle leakage.

1.3.4.2 RQ4 — Candidate, replay, and population support

Do the candidate generator, validity rules, rollout recipes, and replay population provide adequate and diverse support for the RQ1–RQ3 estimands?

The candidate table combines exploration and target-bearing views, records generator provenance, and excludes geometric infeasibility through a hard mask. RQ4 measures candidate-family survival, invalidity, scene and target coverage, replay and horizon support, and resource cost. Scene-disjoint splits and scene-level aggregation prevent dense sampling from masquerading as population coverage.

1.3.5 Scope extensions

1.3.5.1 RQ5 — Online discrete bridge

If offline headroom, replay support, and actor-visible scoring are established, does online interaction over the unchanged discrete candidate set and validity rules improve endpoint target gain or calibration over the offline policy?

RQ5 keeps the target, information, candidate, validity, and endpoint assumptions of RQ1–RQ4 unchanged. It is not required for the offline finite-candidate evaluation.

1.3.5.2 RQ6 — Continuous or simulator-backed control

If the finite-candidate evidence is stable, does a continuous or hierarchical target-then-pose policy provide measurable headroom over the best discrete policy under the same target-specific objective and a comparable acquisition or motion budget?

RQ6 changes the action space and feasibility assumptions and therefore requires separate evidence for support, safety, simulator realism, and comparable cost. It remains deferred; the current implementation implies no continuous, simulator, or real-device result.

1.3.6 Shared evidence constraints

Action validity, actor support, target validity, oracle-label validity, and training eligibility remain distinct; missing evidence is not encoded as low utility. Confirmatory comparisons are paired under the same scene, target, candidates, validity regime, and budget. The scene is the experimental unit, and the analysis plan freezes exclusions, aggregation, uncertainty, and decision rules before policy outcomes are inspected.

1.3.7 Research-to-evidence map

RQ	Question role	Primary evidence	Interpretation gate
RQ1	outcome measurement	target reconstruction endpoint gain	frozen repeatable metric; fixed horizon and budget
RQ2	lookahead and learned gap closure	paired greedy, lookahead, myopic-control, and learned-policy outcomes	meaningful oracle headroom before learned gap closure
RQ3	information boundary	matching, ranking, calibration, and leakage audits	end-to-end actor-visible protocol
RQ4	population and support	candidate, replay, validity, and coverage diagnostics	scene-disjoint aggregation
RQ5	conditional extension	matched online discrete-policy evaluation	offline gates satisfied first
RQ6	deferred extension	continuous or simulator-backed evaluation	separate action space and cost comparison

Table 1: Research-question-to-evidence map.

1.4 Thesis structure

The remaining chapters follow that dependency chain. Chapter 2 derives the objective, support, temporal, and representation concepts needed to state the gap precisely. Chapter 3 constructs the experimental world by defining information access, admissible actions, state transitions, and measured outcomes. Chapter 4 then specifies the one finite-horizon value interface evaluated within that world. Chapter 5 states which evidence admits or blocks each claim. Chapter 6 reports the admitted answers in gate order; Chapter 7 attributes their implications and alternatives; and Chapter 8 synthesizes what those bounded answers establish.

2 Foundations and Related Work

The Introduction located this thesis within active perception: sensing actions change the evidence on which later inference depends [AWB88, Baj88]. Three-dimensional view planning turns that principle into a repeated choice of where to observe next [SRR03]. The unresolved dependency is not another planner implementation but a precise account of what a view is valuable **for**, which actions are available, and what information a sequential score may condition on. By the end of the chapter, the thesis question is reduced to three coupled requirements—utility alignment, temporal dependence, and support/state adequacy—before the experimental world is constructed.

The argument follows the dependencies of the decision problem. It first separates the next-best-view mechanism from the objective used to rank views, then distinguishes target-conditioned reconstruction quality from coverage and uncertainty proxies. It next separates candidate-view support from endpoint, transition, and human-motion feasibility before explaining why partial observability and delayed consequences require a finite-horizon information state. The chapter then derives the geometric properties a candidate scorer should preserve and positions the thesis question against the resulting literature dimensions.

2.1 Active Perception and the Next-Best-View Problem

In active perception, an observation is also an intervention on the future information state. A next-best-view (NBV) method operationalizes this idea by generating possible viewpoints, estimating the benefit of observing from each one, and selecting an action. PB-NBV makes this generate–score–select decomposition explicit for a finite candidate set [Jia+25]. The decomposition identifies three different scientific objects: the proposal mechanism determines which views can be considered, the utility determines how they are ordered, and the selection rule turns that ordering into an action.

How those objects are represented varies across NBV systems. Finite-candidate methods evaluate a bounded set of poses, whereas learned continuous policies predict camera position and orientation directly. GenNBV formulates the latter as a five-degree-of-freedom reinforcement-learning problem, while Hestia factorizes the continuous action into a look-at point followed by a camera position [Che+24, Lu+26]. A finite set makes the evaluated choices explicit and comparable; a continuous or hierarchical policy can search a broader space but couples view proposal and control more tightly.

These action formulations do not determine what makes a view useful. PB-NBV, for example, accelerates candidate evaluation through projected frontier and occupied evidence, so its proposal and scoring mechanism is tied to a coverage objective [Jia+25]. The same finite candidate table could instead be ranked by uncertainty reduction or by reconstruction error. Consequently, changing the utility changes the scientific task even when candidate generation and selection remain unchanged.

The NBV mechanism therefore answers only **how** a sensing choice is made. To interpret the choice, the utility must state **which reconstruction consequence** is valued and **for which spatial support**. This distinction leads from the general active-perception loop to the objective families compared next; only after fixing that objective can candidate support and feasibility be separated from view value [Che+24, Jia+25].

2.2 View Utility and Target-Conditioned Reconstruction

Coverage-based utilities value evidence about previously unobserved surface. SCONE estimates surface-coverage gain from occupancy and visibility fields, MACARONS anticipates coverage gain while reconstructing online, and Hestia rewards newly visible voxel faces [GML22, Gué+23, Lu+26]. These objectives connect a view to geometric completeness, but they weight every counted surface element through the coverage definition; they do not directly ask whether the reconstructed geometry became more accurate.

Information-based utilities instead value a view through its expected reduction of model uncertainty. FisherRF derives an information-gain criterion from the Fisher information of a radiance field [JLD24]. Such a criterion can prefer a view that constrains uncertain model parameters even when its immediate surface coverage is small. The benefit is model-aware exploration; the limitation is that lower parameter uncertainty and lower geometric reconstruction error are related only through assumptions about the model and the evaluated scene.

Direct quality utilities close this proxy gap by evaluating the reconstruction itself. VIN-NBV defines Relative Reconstruction Improvement (RRI) as the reduction in Chamfer distance between reconstructed and ground-truth point clouds after adding a candidate observation, and learns to rank sampled query cameras by this quantity [Fra+25]. The present work follows this direct reconstruction-improvement principle but not VIN-NBV's metric unchanged: it evaluates a target-cropped point–mesh error instead of scene-level point-cloud Chamfer distance. This project-specific definition makes the requested target, rather than all reconstructed geometry, determine which errors count. It does not remove the myopia of one-step scoring, because each candidate is still valued by its immediate improvement rather than by the sequence it may enable.

Target conditioning changes which reconstruction errors contribute to that quality. Object-centric 3DGS planning associates features and confidence with individual Gaussian primitives and gates its information score by the requested object [Jeo+26]. This mechanism demonstrates that a view can be informative for one object without being equally useful for the scene as a whole. It also exposes a separate assumption: supplying an object identifier or region for conditioning is not the same problem as discovering that object and maintaining its identity from observations.

The thesis therefore treats coverage and uncertainty as non-equivalent diagnostics and adopts target-specific reconstruction quality as the endpoint of interest [Fra+25, Jeo+26]. These works motivate direct reconstruction improvement and object conditioning separately; their combination in a target-cropped endpoint is the thesis-specific objective. This choice determines what the decision should improve, but not which viewpoints may be proposed or reached. Candidate support and motion feasibility therefore remain separate from utility before delayed consequences are introduced through the finite-horizon decision problem [Jia+25].

2.3 Candidate-View Support and Motion Feasibility

A finite-candidate policy selects only from the pose rows in its proposed candidate table $\mathcal{Q}_t = \{q_{t,i}\}_{i=1}^{N_t}$. PB-NBV uses a reachability- and camera-conditioned partial hemisphere of target-facing candidates [Jia+25]. Proposal geometry therefore bounds what even a perfect scorer can select.

Target-relative coordinates provide a useful proposal prior without defining the objective. PB-NBV points candidates toward the object centre, while Hestia factorizes a viewpoint into a predicted look-at point and a camera position [Jia+25, Lu+26]. Such parameterizations concentrate support on a requested target, but an orbit or target-facing arc remains a coverage geometry, not evidence that a wearer normally moves that way. Project Aria explicitly contrasts carefully curated scanning motion with natural, unconstrained egocentric activity, so wearable plausibility must be checked against trajectory evidence rather than inferred from a target-relative construction [Eng+23].

Geometric admissibility has at least two levels. Hestia shifts a proposed camera position to its nearest collision-free endpoint, whereas Next Best Sense obtains a list of feasible candidate views but still falls back to the next-ranked view when inverse kinematics or trajectory planning fails [Lu+26, Str+24a]. A collision-free endpoint therefore does not by itself establish that the transition from the current state is executable.

Physical feasibility and wearable plausibility are likewise different requirements. Robot reachability constrains PB-NBV's hemisphere, while Project Aria supplies calibrated device trajectories recorded during natural activities [Eng+23, Jia+25]. Robot-specific reachability can define a hard admissibility test for that platform; it cannot establish a distribution of realistic human head and body motion. For egocentric data generation, that distribution is an empirical prior to be measured, reported, and validated.

These distinctions map the proposed poses $q_{t,i} \in \mathcal{Q}_t$ to the canonical admissible row-index set $\mathcal{A}(s_t) = \{i : m_{t,i} = 1\}$, where $m_{t,i}$ is the hard-valid indicator after endpoint and transition checks. The utility or value function ranks only those admitted rows; a rejected candidate is unavailable rather than merely low-value [Jia+25, Str+24a]. Separating proposal, admission, and ranking makes the state dependence of available actions explicit [SB18, Secs. 3.1 and 3.5, pp. 47–53 and 58–62] and prepares the sequential decision model introduced next.

2.4 Sequential Decision-Making under Partial Observability

An egocentric reconstruction system never observes the complete physical scene. Its decision can depend on the sequence of earlier actions and observations, not only on the latest frame. In a partially observable Markov decision process, a belief state is a sufficient statistic of that action–observation history for policy choice [LHP23]. ARIA-NBV does not attempt general belief inference; the relevant consequence is narrower: any finite representation used for view selection is a hypothesis about which parts of the history are sufficient for predicting future target improvement.

The preceding section defined the admissible choices as the state-dependent row-index set $\mathcal{A}(s_t)$ after proposal and feasibility checks. That set can change as the target-conditioned reconstruction and current pose change, while a hierarchical controller can also condition camera position on an earlier look-at decision [Jia+25, Lu+26]. Sequential value must therefore be conditioned on the currently available action rather than treating every camera pose as permanently selectable.

Sequentiality matters when the best immediate view is not the best first step of a sequence. For target e, candidate i, and requested horizon h, the finite return and corresponding conditional value are

$$G_{t,e}^{(h)} = \sum_{k=0}^{h-1} \gamma^k r_{t+k}^e, \quad 1 \le h \le b_t \le H_{\max}$$

$$Q_{h,e}^{\star}(s_t, i) = \sup_{\pi \in \Pi^{\mathrm{act}}} \mathbb{E}_\pi \left[G_{t,e}^{(h)} \mid s_t, a_t = i \right], \quad i \in \mathcal{A}_t, \quad 1 \le h \le b_t \le H_{\max}, \quad Q_{0,e}^{\star}(s, i) = 0$$

where continuation policies choose only from the admissible candidates at each step. These equations specialize the standard discounted-return and action-value constructions to a requested target and finite horizon [SB18, Secs. 3.3–3.6, pp. 54–67]. Fixed-Horizon TD defines horizon-indexed predictions with the boundary $Q_0 = 0$ and relates horizon h to the shorter horizon $h - 1$ [De +20]. If an episode ends before all h rewards are collected, later rewards are zero. The value thus states the consequence of taking a candidate now and then following a named continuation rule; it is not an intrinsic property of the camera pose alone.

The horizon is distinct from two related bounds. The requested horizon h states how many future acquisition rewards the query represents; the remaining budget b_t limits how many acquisitions are still possible; and $H_{\max}$ marks the largest horizon supported by the chosen model and evidence. Universal Value Function Approximators establish the general possibility of conditioning one approximator on an additional task variable, while Fixed-Horizon TD shares representations across horizon-indexed predictions [De +20, Sch+15]. Neither result licenses extrapolation beyond the horizons actually supported and evaluated.

Learning such values from a fixed data collection introduces an epistemic constraint in addition to the mathematical horizon. Batch-constrained and conservative offline reinforcement-learning methods are motivated by the error that arises when a learned policy assigns high value to actions outside the data distribution [FMP19, Kum+20]. These methods differ in how they control that error, but they share the warning relevant here: a longer horizon does not create evidence for transitions that the data never resolves.

Together, partial observability, state-dependent choices, and finite-horizon return make candidate value explicitly relational [De +20, LHP23]. A camera pose has no context-free value. Its value depends on the causal information state, requested target, currently admissible support, horizon, state-update rule, and continuation policy. The next dependency is therefore representational. The actor state must preserve the distinctions on which those future consequences depend.

2.5 Egocentric and Geometric Representations

The preceding section established that future view value depends on a causal information state rather than on a camera pose alone. Project Aria provides calibrated, time-aligned egocentric streams, and EFM3D lifts posed image features and semidense geometry into a local gravity-aligned representation [Eng+23, Str+24b]. The representation question is therefore not how faithfully to reproduce the entire world, but which distinctions must survive so that target-specific counterfactual return remains predictable.

Pretraining provenance and decision-time role must be distinguished. 3D-NVS uses ImageNet initialization in its NBV classifier, whereas MACARONS reuses pretrained image features within an online mapping pipeline [AKC20, Gué+23]. Next Best Sense instead uses SAM2 and monocular-depth priors to improve 3DGS construction while retaining an analytic Fisher-information selector [Str+24a]. EVL freezes its DINOv2.5 foundation feature extractor, then learns the feature upsampler, 3D U-Net, and task heads that form a local 3D field from posed egocentric streams and semidense geometry [Str+24b]. This structure may preserve semantic and geometric distinctions relevant to target visibility, but whether those distinctions improve candidate value is an empirical property of the downstream task rather than a consequence of pretraining alone.

Four requirements follow. First, **causal sufficiency** requires the actor state to retain the observation history relevant to future return without importing future or unselected evidence. EFM3D supplies strong local evidence but has a finite voxel extent, while Hestia demonstrates that accumulated directional history can alter later choices [Lu+26, Str+24b]. A spatial state is therefore not sufficient merely because it is geometric; sufficiency is a hypothesis to be tested against the task.

Second, **spatial relationality** requires the target, observer, candidates, and observed support to be comparable in a common local geometry. Query-centric models express scene elements relative to the active query, reducing dependence on an arbitrary global origin [Zho+23]. For target-conditioned NBV, the relevant quantity is therefore not candidate position in isolation but candidate geometry relative to the current observer, target, and accumulated evidence. Coordinate choice also determines which changes should be treated as nuisances. Translating the world origin should not change a score, whereas gravity, metric scale, camera direction, target orientation, occlusion, and temporal order can remain informative [Bro+21]. The useful prior is thus relative geometry that removes arbitrary coordinates without erasing task-relevant physical structure.

Third, **candidate-order equivariance** follows from the action set itself. A candidate table denotes physical viewpoints rather than a ranked sequence, so permuting its rows must induce the same permutation of their scores; an invariant output would instead discard which score belongs to which action. Deep Sets provides invariant aggregation for shared set context, while Set Transformer provides permutation-equivariant candidate interaction [Lee+19, Zah+17]. This is a behavioral contract rather than an architecture prescription: independent row scoring, pooled context, and attention can all satisfy it.

Fourth, **physical observability** requires the representation to preserve what the causal sensing process actually distinguishes. EFM3D derives a surface mask from observed semi-dense points and a free-space mask from camera-to-surface rays [Str+24b]. Voxels supported by neither mask remain unobserved; they must not be treated as observed free space or negative surface evidence. This surface/free/unobserved distinction is the observation contract. The choice of carrier is a separate architecture tradeoff: point models retain irregular samples and local relations, sparse-voxel models regularize space at a chosen resolution and extent, and equivariant message passing commits to a chosen symmetry family [CGS19, SHW21, Zha+21]. These families trade spatial fidelity, computational structure, and strength of geometric prior, but none by itself guarantees the observation contract or improves the endpoint utility.

These four requirements—causal sufficiency, spatial relationality, candidate-order equivariance, and physical observability—complete the conceptual dependency chain. The literature synthesis can now compare methods by the scientific distinctions they preserve, while the Method chapter remains responsible for one concrete realization and its tests [Bro+21, Str+24b].

2.6 Related-Work Synthesis and Research Gap

The reviewed approaches differ along six coupled dimensions: utility, target scope, represented evidence, representation provenance and decision-time role, candidate support and admission, and decision horizon [Fra+25, GML22, Jia+25, JLD24]. Comparing only one dimension can obscure the scientific task: a sequential coverage controller does not answer the same question as a greedy reconstruction-quality scorer, and neither comparison is meaningful if the relevant view lies outside candidate support.

Family	Utility and target scope	State evidence and representation role	Support, feasibility, and decision
PB-NBV [Jia+25]	Projected frontier and occupied coverage; reconstruction object	Analytic classified voxel clusters and compact projection proxies	Reachability- and camera-conditioned partial hemisphere; greedy score
3D-NVS [AKC20]	Pairwise voxel reconstruction; object	ImageNet-pretrained VGG16 directly scores fixed next-view classes	Eleven fixed view classes; one-step classifier
SCONE / MACARONS [GML22, Gué+23]	Expected surface coverage; scene-wide	Task-trained occupancy and visibility; pretrained ResNet features support MACARONS depth	Iterative candidate selection
GenNBV / Hestia [Che+24, Lu+26]	Coverage gain; reconstruction object	Task-trained geometric, history, and shallow-image encoders; Hestia uses no external pretraining	Continuous or hierarchical policy; collision handling
FisherRF [JLD24]	Information gain; radiance-field scene model	Per-scene model uncertainty enters an analytic score	Candidate views; greedy score
VIN-NBV [Fra+25]	Direct reconstruction improvement; reconstructed object	Task-trained CNN over candidate-conditioned projections of current reconstruction	Sampled candidates; greedy score
Next Best Sense [Str+24a]	Color–depth information gain; radiance-field scene model	SAM2 and depth priors support 3DGS; Fisher selection remains analytic	Feasible candidates; fallback after kinematic or planning failure
Object-centric 3DGS [Jeo+26]	Object-conditioned information gain; requested object	Foundation-model masks and CLIP support object state and selection; target score remains analytic	Candidate views; greedy score

Table 2: Concept-centred comparison of the literature families that bound the thesis question. Rows are distinguished by utility, target scope, state evidence and representation role, candidate support and feasibility, and decision structure; they are not ranked by reported performance across incompatible settings.

Pretraining alone does not determine the scientific role of a representation. 3D-NVS uses ImageNet initialization directly in a fixed-view classifier, while MACARONS uses pretrained image features inside an online mapping pipeline [AKC20, Gué+23]. Next Best Sense places foundation-assisted masks and depth upstream of an analytic Fisher selector [Str+24a]. By contrast, VIN-NBV, GenNBV, and Hestia learn their decision state from task-specific reconstruction cues or shallow encoders [Che+24, Fra+25, Lu+26]. These approaches establish different uses for transferred visual features, but they do not isolate the value of an egocentric 3D representation that combines frozen 2D foundation features with learned 3D processing for target-conditioned endpoint prediction.

The comparison yields two requirements for the value definition. First, **utility alignment** requires the score to reflect the endpoint of interest: coverage and information criteria reward proxies for reconstruction, whereas VIN-NBV evaluates realized reconstruction error [Fra+25, GML22, JLD24]. Second, **temporal dependence** requires the score to represent delayed consequences. Sequential policies retain history and optimize delayed reward, but the reviewed examples optimize coverage rather than the reconstruction quality of a requested target [Che+24, Lu+26]. Existing work therefore establishes direct quality, target conditioning, and sequentiality separately under different assumptions.

A third requirement is **support/state adequacy**: valuable actions must be available, and the causal state must retain the consequences needed to value later actions. PB-NBV constrains proposal support; Hestia carries directional history while separately repairing infeasible endpoints; and Next Best Sense distinguishes feasible candidates from execution fallback [Jia+25, Lu+26, Str+24a]. Even a perfect scorer cannot exploit a view outside its support, and a longer horizon cannot recover a distinction erased from the state.

Within the actor-state question, representation transfer is a downstream diagnostic. A pretrained carrier can help only if it preserves distinctions aligned with target-specific return; semantic strength in 2D does not by itself provide causal scene memory, metric support, or candidate-relative geometry. EFM3D/

EVL supplies a foundation-derived local field in which a frozen 2D feature extractor feeds learned upsampling, 3D processing, and task heads alongside observed surface and free-space evidence [Str+24b]. Whether that field improves target-conditioned candidate value must therefore be measured against matched actor-visible geometric controls after the core evidence gates, rather than inferred from pretraining.

This thesis studies the core requirements jointly within a bounded setting: whether a finite-candidate policy can improve endpoint reconstruction quality for a specified target by valuing more than the next observation. Actor-visible foundation-derived egocentric evidence is then a conditional representation diagnostic for the corresponding value model [De +20, Fra+25, Str+24b]. Among the reviewed reconstruction-NBV methods, none establishes this conjunction: direct-quality prediction, target conditioning, bounded lookahead, and a matched test of a foundation-derived egocentric representation with frozen 2D features and learned 3D processing. The literature therefore cannot determine whether non-myopic headroom exists or whether EFM3D/EVL-derived evidence improves its recovery; the latter remains a conditional diagnostic rather than another principal question.

The resulting principle is relational rather than architectural: view value is defined by a target, a causal state, admissible support, a horizon, a state update, and a continuation rule. The thesis does not generalize this bounded relation to exhaustive novelty priority, continuous control, actor-visible target discovery, closed-loop motion execution, or natural human movement [Che+24, Eng+23, Lu+26, Str+24a]. Chapter 3 makes the bounded relation operational by separating observable state from the privileged geometry used to construct tasks and evaluate outcomes.

3 Oracle and Data Generation

This chapter defines the experimental objects from which ARIA-NBV constructs training and evaluation data. A logged snippet first induces an actor-side information state s_t^{hist}. A target task then turns that state into a finite candidate table $\mathcal{Q}_t$ the hard action mask $m_{t,i}^{\mathrm{act}}$ selects its feasible subset. Privileged rendering assigns target-specific reconstruction outcomes, while a selected action creates the only factual successor that may extend the causal history. Replay storage preserves these relations. Final padded tensors are model-specific projections.

State determines what may be conditioned on. The target and proposal mechanism determine which actions can be compared. The oracle defines their supervision, and lineage determines which multi-step returns can be reconstructed from factual transitions. These dependencies connect the chapter's information, action, measurement, and storage contracts to the learned scorer in the Method chapter.

3.1 State as an Information Contract

An NBV state contains exactly the information on which an action value may depend. ARIA-NBV therefore distinguishes three nested roles: logged evidence from the observed snippet, causal evidence produced by actions that were actually selected, and privileged quantities used only to construct or evaluate counterfactuals. ASE provides both sensor-like streams and ground truth, whereas EFM3D transforms only logged image evidence into a local 3D representation [Met25, Str+24b]. Provenance and acquisition time determine information role independently of file location and tensor shape.

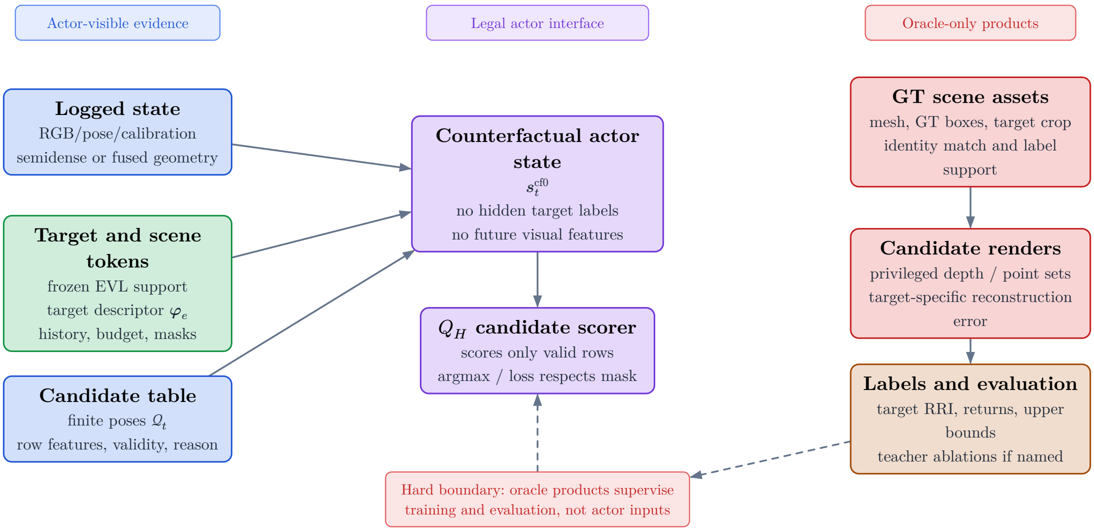


Figure 1: Information roles in the offline experiment. The actor side contains calibrated logged observations, derived EFM3D evidence, an explicitly sourced target instruction, the selected-view prefix, remaining budget, candidates, and the hard action mask. The oracle side adds GT geometry, counterfactual renders, labels, and endpoint evaluation. Every stored field keeps its declared information role.

The boundary is also temporal. A render from an unselected candidate is a counterfactual oracle query. After that candidate is selected, an observation from its pose may enter the next state, but only under a declared acquisition model: a real observation, an admitted sensor simulation, or a privileged mesh render. Actor visibility requires both factual selection and an admitted source role.

3.1.1 Logged Information State

The logged historic state s_t^{hist} groups the evidence available before any counterfactual acquisition:

$$s_t^{\mathrm{hist}} = \left(\boldsymbol{I}^{\mathrm{rgb}}, \boldsymbol{X}, \mathcal{P}^{\mathrm{semi}}, \boldsymbol{F}_v, e_t, b_t\right) \in \mathcal{S}^{\mathrm{hist}}$$

Calibrated images and trajectory poses establish where and when evidence was acquired. Semi-dense points provide sparse surface observations, and the EFM3D field $\boldsymbol{F}_v$ supplies a learned local representation. EFM3D lifts posed image features into a finite gravity-aligned local volume [Str+24b]. Absence from that volume denotes missing representation support. Empty-space evidence requires measured rays, and pose feasibility requires the geometric and motion checks defined later in this chapter.

Semi-dense points $\mathcal{P}^{\text{semi}}$ retain uncertainty, timestamps, and observing-camera lineage. EFM3D uses that lineage to distinguish surface samples from free-space samples along measured rays [Met25, Str+24b]. A point set supplies surface samples; camera rays additionally supply free-space evidence. Missing surface samples carry no free-space meaning without that lineage.

The target e_t and remaining budget b_t are task variables. They make the value conditional, conceptually $Q(s_t, e_t, a_t)$, and distinguish two otherwise identical observations posed with different requested objects or acquisition budgets. In the present oracle task, e_t is derived from a selected GT box. A deployable task derives the target instruction from an observation-side proposal and association path.

3.1.2 Causal Successor State

After an action is selected, the state may grow only with evidence generated by that factual choice. The conceptual counterfactual actor state is

$$s_t^{\text{cf0}} = \big(\boldsymbol{F}_v^{\text{root}}, \mathcal{P}_t, \mathcal{Q}_t, m_{t,i}, \rho_{t,i}, e_t, b_t\big) \in \mathcal{S}^{\text{cf0}}$$

and is read together with the ordered selected-view history:

$$\boldsymbol{h}_{t,e,i} = \text{Read}\big(\boldsymbol{\Phi}_t^{\text{scene}}, \boldsymbol{h}_e^{\text{tgt}}, q_{t,i}, \boldsymbol{H}_t, t, H\big)$$

Here the root field is immutable, while the accumulated evidence may grow only through selected observations. Candidate rows $q_{t,i}$ encode possible poses and proposal provenance. Observations from those poses become available only after selection. The hard mask $m_{t,i}^{\text{act}}$ defines the action set. Invalid-reason codes explain exclusions as diagnostic fields outside the value-model input. This separation prevents the proposal table from leaking the consequences it is supposed to predict.

The implemented `S0-pose` baseline materializes the weaker state

$$s_t^{\text{S0-pose}} = \big(\mathcal{S}_{\text{root}}^{\text{VIN}}, \boldsymbol{F}_v^{\text{root}}, \big(\boldsymbol{T}_{r,e}, \boldsymbol{l}_e\big), \mathcal{Q}_t, m_{t,i}, \boldsymbol{H}_t^{\text{pose}}, b_t\big)$$

It carries root evidence, target context, the current candidate table and mask, the selected-pose prefix, and the remaining budget. This state represents the actor's motion history. A geometry-enriched successor additionally retains what the selected observations revealed:

$$s_t^{\mathrm{cf+}} = \left(s_t^{\mathrm{cf0}}, \left(\boldsymbol{D}, \boldsymbol{V}, \mathcal{P}^{\mathrm{cf}}, \boldsymbol{n}\right)_{1:t}\right)$$

The implemented richer control is the intermediate carrier

$$s_t^{\text{CF-GT-carrier}} = \left(s_t^{\text{S0-pose}}, \left(\boldsymbol{D}, \boldsymbol{V}, \mathcal{K}, \boldsymbol{T}_{\text{root<-cam}}\right)_{1:t}^{\text{sel}}\right)$$

It retains calibrated depth only for selected actions. This makes the carrier causal with respect to the selected path, while its mesh-rendered source keeps the control privileged. Fusion into surface, free-space, uncertainty, and EFM3D evidence defines the fuller geometry state. The implemented carrier isolates the value of selected depth before that fusion.

More generally, the transition from t to $t+1$ may consume only the observation produced by a_t. RGB and EFM3D updates require the corresponding calibrated image stream; depth updates the declared geometry channels. s_t^{cf0} is an **information state** for value prediction. Its sufficiency for the physical process is an empirical question: it is adequate only if it preserves the distinctions needed to predict future target-specific return.

3.1.3 Oracle Augmentation

The oracle augments the causal state with quantities needed to answer counterfactual questions:

$$s_t^{\mathrm{oracle}} = \left(s_t^{\mathrm{cf+}}, \mathcal{M}^{\mathrm{GT}}, \mathcal{M}_e^{\mathrm{GT}}, \boldsymbol{D}_q, \mathcal{P}_q, \mathrm{RRI}\right) \in \mathcal{S}^{\mathrm{oracle}}$$

ASE provides per-frame metric GT depth aligned with RGB, per-pixel instance identifiers, class mappings, and the scene trajectory. The EFM3D release adds ASE OBB metadata and validation meshes for object-detection and surface-reconstruction supervision [Met25, Str+24b]. EFM3D uses depth to supervise occupancy at sampled free, surface, and behind-surface points, while visible OBBs supervise centerness, class, and box geometry [Str+24b]. These uses establish label provenance; the information source determines actor observability.

For ARIA-NBV, the scene mesh $\mathcal{M}^{\mathrm{GT}}$ supports candidate rendering, and the target crop $\mathcal{M}_e^{\mathrm{GT}}$ fixes the surface on which error is measured. All-candidate depth $\boldsymbol{D}_q$ and backprojected points $\mathcal{P}_q$ instantiate the hypothetical observation associated with each candidate. The resulting RRI labels the consequence of an action and becomes available only to the oracle.

A GT OBB defines oracle identity, pose, extent, crop, and target bearing. Actor detection and association require an observation-derived proposal. In a deployable protocol, that proposal defines the target hypothesis, while ASE identity and geometry serve as matching and evaluation references.

The three states form a controlled augmentation chain: s_t^{hist} fixes logged evidence, s_t^{cf0} adds only factual selected history, and s_t^{oracle} adds counterfactual supervision. This ordering is the conceptual data contract inherited by candidate generation and replay.

3.1.4 Visibility, Feasibility, and Utility

Three predicates apply to a candidate and answer different questions. **Visibility** asks whether a view supplies evidence about a spatial element. **Feasibility** asks whether the pose belongs to the admissible action set. **Utility** asks how the target reconstruction changes after an admissible action. A candidate can be physically feasible yet outside the local EFM3D volume, or it can frame the target yet yield little reconstruction gain. These cases coexist because the predicates refer to different structures.

Accordingly, $m_{t,i}^{\mathrm{act}}$ owns infeasibility, and reward is defined only on the feasible action set. A feasible row may have negative gain. Oracle evaluation can fail for a separate reason—for example, an unusable target crop or render—in which case the label mask $m_{t,i}^{Q,h}$ is false while the geometric reason code is unchanged. Keeping the masks distinct preserves the difference between actions outside the feasible set and feasible actions lacking an oracle training label.

Actor visibility is a property of the complete pipeline. It includes the scorer tensor, the source of the target, the construction and masking of candidate support, the selected observation used for the successor, and the fields passed to the model. A scorer with actor-only tensors can still depend indirectly on privileged support construction. Such a protocol supports an oracle-assisted experiment. Claims about independently actor-constructed actions require an actor-side proposal and feasibility path.

3.2 Task and Action Construction

An action has value only relative to a target and to the alternatives available in the same state. Data generation therefore begins by constructing a **target task** and a finite **candidate shell**. The target task fixes the requested entity e, its evaluation support $\mathcal{M}_e^{\mathrm{GT}}$, and the instruction used by target-bearing proposal families. Its reusable descriptor is

$$\boldsymbol{\varphi}_e = \mathrm{Enc}_{\mathrm{tgt}}\left(\hat{\boldsymbol{B}}_e, \hat{\boldsymbol{y}}_e, \hat{\pi}_e, A_e^{\mathrm{proj}}, n_e^{\mathrm{semi}}, n_e^{\mathrm{EVL}}, \omega_e^{\mathrm{EVL}}, \ell_e^{\mathrm{src}}, \boldsymbol{T}_{r_t,e}, \boldsymbol{T}_{c_t,e}\right)$$

where geometry, class evidence, observation support, source role, and target-relative transforms stay explicit. In the present oracle protocol these fields originate from a selected GT OBB. The descriptor therefore defines a controlled target-conditioned task. Actor-visible target discovery is evaluated through the separate proposal-association path below.

At decision step t, the proposal mechanism produces $\mathcal{Q}_t$. Each row $q_{t,i}$ represents one possible camera pose together with its proposal provenance. The hard mask $m_{t,i}^{\mathrm{act}}$ then induces the feasible action set

$$\mathcal{Q}_t = \left\{q_{t,i}\right\}_{i=1}^{N_q}, \quad \mathcal{A}_t = \left\{i \in \left\{1, ..., N_q\right\} : m_{t,i}^{\mathrm{act}} = 1\right\}$$

Candidate generation and feasibility are deliberately separate operations. The proposal distribution determines which parts of the local action space are represented; the mask determines which proposed rows may be selected. Invalid rows are retained for action-space coverage and failure analysis. Selection, stochastic normalization, loss targets, and bootstrap maximization use the feasible subset. A feasible row with poor target gain is a valid low-utility example.

3.2.1 Target Selection

The oracle task sampler enumerates non-padding GT OBBs with finite positive geometry and samples them uniformly without replacement up to the manifest-defined per-snippet cap. This choice makes task selection independent of candidate reward: near-solved, difficult, and negative-gain targets are eligible. The stored `matched` status records a geometry-valid oracle task. Successful actor proposal-to-identity matching is a separate observed-target event.

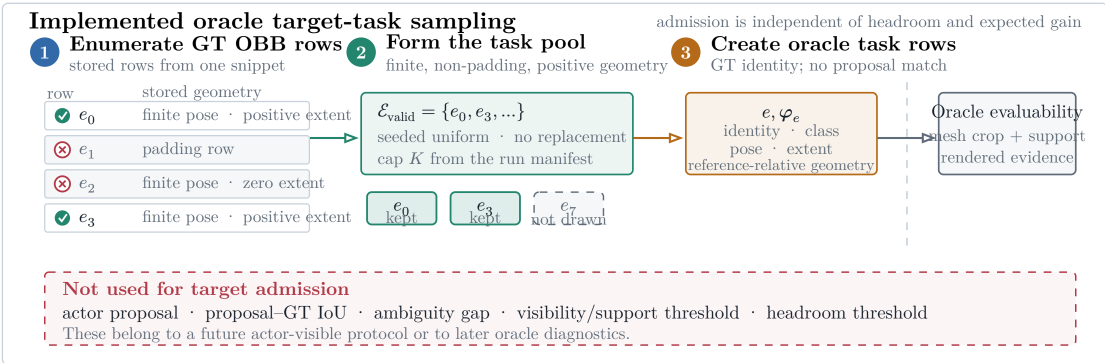


Figure 2: Oracle target-task construction. Geometry-valid GT OBB rows form the task population; seeded uniform sampling without replacement applies the manifest-defined cap. Candidate scoring later determines whether the selected target crop is evaluable.

This sampler controls computational cost and keeps the task population independent of oracle headroom. Actor-visible target selection uses observation-derived proposals and association as a distinct stage. Oracle task admission supplies the controlled reference population for that evaluation.

3.2.2 Observed-Target Admission

The observed-target matcher implements that distinct stage. Given an observed or predicted proposal $\hat{e}$, it compares only same-class reference OBBs using oriented three-dimensional IoU,

$$\mu_{\text{IoU}}(\hat{e}, e) = \text{IoU}_{3\text{D}}\left(\hat{\boldsymbol{B}}_{\hat{e}}, \boldsymbol{B}_e\right)$$

$$\tau_{\text{IoU}} = 0.20$$

A reference row qualifies only when its IoU is strictly greater than 0.20. The proposal is admitted exactly when one reference row qualifies:

$$n_{\mathrm{qual}}(\hat{e}) = \mathrm{card}(\{e \in \mathcal{E} : \kappa(\hat{y}_{\hat{e}}, y_e) = 1 \wedge \mu_{\mathrm{IoU}}(\hat{e}, e) > \tau_{\mathrm{IoU}}\})$$

$$a_{\mathrm{id}}(\hat{e}) = 1 \text{ iff } n_{\mathrm{qual}}(\hat{e}) = 1$$

Equality at the threshold, no qualifying row, and multiple qualifying rows leave the proposal unresolved. The rule is intentionally simple: it turns an observation proposal into an unambiguous evaluation identity. Association confidence is a separate diagnostic, and the current oracle task sampler operates directly on GT rows.

3.2.3 Candidate View Generation

Candidate generation is a proposal distribution over a bounded local camera motion space. It determines the alternatives over which the NBV scorer ranks actions. At step t, it constructs the shell

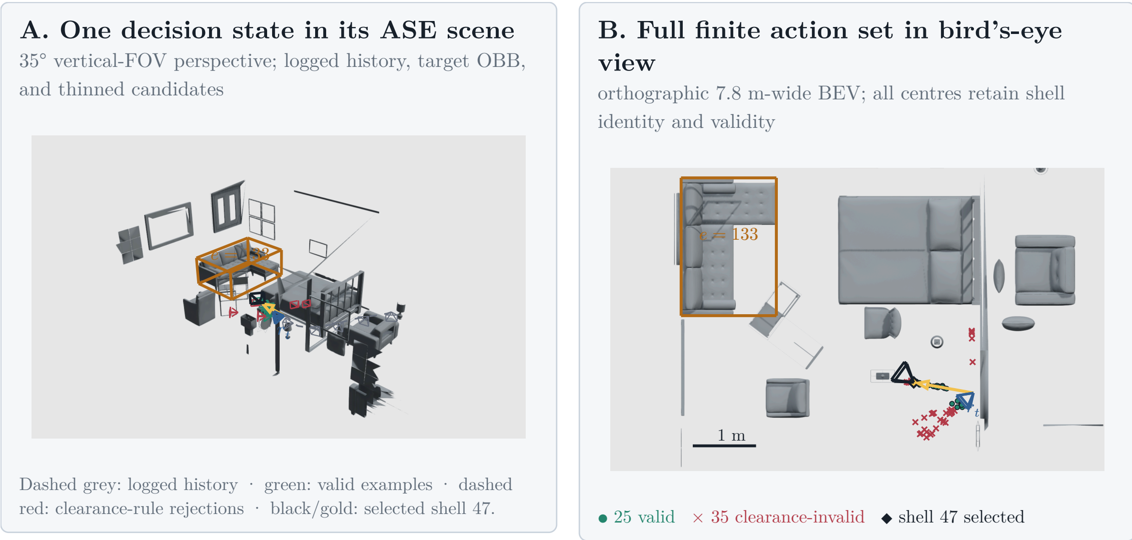
$$\mathcal{Q}_t = \{q_{t,i}\}_{i=1}^{N_q}, \quad N_q = 60$$

and records a family label $k(i)$ for every row. The frozen `realistic_core_60` profile allocates 24 forward-local, 24 target-bearing-local, and 12 lateral-target-bypass rows. The three families encode complementary geometric hypotheses:

Family	Geometric role	Profile support
Forward-local	Preserve egocentric motion continuity while perturbing the current forward direction.	24 rows; radius $[0.25, 1.1]$ m
Target-bearing-local	Move approximately along the current target bearing and orient the camera towards the target.	24 rows; radius $[0.4, 1.1]$ m
Lateral-target-bypass	Introduce side-step parallax while retaining a component towards the target.	12 rows; radius $[0.25, 1.1]$ m

Table 3: Geometric roles of the three families in the frozen `realistic_core_60` intervention. Counts and radii are specific to this profile.

Optional refinement, revisit, and target-orbit families activate only in challenger profiles. Because a profile changes the represented action support, its family counts, radii, angular caps, and pruning parameters belong to the resolved run manifest. A comparison fixes that manifest across methods.



Scene 81286, sample `ASE_81286_Atek_000035`, rollout row 73, step row 121 24 forward-local + 24 target-bearing
1. 12 lateral-bypass candidates. Dense mesh layers are
z-buffered raster; OBBs, paths, centres, and wire frusta remain vector geometry.

Figure 3: One stored candidate shell from ASE scene 81286 and sample `ASE_81286_Atek_000035`. The perspective panel relates the shell to the logged camera history, target OBB, and selected prefix. The bird's-eye panel retains all 60 candidate centres: 25 are feasible, 35 fail clearance, and oracle-greedy selects row 47. The example's evidential role is proposal support and hard feasibility.

Each row begins with a random unit direction in the reference rig frame. The frozen profile uses a forward-biased Power Spherical distribution with $\kappa = 8$ [DAT20]:

$$\boldsymbol{u}_i \sim \mathrm{PS}(\boldsymbol{e}_z, \kappa), \quad p(\boldsymbol{u}) = c_\kappa \left(1 + \boldsymbol{e}_z^T \boldsymbol{u}\right)^\kappa, \quad \kappa = 8$$

The sampled direction is then mapped into the configured azimuth and elevation caps. With $\psi = \operatorname{atan2}(u_x, u_z)$ and $u_y = \sin\theta$, the transform is

$$\psi' = \psi \frac{\Delta_\psi}{2\pi}, \quad y' = \sin\theta_{\min} + \frac{u_y + 1}{2} \cdot (\sin\theta_{\max} - \sin\theta_{\min})$$

$$\boldsymbol{d}_i^0 = \left(\sqrt{1 - y'^2} \sin\psi', y', \sqrt{1 - y'^2} \cos\psi'\right), \quad \left\|\boldsymbol{d}_i^0\right\|_2 = 1$$

The cap controls angular extent without changing the row count. Family-specific maps then reinterpret the same bounded random variable relative to rig forward, the target bearing, or the lateral direction. This factorization separates **diversity within a family** from **the geometric meaning of the family**.

Let $\boldsymbol{f} = \boldsymbol{e}_z$ denote rig forward, $\boldsymbol{b}_e$ the supplied target bearing, $\boldsymbol{l}_e$ its normalized horizontal lateral direction, and $\boldsymbol{e}_y$ world up in the sampling frame. The three core maps are

$$\boldsymbol{g}_i^{\text{forward}} = \boldsymbol{f} + \alpha_f\big(\boldsymbol{d}_i^0 - (\boldsymbol{d}_i^0 \cdot \boldsymbol{f})\boldsymbol{f}\big), \quad \boldsymbol{d}_i^{\text{forward}} = \frac{\boldsymbol{g}_i^{\text{forward}}}{\left\|\boldsymbol{g}_i^{\text{forward}}\right\|_2}, \quad \alpha_f = 0.45$$

$$\boldsymbol{g}_i^{\text{target}} = \boldsymbol{b}_e + \alpha_t\big(\boldsymbol{d}_i^0 - (\boldsymbol{d}_i^0 \cdot \boldsymbol{b}_e)\boldsymbol{b}_e\big), \quad \boldsymbol{d}_i^{\text{target}} = \frac{\boldsymbol{g}_i^{\text{target}}}{\left\|\boldsymbol{g}_i^{\text{target}}\right\|_2}, \quad \alpha_t = 0.4$$

$$\boldsymbol{g}_i^{\text{bypass}} = 0.55\boldsymbol{b}_e + 0.85\,\text{sign}\big(d_{i,x}^0\big)\boldsymbol{l}_e + \text{clip}\Big(d_{i,y}^0, -0.35, 0.35\Big)\boldsymbol{e}_y, \quad \boldsymbol{d}_i^{\text{bypass}} = \frac{\boldsymbol{g}_i^{\text{bypass}}}{\left\|\boldsymbol{g}_i^{\text{bypass}}\right\|_2}$$

A family-conditioned radius turns direction into translation; the reference pose then maps that offset into world coordinates:

$$r_i \sim \mathcal{U}\left(r_{\min}^{k(i)}, r_{\max}^{k(i)}\right), \quad \boldsymbol{c}_i^w = \boldsymbol{T}_r^w\left(r_i \boldsymbol{d}_i^{k(i)}\right)$$

Forward-local rows retain the reference orientation. Target-looking rows use the supplied target centre $\boldsymbol{p}_e$ to construct an orthonormal look-at frame:

$$\boldsymbol{z}_i^w = \frac{\boldsymbol{p}_e - \boldsymbol{c}_i^w}{\|\boldsymbol{p}_e - \boldsymbol{c}_i^w\|_2}, \quad \boldsymbol{g}_i^{\text{up}} = \boldsymbol{e}_y - \big(\boldsymbol{e}_y^T \boldsymbol{z}_i^w\big)\boldsymbol{z}_i^w, \quad \boldsymbol{y}_i^w = \frac{\boldsymbol{g}_i^{\text{up}}}{\|\boldsymbol{g}_i^{\text{up}}\|_2}, \quad \boldsymbol{x}_i^w = \boldsymbol{y}_i^w \times \boldsymbol{z}_i^w$$

The target centre in these equations is privileged under the present oracle protocol. A deployable candidate generator must replace it with an observation-derived target estimate. These equations specify the sampling prior; oracle opportunity and downstream policy evidence evaluate the utility of the actions that survive feasibility pruning.

The optional `target_orbit` challenger makes this distinction explicit. It proposes paired signed angular offsets at the current horizontal target standoff, thereby testing bilateral partial-orbit coverage. Pruning can remove one side, so proposal symmetry and feasible symmetry are measured separately. The family serves as a geometric ablation and contains no learned human-motion model.

Pruning evaluates each proposal independently of its oracle reward. A row is feasible only if it lies in the snippet support, stays clear of the GT mesh, admits a collision-free straight path, and satisfies the local motion bounds

$$\|\boldsymbol{o}_i\|_2 \leq 1.0 \text{ m}, \quad |\Delta h_i| \leq 0.25 \text{ m}, \quad \max\big(0, -o_{i,z}\big) \leq 0.25 \text{ m}, \quad \Delta\psi_i \leq 70 \text{ deg}$$

The full shell retains proposal probability, family, rule masks, diagnostics, and invalid-reason bitsets. The hard mask then produces $\mathcal{A}(s_t)$ while retaining rejected rows for analysis. A separate root-support gate rejects a task if pruning leaves too few alternatives:

$$N_{\text{valid}} \geq \max\big(12, \text{ceil}\big(0.25N_q\big)\big)$$

This gate protects the statistical support of the decision problem: one or two surviving actions provide insufficient variation for a candidate-ranking test. Oracle opportunity measures the utility available among the surviving candidates. Rejected roots and family-specific failure reasons enter the dataset population report.

3.2.4 Candidate-Support and Geometric-Coverage Diagnostics

Learning can rank only the alternatives supplied by candidate generation. The proposal profile is therefore part of the experimental design, and its support must be characterized before reward or policy results are interpreted. Let $\mathcal{I}_s$ contain all attempted non-padding rows, let $\mathcal{V}_s \subset \mathcal{I}_s$ contain the feasible rows selected by $m^{\text{act}}_{t,i}$, and let $\mathcal{L}_s \subset \mathcal{V}_s$ contain feasible rows with finite oracle labels. These nested populations separate three questions: what was proposed, what could be selected, and what could supervise learning.

The actor-valid fraction

$$f_{\text{actor}(s)} = \frac{\sum_{i \in \mathcal{I}_s} m^{\text{act}}_{s,i}}{|} \mathcal{I}_s|$$

measures pruning severity, whereas

$$n_{\text{valid}(s)} = \big|\big\{i \in \mathcal{I}_s : m^{\text{act}}_i = 1\big\}\big|$$

counts the number of alternatives that survive pruning. The distinction matters because two profiles can retain the same fraction while presenting very different numbers of choices. Family collapse is measured separately by

$$z_{\mathrm{family}(s)} = \left(\frac{1}{|}\mathcal{F}_s|\right) \sum_{f \in \mathcal{F}_s} \mathbb{1}\left[n_{\mathrm{valid}(s,f)} = 0\right]$$

which is one exactly when every configured family has zero feasible rows and zero only when every family survives. It detects a failure that a pooled valid fraction can hide: a large forward family may dominate the row count even when all target-conditioned families disappear.

Because many states can originate from one scene, row-level pooling would let large scenes dominate the study. Every state statistic $q(s)$ is therefore averaged within scene and then macro-averaged across scenes:

$$\mathcal{S}_{c,q} = \{s \in \mathcal{S}_c : q(s) \ \text{ is defined}\}, \quad |(q)_c = \left(\frac{1}{|}\mathcal{S}_{c,q}|\right) \sum_{s \in \mathcal{S}_{c,q}} q(s), \quad \mathcal{C}_q = \left\{c \in \mathcal{C} : |\mathcal{S}_{c,q}| > 0\right\}, \quad |(q) = \left(\frac{1}{|}\mathcal{C}_q|\right) \sum_{c \in \mathcal{C}_q} |(q)$$

The metric-specific set $\mathcal{S}_{c,q}$ makes undefinedness explicit. Failed generation roots contribute zero support and complete family collapse, while a directional statistic is undefined if no relevant direction was attempted. Undefined states and scenes are reported separately, and paired profile contrasts use the shared eligible-scene intersection. This preserves the difference between absence of support and absence of a defined measurement.

Support geometry is compared in the target-relative normalized frame $d_{t,e}^{\mathrm{current}} = \left\|\boldsymbol{p}_e^w - \boldsymbol{c}_{r_t}^w\right\|_2$, $\tilde{\boldsymbol{c}}_{t,i}^{\mathrm{support}} = \left(\boldsymbol{B}_{r,t}^{\mathrm{Z\text{-}up}}\right)^\top \frac{\boldsymbol{c}_{t,i}^w - \boldsymbol{c}_{r_t}^w}{d_{t,e}^{\mathrm{current}}}$, $\tilde{\boldsymbol{p}}_{t,e}^{\mathrm{support}} = \left(\boldsymbol{B}_{r,t}^{\mathrm{Z\text{-}up}}\right)^\top \frac{\boldsymbol{p}_e^w - \boldsymbol{c}_{r_t}^w}{d_{t,e}^{\mathrm{current}}}$, $\left\|\tilde{\boldsymbol{p}}_{t,e}^{\mathrm{support}}\right\|_2 = 1$. The factual expansion pose is the origin, the target lies at unit distance, and world up fixes the vertical axis. The normalization removes absolute standoff while preserving whether candidates move towards, around, above, or behind the target. This coordinate system describes candidate motion; calibrated projection and visibility are measured in their respective image and observation domains.

For attempted target-conditioned rows, side balance

$$> \varepsilon\Big], \quad n_-(s) = \sum_{i \in \mathcal{J}_s^{\text{target}}} \mathbb{1}\Big[y_{s,i}^{\text{target}} < -\varepsilon\Big], \quad n_0(s) = \sum_{i \in \mathcal{J}_s^{\text{target}}} \mathbb{1}\Big[\Big[y_{s,i}^{\text{target}}| \leq \varepsilon\Big], \quad b_{\text{side}(s)} = 1 - |n_+(s) - n_-(s)\frac{|}{n_+(s) + n_-(s}$$

compares positive and negative target-relative sides after excluding the neutral band $|y_{s,i}^{\text{target}}| \leq 10^{-9}$. Circular orbit span

$$\alpha_{s,1} \leq \ldots \leq \alpha_{s,n_s}, \quad \alpha_{s,n_s+1} = \alpha_{s,1} + 2\pi, \quad s_{\text{orbit}(s)} = 2\pi - \max_{1 \leq j \leq n_s} \left(\alpha_{s,j+1} - \alpha_{s,j}\right)$$

measures the smallest angular arc covering the attempted directions. These statistics characterize proposal diversity before pruning. Family survival and oracle opportunity quantify the later feasibility and utility stages.

For evaluated candidates, the projection fraction

$$f_{\text{proj}(s)} = \frac{\sum_{i \in \mathcal{J}_s^{\text{evaluated}}} \mathbb{1}\left[i \in \mathcal{J}_s^{\text{projected}}\right]}{|}\mathcal{J}_s^{\text{evaluated}}|, \quad |\mathcal{J}_s^{\text{evaluated}}| > 0$$

records whether the target centre falls inside the calibrated image domain. It measures geometric framing. Visibility additionally requires object extent and occlusion evidence. The finite-support oracle opportunity

$$o_{\text{oracle}(s)} = \max_{i \in \mathcal{L}_s} g_{s,i}^{\text{target-root}}, \quad |\mathcal{L}_s| > 0$$

is the best target-root gain present in $\mathcal{L}_s$. It measures whether the finite shell contains a useful action, independent of which policy selects it. Low opportunity identifies an action-support limitation; high opportunity with poor policy return instead leaves room for a ranking limitation.

Finally, jitter compliance checks whether within-family perturbations obey their declared angular caps:

$$f_{\text{jitter}(s)} = \frac{\sum_{i \in \mathcal{J}_s^{\text{jitter}}} \mathbb{1}[b_i = 1] \cdot \mathbb{1}[|\Delta\psi_i| \leq |(\psi)_i] \cdot \mathbb{1}[|\Delta\theta_i| \leq |(\theta)_i]}{\sum_{i \in \mathcal{J}_s^{\text{jitter}}} \mathbb{1}[b_i = 1]}, \quad \sum_{i \in \mathcal{J}_s^{\text{jitter}}} \mathbb{1}[b_i = 1] > 0$$

Uncapped spherical profiles form a different support class and are reported separately. Jitter compliance provides generator quality control; visibility and policy quality use their own measurements.

Quantity	Population	Unit	Interpretation boundary
Actor-valid fraction	Attempted rows per state	fraction	Hard feasibility before label and training filters.
Valid support	Actor-valid rows per state	rows	Report the distribution, lower tail, and failed roots.
Configured-family zero rate	Configured families per state	fraction	Retains absent and zero-support families; state values are scene-macro reduced.
Side balance / circular span	Non-neutral attempted target-conditioned rows	fraction / angle	Neutral rows and undefined states are reported separately; proposal coverage precedes pruning.
Target-centre projection	Evaluated candidate views per state	fraction	Calibrated framing; zero-evaluated states are reported as undefined.
Oracle opportunity	Finite actor-valid rewards per eligible state	gain	Available one-step headroom; undefined when no finite actor-valid label exists.
Jitter compliance	Bounded-jitter rows per state	fraction	State–scene–cohort macro; report undefined, nonzero, and uncapped support separately.

Table 4: Candidate-generation diagnostics and their aggregation populations. Every state-level quantity is reduced before scene and cohort aggregation.

3.2.5 Rollout Branch Sampling and Dataset Impact

Rollout recipes convert feasible candidate shells into factual training chains. Uniform sampling explores support without a utility preference; one-step oracle greedy selects the largest immediate target gain; bounded oracle lookahead ranks finite action sequences; and temperature-softmax samples according to oracle score while retaining stochastic diversity. These recipes define the behavior distribution from which replay is collected. They serve as data-generation policies and reference strategies. Chapter 5 evaluates the learned policy on held-out tasks.

Lookahead can select a different first action from one-step greedy because an action changes both the causal state and the next candidate shell (Figure 4). It ranks the return of retained chains, so future rewards can outweigh the first reward. This is decision-time planning: computation expands possible consequences to improve the current decision [SB18, Sec. 8.8, pp. 180–181]. The store preserves selected and beam-retained chains with their per-state shells. Search expands the other counterfactual branches transiently.

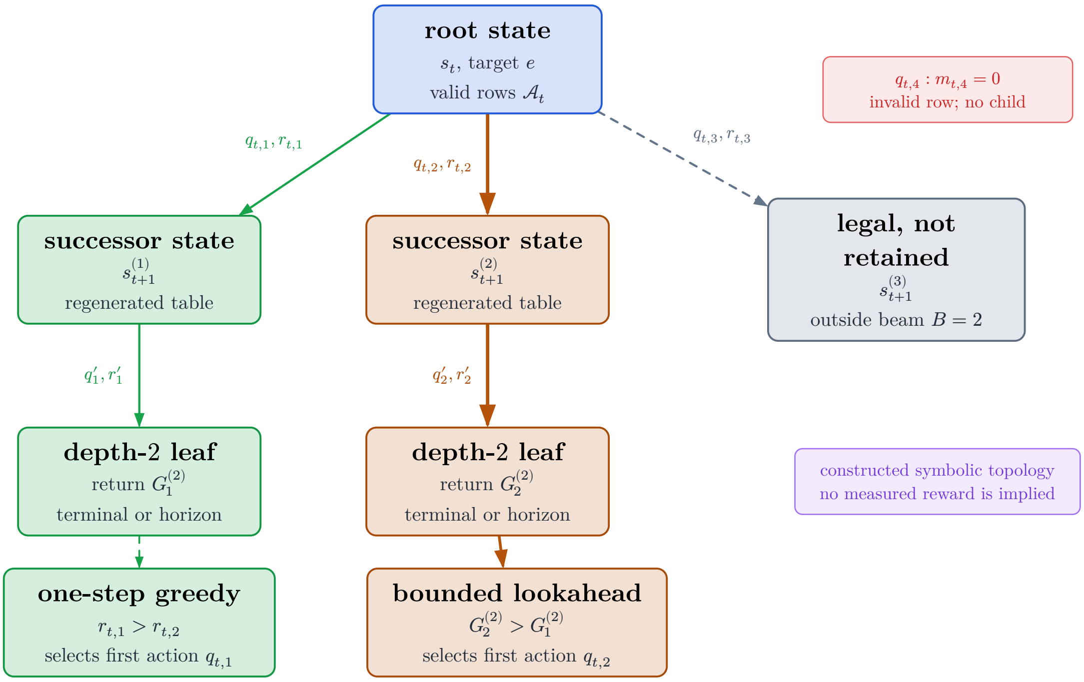


Figure 4: Conceptual topology of bounded oracle lookahead. Nodes are counterfactual states; edges are feasible actions labelled by immediate reward. The beam retains only a bounded set of prefixes. The symbolic inequalities show how a lower immediate reward can lead to a larger finite-horizon return.

For stochastic collection, the temperature-softmax recipe converts each finite oracle score s_i into the robust logit

$$\ell_i = \frac{s_i - \mathrm{median}(s)}{\mathrm{IQR}(s)\tau}, \quad P(i \mid m_i = 1) = \frac{\exp(\ell_i)}{\sum_{j:m_j=1} \exp(\ell_j)}$$

Median and interquartile-range normalization reduce sensitivity to the absolute score scale, while temperature controls concentration over the feasible rows. The resulting replay distribution is determined jointly by target sampling, candidate support, pruning, and rollout recipe. Reporting these factors isolates the population and behavior distribution on which learned ranking is assessed. Train-only pilots provide generation-cost and support-diversity evidence; held-out evaluation supplies policy evidence.

3.3 Measurement and Outcomes

The candidate shell defines possible interventions; the measurement operator assigns their target-specific consequences. For each feasible row, the oracle renders depth from $\mathcal{M}^{\mathrm{GT}}$, backprojects $\mathcal{P}_q$, fuses the candidate evidence with the root evaluation cloud, crops both points and mesh to target e, and computes reconstruction error [Fra+25]. Applying the same operator before and after the hypothetical acquisition yields Δ_t^e and $\Delta_{t|i}^e$. The candidate label depends jointly on camera pose, current state, target crop, and update rule.

Validation TODO: Validate repeatability, numerical-failure handling, construct interpretation, root normalization, and matched endpoint re-evaluation before treating target gain as a reliable study outcome.
Source: docs/literature/tex-src/arXiv-VIN-NBV; metric-validation artifacts
Gate: frozen metric protocol, repeatability table, and claim-level source locators

3.3.1 Operational Reconstruction Error

Let C_e crop geometry to the selected target region. The evaluation point set $P_t^e = C_e(\mathcal{P}_t)$ is reconstructed from the observed prefix of ASE GT depth. Candidate i contributes renderer-derived depth, which is backprojected and fused with the same root cloud to obtain $P_{t|i}^e$. MPS semi-dense points serve as actor-side representation input and as a legacy diagnostic source. ASE depth defines the current oracle evaluation cloud.

For target mesh $M_e = \mathcal{M}_e^{\mathrm{GT}}$, the point-to-mesh term is the mean squared distance from each reconstructed point to its closest reference triangle,

$$D_{P\to M}(\mathcal{P}, \mathcal{M}^{\mathrm{GT}}) = \frac{1}{\|\mathcal{P}\|} \sum_{\boldsymbol{p}\in\mathcal{P}} \min_{\boldsymbol{f}\in\mathcal{F}^{\mathrm{GT}}} d(\boldsymbol{p}, \boldsymbol{f})^2$$

It measures how closely the reconstructed evidence lies on the target surface and is therefore sensitive to extra, noisy, or misregistered points. The reverse mesh-to-point term is

$$D_{M\to P}(\mathcal{P}, \mathcal{M}^{\mathrm{GT}}) = \frac{1}{\|\mathcal{F}^{\mathrm{GT}}\|} \sum_{\boldsymbol{f}\in\mathcal{F}^{\mathrm{GT}}} \min_{\boldsymbol{p}\in\mathcal{P}} d(\boldsymbol{p}, \boldsymbol{f})^2$$

and measures which target faces lack nearby reconstructed evidence. Their sum is the target error

$$\Delta_t^e = d\big(C_e(\mathcal{P}_t), \mathcal{M}_e^{\mathrm{GT}}\big) = D^e_{P\to M,t} + D^e_{M\to P,t}$$

with units of square metres. Root and candidate-augmented clouds pass through the same deterministic fusion and point-cap operator, so the difference isolates the candidate evidence under a fixed update rule. The resulting estimand is defined on the capped point set, fixed target mesh, and declared fusion operator.

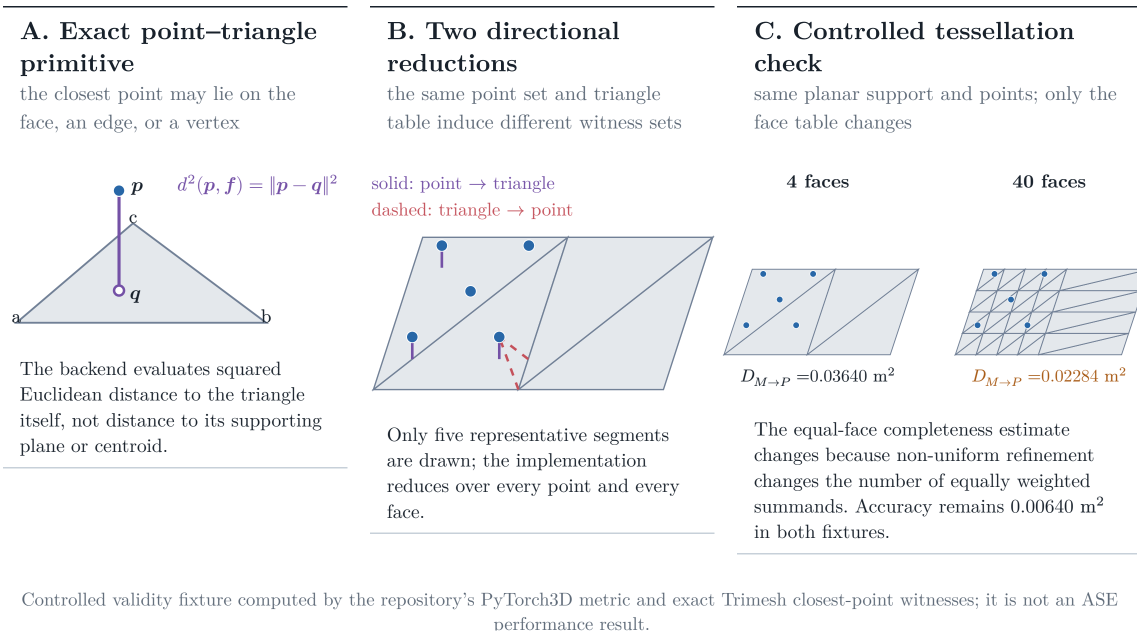


Figure 5: Geometry of the operational point–mesh error. Panel A isolates the point-to-triangle primitive; panel B separates point-to-face accuracy from face-to-point completeness. Panel C keeps the surface and point set fixed but changes its tessellation, changing equal-face completeness from 0.03640 to 0.02284 m^2. The synthetic fixture provides evidence for metric sensitivity.

Because completeness weights faces equally, its value depends on mesh tessellation and differs from an area-weighted surface integral. Subdividing one region changes its contribution even when the geometric surface is unchanged (Figure 5). Comparisons are internally consistent when they share one fixed target mesh and scoring protocol, but the numeric error is specific to that mesh representation.

3.3.2 Reliability under the Frozen Protocol

For fixed source state, target, candidate, and manifest, repeated oracle evaluation should reproduce both $\Delta^e_{t|i}$ and the induced candidate ordering. This requirement covers the complete measurement operator—rendering, backprojection, target cropping, fusion, capping, and numerical failure handling—because instability in any component changes the label. Deterministic code and a fixed mesh make exact replay possible, but RQ1 still requires repeatability measurements over the intended scene and target population.

Metric family	**Question answered**	**Dependency and thesis role**
Equal-face bidirectional point–mesh error	Are reconstructed points close to GT faces, and does every GT face have nearby point evidence?	Depends on point fusion and mesh tessellation; selected operational RRI error.
Point-sampled Chamfer / accuracy–completeness	Are two sampled surface point sets mutually close?	Depends on sampling density, randomness, norm, and squaring convention; alternative operationalization.
Thresholded precision, recall, and F-score	What fraction of predicted and reference samples fall within tolerance τ, and how do the two fractions balance?	Interpretable at a physical tolerance but threshold-dependent; secondary diagnostic alongside scalar RRI.
TSDF, occupancy, coverage, or information gain	How much volumetric surface, free space, or previously unknown space is reconstructed or observed?	Depends on grid, truncation, and visibility definitions; answers a different question from target surface quality.

Table 5: Reconstruction measures answer different geometric questions. The equal-face point–mesh error is the selected outcome; the other families are diagnostics or alternative operationalizations.

Point-sampled Chamfer distance draws a finite reference set Q_e from the mesh and averages nearest-neighbour distances in both directions:

$$D_{\text{PS-Chamfer}(\mathcal{P},\mathcal{Q})} = \frac{1}{\|\mathcal{P}\|}\sum_{\boldsymbol{p}\in\mathcal{P}}\min_{\boldsymbol{q}\in\mathcal{Q}}\|\boldsymbol{p}-\boldsymbol{q}\|_2^2 + \frac{1}{\|\mathcal{Q}\|}\sum_{\boldsymbol{q}\in\mathcal{Q}}\min_{\boldsymbol{p}\in\mathcal{P}}\|\boldsymbol{q}-\boldsymbol{p}\|_2^2$$

Sampling density, randomness, norm, and squaring convention determine its finite value despite the symmetric formula. Thresholded precision, recall, and F-score add a physically interpretable tolerance τ, but discard error magnitude on either side of that threshold. Volumetric coverage and information gain measure observed space or uncertainty reduction. These are useful complementary quantities. Each operationalization maps a different geometric construct.

3.3.3 Construct Validity

The operational construct is **target-surface evidence after a declared acquisition and update**. ASE depth, the selected target crop, equal-face mesh reduction, and the fusion operator jointly define it. Perceived object quality and whole-scene completeness require different constructs. Two systems that share the final point–mesh formula but apply different state updates estimate different intervention effects.

The implementation includes a mesh face in the target crop when any vertex lies inside the target Oriented Bounding Box (OBB). A geometry-invalid candidate is excluded by $m^{\text{act}}_{t,i}$. An empty crop, insufficient root support, or unusable render makes the oracle outcome undefined and clears $m^{Q,h}_{t,i}$. This preserves whether failure occurred in the action domain or in the measurement operator.

Comparative policy evidence therefore requires the same target mesh and crop, ASE-depth source, renderer, backprojection, fusion, point cap, and metric configuration. Replacing ASE depth with MPS semi-dense evidence changes both the observation process and the error population and therefore defines a separate experimental outcome.

3.3.4 RRI, Training Reward, and Endpoint Gain

The learning and evaluation quantities are derived from the same error sequence $\Delta^e_0, \Delta^e_1, ..., \Delta^e_H$, but answer different questions. State-relative RRI asks how much one action improves the **current** state; target-root reward assigns additive credit along a chain; endpoint gain asks how much the target improved after a fixed budget. Their denominators determine whether they can be accumulated or compared across states.

The VIN-compatible marginal diagnostic is

$$\mathrm{RRI}^e_{t,i} = \frac{\Delta^e_t - \Delta^e_{t|i}}{\max(\Delta^e_t, \varepsilon)}$$

Its selected-chain running sum is

$$C_t^{\mathrm{RRI},e} = \sum_{k=0}^{t-1} \mathrm{RRI}_{k,a_k}^e$$

Because each term is normalized by its own Δ_t^e, the sum describes accumulated local relative improvements. Fixed-budget policy outcomes use the root-to-endpoint definition below.

The training reward normalizes every step by the rollout-root target error [Fra+25]:

$$r_t^e = \frac{\Delta_t^e - \Delta_{t+1}^e}{\max(\Delta_0^e, \varepsilon)}$$

With unit discount, summing these rewards over a factual chain telescopes:

$$J_t^e = \sum_{k=0}^{t-1} r_k^e = \frac{\Delta_0^e - \Delta_t^e}{\max(\Delta_0^e, \varepsilon)}$$

The horizon-conditioned return used to supervise Q_H is

$$G_{t,e}^{(h)} = \sum_{k=0}^{h-1} \gamma^k r_{t+k}^e, \quad 1 \leq h \leq b_t \leq H_{\max}$$

This return evaluates a candidate first action under the continuation rule that generated the remaining factual chain. Its scope is offline finite-candidate control.

The primary policy outcome is endpoint gain

$$J_e^{(H)} = \frac{\Delta_0^e - \Delta_H^e}{\Delta_0^e + \varepsilon}$$

because it compares target error after the same acquisition budget H. The log-gain variant

$$J_{e,\log}^{(H)} = \log(\Delta_0^e + \varepsilon) - \log(\Delta_H^e + \varepsilon)$$

is retained as a scale-sensitivity diagnostic. Thus $\mathrm{RRI}_{t,i}^e$ describes local candidate improvement, $G_t^{(H)}$ supplies the learning target, and $J_e^{(H)}$ supplies the fixed-budget evaluation outcome. The endpoint denominator $\Delta_0^e + \varepsilon$ differs from the clamped root denominator $\max(\Delta_0^e, \varepsilon)$ used by the additive reward. Near zero root error, each quantity must be interpreted under its own denominator. The resolved manifest must therefore freeze the discount, clipping, crop, point update, and all evaluation-geometry parameters for each reported experiment.

3.4 Dataset and Storage Semantics

The replay dataset is organized around decision-state relations. Each state binds an information state s_t and target e to a candidate table $\mathcal{Q}_t$, row-specific feasibility and label masks, oracle outcomes where defined, and the selected successor. Its normalized lineage connects a source to target tasks, retained chains, decision states, and candidate rows.

One logged source can support several target tasks. Each task can be explored by several rollout recipes. A retained chain contains ordered selected steps, and each step owns the full finite candidate shell. This layout preserves two forms of supervision. Candidate rows record the counterfactual outcomes available at one state; the selected row a_t links that state to the factual successor s_{t+1}. The first supports dense one-step ranking, while the second supports finite-horizon returns along a causal trajectory.

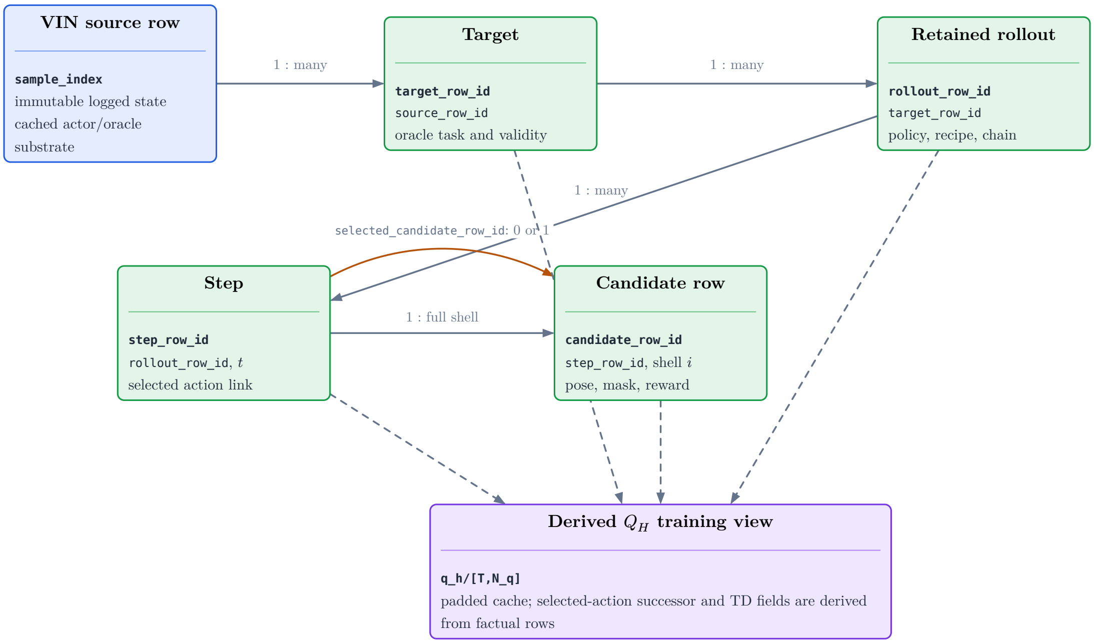


Figure 6: Normalized replay lineage. A source defines target tasks; each task produces recipe-specific retained chains; each chain orders factual decision states; and each state owns one complete candidate shell. Model-ready tensors are derived by padding and batching these relations.

Stored relation	Mathematical role	Scientific meaning
Source–target	s_0, e	Binds a task to one immutable logged substrate and one declared target source.
State–candidate	$s_t, q_{t,i}$	Preserves the full proposal support, hard feasibility, label availability, and proposal provenance.
Selected transition	$(s_t, a_t, r_t^e, s_{t+1})$	Records the causal edge created by the selected action.
Retained chain	$(s_0, a_0, \ldots, s_H)$	Identifies the recipe-specific trajectory from which finite-horizon returns may be derived.
Derived training view	$(\boldsymbol{X}, \boldsymbol{m}, \boldsymbol{y})$	Pads and batches factual relations without changing their information or missingness semantics.

Table 6: Scientific roles of the normalized replay relations. State, target, action support, supervision, and lineage are bound before model-specific tensorization.

3.4.1 Immutable Sources and Causal Replay

An immutable source store owns the logged actor substrate and one-step oracle products. A replay store references those identities and adds target tasks, retained chains, per-state candidate shells, selected-action successors, and recipe provenance. Changing a rollout recipe creates a new replay dataset over the same source facts. Observations and one-step oracle products keep stable identities across those experiments, which enables paired comparison.

For each selected action, replay stores one calibrated depth raster. The ordered rasters reconstruct the selected-observation prefix without duplicating every candidate render inside each chain. Source role stays attached to each raster: mesh-rendered depth defines the privileged causal control, and sensor-like or observed depth defines an actor-side successor. Candidate renders for unselected rows stay with the one-step oracle products used to construct labels.

The stored data-generation transition is

$$(x_{t+1}, \boldsymbol{H}_{t+1}, b_{t+1}, \mathcal{Q}_{t+1}) = \mathrm{Step}\Big(x_t, \boldsymbol{H}_t, b_t, q_{t,a_t}, \xi_t\Big)$$

It updates reference pose, selected history, budget, and the next candidate table under generation context ξ_t. A visual successor adds a separately specified observation operator and state update for RGB, depth, EFM3D features, or surface memory. The rollout recipe retains the chains it selects or keeps in its beam. Matched oracle endpoint re-evaluation then supplies a common fixed-budget outcome for policy comparison.

3.4.2 Missingness and Derived Views

Every candidate row carries separate predicates for materialization, actor-feasible action $m^{\mathrm{act}}_{t,i}$, finite value label $m^{Q,h}_{t,i}$, and factual successor m^{succ}_{t}. These predicates describe structure, action support, supervision support, and trajectory heritage. A training objective may select their intersection; the stored fields preserve the reason each row enters or leaves that population.

Model-ready arrays are deterministic projections of the normalized store. They pad candidate shells, expand a chain into state queries, and cache features while preserving source identity, target identity, state order, masks, requested horizon, and label provenance. Several states from one chain may share a batch, but each query receives only its own causal prefix.

Reproducibility requires content identity and transformation identity. The source manifest fixes scene, snippet, target, geometry, and oracle products; the rollout manifest fixes proposal profile, pruning, recipe, seed, and retention; the training view fixes tensorization and supported horizons. Coverage, failed roots, family survival, undefined metrics, source-role counts, and exclusions define the population to which a learned value claim applies. Compression, chunking, hashes, and execution commands complete the reproducibility record. Development bandwidth pilots contribute cost estimates; held-out policy evidence comes from the evaluation protocol in Chapter 5.

4 Method

Chapter 3 fixed the experimental world: a target-specific task, a finite and hard-masked action set, a causal replay transition, and a root-normalized reconstruction-gain outcome. This chapter fixes the learner that operates inside that world. It distinguishes two levels throughout. The **current realization** is the executable baseline used to validate interfaces and expose representation limits. The **scientific target** states the actor-visible information and evidence required to answer the research questions. It is a promotion contract, not a claim about code or results already obtained.

Both levels share a scalar-horizon, target-conditioned candidate-value interface. The scorer reads admitted scene and target evidence, relative candidate geometry, causal history, and remaining budget; it predicts a conditional value for every materialized candidate before the authoritative hard mask is applied. What differs is whether the admitted actor state retains the target-specific information on which future return can depend.

The selected executable configuration uses the `S0-pose` state, H0 mean-pooled pose history, A1 candidate-to-state cross-attention, and direct continuous Huber regression. A0, an independent-row MLP with the same inputs and decoder, is the matched interaction control. Both execute the same value task; their held-out comparison remains pending. Richer selected-depth state, ordered history, ordinal decoding, and candidate-set interaction remain alternatives because none has yet passed a frozen comparative evaluation.

The value query distinguishes factual remaining budget b_t from requested residual horizon h. Current supervision supports dense one-step queries and recursive queries on the factual budget diagonal; wider executable inputs do not establish wider learned support. Exact horizon two is therefore the first epistemic test of learned lookahead: it can compare learned recursion with an exact finite-support endpoint without trusting a learned longer-horizon continuation. Passing that test is necessary but not sufficient for a policy claim, which additionally requires positive oracle headroom and held-out endpoint recovery.

The chapter proceeds from the selected state and encoding, through finite action and replay semantics, to architectural acceptance properties and the finite-horizon learning objective. This order keeps failures interpretable: lost state information, malformed action or replay semantics, and value-learning error remain distinct explanations rather than an undifferentiated model defect.

4.1 Current and Target Actor State

Implementation: partial **Evidence: pending**

The `S0-pose` baseline and the independent `v1_observed` descriptor path are implemented and tested. The current selected experiment still uses privileged `v0_gt_input`; no frozen `v1_observed` evaluation or causal observation-updated actor state yet supports the scientific target.

Literature: [LHP23, Str+24b]

Promotion gate: freeze and evaluate an actor-visible target corpus; implement the causal state-update contract; pass source-dropout, no-future-observation, leakage, and held-out task-sufficiency tests

Development source: aria_nbv/aria_nbv/targets/protocol.py; aria_nbv/aria_nbv/targets/selection.py; aria_nbv/aria_nbv/oracle/pipelines/campaign.py; aria_nbv/aria_nbv/oracle/pipelines/rollout_dataset.py; aria_nbv/aria_nbv/data_handling/qh_data/views.py; aria_nbv/aria_nbv/vin/modules/qh_scene_encoders.py; aria_nbv/aria_nbv/rollouts/qh_reader.py; aria_nbv/aria_nbv/vin/models/target_finite_horizon.py

The actor state is the information available before selecting action a_t, not every quantity co-located in a replay row. Oracle target gains, mesh crops, ground-truth associations, current all-candidate renders, and labels are excluded from the scorer graph. Two independent protocol axes matter. The target-source axis distinguishes non-deployable `v0_gt_input` geometry from a `v1_observed` descriptor constructed from actor-visible observations with bound source and construction provenance. The dynamic-state axis distinguishes the pose-only `S0-pose` carrier from a state updated with selected observations. Sharing tensor shapes across either axis does not make their scientific claims interchangeable.

Both the current and target states retain the same decision interface,

$$\boldsymbol{h}_{t,e,i} = \mathrm{Read}\big(\boldsymbol{\Phi}_t^{\mathrm{scene}}, \boldsymbol{h}_e^{\mathrm{tgt}}, q_{t,i}, \boldsymbol{H}_t, t, H\big)$$

but differ in what the state makes available to that read. The current root component contains semidense evidence and lossy global moments of locally supported

EFM3D features. Its dynamic component is only the strictly causal selected-pose prefix. Candidate regeneration updates the reference pose, prefix, remaining budget, and action table; it does not imply a new RGB observation, a refreshed EFM3D field, or fused selected depth.

Component	Current realization	Scientific target
Root scene	immutable semidense points and global moments of locally supported EFM3D features	immutable actor-visible evidence with explicit finite support and missingness; absent evidence must not mean observed free space
Target	selected experiment: privileged `v0_gt_input`; an independent `v1_observed` admission and I/O path is implemented but not frozen or evaluated	`v1_observed`: actor-visible identity and geometry with support, source, construction provenance, and explicit matching failures
Dynamic state	selected poses $\boldsymbol{H}_t$ only	a strictly causal update from the selected observation that preserves observed surface, observed free, unknown, support, uncertainty, source, and recency
Decision context	candidate rows, hard mask, b_t, and h	the same finite support plus target- and candidate-relative access to the actor-visible state needed for target-specific return

Table 7: Current baseline and scientific target. The right column is a promotion contract: it states information that must be represented and tested, not an architecture already implemented or a result already obtained.

The scientific target is intentionally a semantic contract rather than a premature choice among points, voxels, rays, or another carrier. A proposed carrier is admissible only if its deterministic fusion rule exposes source and availability, uses the selected actor-visible observation and no future or unselected render, and preserves the distinctions in Table 7. Source-dropout, no-future-observation, and leakage tests establish the information boundary; held-out task-sufficiency evidence must then show that the retained distinctions are useful for the target-specific value task.

4.1.1 Sufficiency and the local-evidence limit

A decision state is task-sufficient for this study only if it preserves the distinctions needed to predict target-specific future return. Under partial observability, this is an empirical representation requirement rather than a claim that the study solves a general belief-state POMDP [LHP23]. `S0-pose` cannot satisfy that condition by construction when two histories share poses but reveal different surfaces, occlusions, or free-space evidence. It is therefore a deliberately compact baseline and an interface test, not a claim that pose history is a complete reconstruction state.

EFM3D reinforces this limit. Its gravity-aligned voxel field has an explicit pose and finite extent [Str+24b]. A target or candidate outside that extent may remain physically valid while lacking local lifted evidence. Global moments further remove spatial correspondence. The selected method can test whether this compact context contains usable signal, but a failure cannot be attributed to value learning alone, and a success cannot establish spatial state sufficiency. Richer causal surface or ray memories are kept outside the selected method until a frozen comparison isolates their added information.

4.2 Target, Candidate, and History Encoding

Implementation: implemented **Evidence: pending**

The selected encoder materializes one target-conditioned query per candidate from root evidence, local relative geometry, H0 selected-pose history, remaining budget, and requested horizon. Its tensor contract is tested; scientific usefulness remains pending.

Literature: [Bro+21, Sch+15]

Promotion gate: preserve local-frame, mask, source, and strictly causal prefix tests; evaluate held-out ranking

Development source: aria_nbv/aria_nbv/data_handling/qh_data/views.py; aria_nbv/aria_nbv/vin/models/target_finite_horizon.py; aria_nbv/aria_nbv/vin/modules/qh_history_encoders.py; aria_nbv/tests/vin/test_target_finite_horizon.py; aria_nbv/tests/vin/test_qh_history_encoders.py

For target e and candidate $q_{t,i}$, the scorer input is the structured relation

$$\mathcal{I}_{t,e} = \left(\boldsymbol{\Phi}_t^{\text{scene}}, \boldsymbol{T}_{r\leftarrow e}, \boldsymbol{a}_e, \left\{\boldsymbol{T}_{r\leftarrow c_i}, \boldsymbol{T}_{c_t\leftarrow c_i}, \boldsymbol{T}_{c_i\leftarrow e}, m_{t,i}^{\text{cand}}\right\}_{i=1}^{N_q}, \left\{\boldsymbol{T}_{c_t\leftarrow c_j}\right\}_{j<t}, b_t, h\right)$$

rather than a flat replay row. Source and availability masks distinguish absent evidence from measured zeros; padding is separate from physical invalidity; supervision and audit lineage never enter the learned query.

4.2.1 Target-conditioned query

The selected target protocol supplies a root-relative target pose and metric OBB extents. The target token is therefore

$$\boldsymbol{h}_e^{\mathrm{tgt}} = \mathrm{TargetProj}(\mathrm{concat}(\mathrm{PoseEnc}(\boldsymbol{T}_{r\leftarrow e}), \boldsymbol{a}_e))$$

The source protocol that produced this geometry is frozen experiment identity, not another model input. Current oracle tasks use ground-truth-derived target geometry; this privileged source limits deployability even though gains, meshes, associations, and audit lineage remain outside the prediction graph.

4.2.2 Relative candidate geometry

Canonical world poses remain reproducibility facts, but the selected scorer encodes each candidate both relative to the rollout root and relative to the factual current camera. Its physical trunk is

$$\boldsymbol{h}_{t,i}^{\mathrm{phys}} = \mathrm{PhysicalProj}\Big(\mathrm{concat}\Big(\mathrm{PoseEnc}\Big(\boldsymbol{T}_{r\leftarrow c_i}\Big), \mathrm{PoseEnc}\Big(\boldsymbol{T}_{c_t\leftarrow c_i}\Big), \boldsymbol{\Phi}_t^{\mathrm{scene}}\Big)\Big)$$

The conditional-value query then adds the target pose expressed from that candidate:

$$\boldsymbol{x}_{t,i} = \mathrm{ValueQueryProj}\Big(\mathrm{concat}\Big(\boldsymbol{h}_{t,i}^{\mathrm{phys}}, \mathrm{PoseEnc}\Big(\boldsymbol{T}_{c_i\leftarrow e}\Big)\Big)\Big)$$

This factorization removes arbitrary world origin while retaining the complete relative rotations and translations represented by the shared PoseTW encoder. It does not append handcrafted range, bearing, height, frustum, sampler-family, or generator-provenance features. The physical token feeds feasibility without target or horizon information; only the conditional-value query receives the candidate-from-target transform.

4.2.3 Strictly causal pose history

For every realized state, the reader supplies the complete selected-pose prefix $j < t$ expressed from the factual current camera. The selected H0 encoder applies the shared pose map and a padding-aware arithmetic mean:

$$
\begin{aligned}
\boldsymbol{p}_{t,j}^{\text{hist}} &= \operatorname{PoseEnc}\Big(T_{c_t \leftarrow c_j}\Big), \quad j < t \\
a_{t,j}^{\text{hist}} &= \frac{t-1-j}{H_{\max}} \\
\boldsymbol{h}_t^{\text{hist}^{\text{H0}}} &= \operatorname{HistProj}\left(\frac{1}{\max(1,t)} \sum_{j<t} \boldsymbol{p}_{t,j}^{\text{hist}}\right) \\
\boldsymbol{h}_t^{\text{hist}^{\text{H1}}} &= \operatorname{HistProj}\Big(\operatorname{LastValid}\Big(\operatorname{CausalTransformer}\Big(\Big[\boldsymbol{e}_{\text{empty}}, \Big\{\boldsymbol{p}_{t,j}^{\text{hist}} + g\Big(a_{t,j}^{\text{hist}}\Big)\Big\}_{j<t}\Big]\Big)\Big)\Big)
\end{aligned}
$$

Only the H0 branch of this equation belongs to the selected method. The root state uses a learned empty token; padded entries cannot affect the history summary. Prefix cardinality and chronology are not substitutes for remaining budget b_t or requested horizon h, which remain separate tokens. Because H0 observes poses rather than selected surfaces or free-space evidence, it inherits the `S0-pose` sufficiency limitation identified above.

The current encoder implements the `S0-pose` actor-state carrier described above. The implemented `v1_observed` protocol can supply its target token without adding selected-observation state; target provenance and dynamic state are separate axes. Reaching the scientific target therefore does not merely append a larger scene token: the encoder must preserve source and availability while letting each candidate query both the actor-visible target and the causal dynamic state in relative geometry. Whether points, voxels, rays, or another carrier best meets that requirement remains an empirical comparison.

The maximum horizon $H_{\max}$ binds data, model, and checkpoint. Remaining budget records the factual state and h selects one return from $1 \leq h \leq b_t \leq H_{\max}$. Current bundles record trained horizons rather than budget–horizon pairs, so deployed inference requests the factual diagonal $h = b_t$. A manifest-bound pair gate is required before any off-diagonal $h > 1$ query is promoted beyond the syntactic scorer interface. The public interface scores one scalar horizon at a time and preserves candidate order in its $\big[B, S, N_q\big]$ output.

4.3 Finite Action and Causal Replay

Implementation: partial **Evidence: pending**

Finite candidate tables, hard masks, and the `S0-pose` replay transition are implemented and tested. The scientific target additionally requires a causal actor-visible observation update, which remains pending.

Promotion gate: retain deterministic row identity, source roles, hard-mask isolation, and selected-transition validation; add selected-observation fusion and no-future-observation tests

Development source: aria_nbv/aria_nbv/targets/protocol.py; aria_nbv/aria_nbv/rollouts/replay/engine.py; aria_nbv/aria_nbv/rollouts/zarr_store.py; aria_nbv/aria_nbv/rollouts/qh_reader.py; aria_nbv/tests/rollouts/test_qh_reader.py

At step t, the generator returns the full candidate table $\mathcal{Q}_t$, a hard-valid mask $\boldsymbol{m}_t$, and versioned failure reasons. The admissible action set is

$$\mathcal{Q}_t = \{q_{t,i}\}_{i=1}^{N_q}, \quad \mathcal{A}_t = \{i \in \{1, \dots, N_q\} : m_{t,i}^{\mathrm{act}} = 1\}$$

Invalid rows remain auditable but cannot enter selection, Q supervision, or a bootstrap maximum. A feasible row may have negative utility; infeasibility is therefore never encoded as a small value. Padding, materialization, `valid_action_mask`, `q_train_mask`, and any predicted feasibility remain distinct. Under the dense-valid supervision profile, the writer proves and the reader revalidates equality of Q-label support and hard action support on every realized state. Unknown or legacy label-density profiles fail closed for this claim.

Candidate orientations factor a component-specific base gaze from bounded yaw–pitch perturbations. Paired proposal components may reuse a camera centre across two base-gaze families, but both remain separate actions. This makes translation and viewing direction distinguishable without collapsing their values. Generator family, sampled support, hard rejection, and stable shell identity remain attached to every row so proposal coverage can be audited before policy quality is interpreted.

4.3.1 Supervision and audit contracts

The persisted supervision profile makes label density an auditable contract rather than an inference from array shape. Historical stores declare `legacy_unspecified` together with `subset_of_action_v1`: their Q-label support may be any subset of hard-valid materialized rows and cannot establish dense one-step supervision. A fitted-Q store may instead declare `dense_valid` together with `equals_action_on_realized_steps_v1`. The writer then proves, and the reader revalidates, exact equality between `q_train_mask` and `valid_action_mask` on every realized state. Padding is excluded from both masks. Hard-invalid materialized rows remain useful binary examples for the physical feasibility head, but receive neither a fabricated Q target nor bootstrap support. Unknown or crossed profile pairs fail closed.

For benchmark reporting, **proposal support** is the persisted set of candidate rows that a declared generator attempted, with applicability and validity masks retained per family. **Proposal regret** is a descriptive comparison between the selected row and the best admissible row under the same persisted oracle contract; it is unavailable when no compatible oracle labels exist. Scene-paired aggregation first computes state-level quantities and then gives each scene equal weight, preventing scenes with more states from dominating a macro estimate. An immutable evidence bundle is the hash-bound JSON manifest and Parquet projection of these facts. Readers reject incomplete, stale, schema-mismatched, fixture, or hash-mismatched bundles; presentation layers consume the reader and do not recompute scientific quantities.

The candidate-family preflight separates two estimands. Per factual state, the root-support floor is $\max\big(12, \lceil 0.25 N_q \rceil\big)$; thus 14 hard-valid rows fail and 15 pass for $N_q = 60$. Independently, the versioned diagnostic family floor requires at least one final-shell row from every applicable family in each audited state and at least three final-shell rows in total from applicable non-forward target-aware families. Forward-local selections cannot satisfy the second requirement. Inapplicable family/state cells remain explicit and do not fail the gate, whereas missing legacy applicability is an unknown provenance state and fails closed for deployment. These rules diagnose proposal support; they do not assign low utility to invalid or absent candidates. Here selection means admission into the compact valid action shell, not the single action later chosen by the rollout policy.

Reward variation is label-conditional. When finite target-root-gain labels exist under one manifest-bound oracle contract, the preflight compares their observed range with a versioned tolerance and reports the exact label-support denominator. A Phase-A audit has no such reward labels, so `flat_gain` is unavailable with both its label and eligible-state denominators reported as zero rather than being inferred from geometric dispersion. It is specifically a no-render, no-reward-label proposal-support audit with privileged ground-truth target instruction and mesh validity, not an oracle-free audit. A passing Phase-A family gate is necessary but not sufficient for broad rollout generation; the final hash-bound pre-scale decision remains a later issue-120 gate.

All persisted Phase-A geometry uses the proposal-support normalization

$$d_{t,e}^{\text{current}} = \left\|\boldsymbol{p}_e^w - \boldsymbol{c}_{r_t}^w\right\|_2, \quad \tilde{\boldsymbol{c}}_{t,i}^{\text{support}} = \left(\boldsymbol{B}_{r,t}^{\text{Z-up}}\right)^\top \frac{\boldsymbol{c}_{t,i}^w - \boldsymbol{c}_{r_t}^w}{d_{t,e}^{\text{current}}}, \quad \tilde{\boldsymbol{p}}_{t,e}^{\text{support}} = \left(\boldsymbol{B}_{r,t}^{\text{Z-up}}\right)^\top \frac{\boldsymbol{p}_e^w - \boldsymbol{c}_{r_t}^w}{d_{t,e}^{\text{current}}}, \quad \left\|\tilde{\boldsymbol{p}}_{t,e}^{\text{support}}\right\|_2 = 1$$

. The columns of $\boldsymbol{B}_{r,t}^{\text{Z-up}}$ are the horizontal expansion-to-target direction, its world-Z-up left direction, and world up. Candidate centres and target-relative vectors are divided by the current three-dimensional target distance, while camera-forward directions are rotated by the same basis but remain unit vectors. Hence target-forward and target-lateral plots are invariant to factual rig yaw; rig-frame components cannot carry those axis labels directly.

The authenticated Phase-A outcome belongs to Results rather than to this definition of the audit. Method fixes the estimand, denominators, coordinate normalization, and failure semantics; Chapter 6 reports what the frozen evidence actually showed.

4.3.2 Current replay transition

After selecting an admitted row, the replay engine applies

$$(x_{t+1}, \boldsymbol{H}_{t+1}, b_{t+1}, \mathcal{Q}_{t+1}) = \text{Step}\Big(x_t, \boldsymbol{H}_t, b_t, q_{t,a_t}, \xi_t\Big)$$

where x_t is the current reference pose, $\boldsymbol{H}_t$ the selected-pose prefix, b_t the remaining budget, and ξ_t the deterministic generation context. The next candidate table is regenerated around the selected pose under the same target task and versioned generator. Proposal and action-selection randomness use separate streams keyed by the factual selected-action history, so reordering retained trajectories cannot change a previously defined state.

This is the complete transition of the selected `S0-pose` method. It changes reference pose, selected-pose history, remaining budget, and finite action support. It does not claim that the actor received RGB, fused depth, or updated a spatial reconstruction. Selected mesh-rendered depth may be persisted as privileged counterfactual evidence for a separate control, but unselected candidate renders never enter a successor state.

4.3.3 Scientific target transition

The scientific target retains the same factual action, budget, and candidate-regeneration semantics, but its dynamic state must also update from the observation acquired at the selected pose. That update is strictly causal: it may use the previous actor-visible state, the selected action, and the selected observation, but neither a future observation nor any unselected candidate render. Its carrier must preserve observed surface, observed free, unknown, finite support, uncertainty, source, and recency so that two pose-identical histories with different observations need not collapse to one state.

This target transition is a scientific requirement, not the behavior of the current replay engine. Promotion therefore requires deterministic fusion, source-dropout and no-future-observation tests, target-source leakage checks, and held-out evidence that the added state improves the target-specific value task. Persisted mesh depth remains a privileged control until an actor-visible observation path satisfies those gates.

The normalized replay view expands one retained chain into its realized decision states. Each state sees only the prefix available before its action. Several states from one chain may share a batch, but that tensor colocation does not create temporal communication. The factual selected action supplies the only successor link used by finite-horizon learning; no counterfactual transition is fabricated for unselected rows.

Training, validation, and test populations must share reward, mask, target protocol, actor-state protocol, horizon, and label-support semantics. They may use different candidate generators or behavior policies because those fields describe population shift rather than alter the value definition. Each split therefore binds its own support provenance, and acquisition remains scene-disjoint and independent of oracle outcomes or model error.

4.4 Selected Interaction and Acceptance Conditions

Implementation: implemented **Evidence: pending**

A1 candidate-to-state cross-attention is the selected interaction; A0 is its feature-matched independent-row control. Both pass the same executable acceptance tests. Their scientific comparison remains pending.

Literature: [Bro+21]

Promotion gate: retain row-equivariance, invalid-row isolation, frame, source, mask-independence, and horizon tests; measure the A0/A1 control

Development source: aria_nbv/aria_nbv/vin/models/target_finite_horizon.py; aria_nbv/aria_nbv/vin/modules/qh_state_fusion.py; aria_nbv/tests/vin/test_qh_state_fusion.py; aria_nbv/tests/vin/test_target_finite_horizon.py

Each materialized candidate is an independent query over shared scene, target, history, budget, and horizon tokens. A1 uses the candidate as query and those five state tokens as keys and values; candidates never attend to other candidates. A0 flattens the same ordered tokens and applies a row-shared MLP. Both expose the same-width context to the same value-decoder seam.

The A0/A1 choice is orthogonal to the state distinction in Table 7. A1 is the current interaction choice, not the scientific target itself: the same row-equivariant interface can read a richer causal state without introducing candidate-to-candidate communication. Architecture should escalate only if a frozen comparison shows that an admitted state is informative but A0/A1 cannot recover its value.

Control	Interaction	Interpretive role
A0	independent-row MLP	Tests whether the frozen descriptors and objective are learnable without attention.
A1	candidate-to-state cross-attention	Selected model; tests whether query-dependent reading of shared state improves the same value task.

Table 8: Admitted interaction pair. Inputs, state protocol, decoder, objective, and hard-mask owner remain fixed; parameter count and runtime must still be reported.

Candidate order has no semantic meaning. Jointly permuting aligned rows by Π must permute predictions identically:

$$f_\theta\big(\Pi\boldsymbol{X}_t, \Pi\boldsymbol{m}_t^{\text{cand}}\big) = \Pi f_\theta\big(\boldsymbol{X}_t, \boldsymbol{m}_t^{\text{cand}}\big)$$

Changing only hard-invalid row contents must not alter valid-row predictions:

$$\operatorname{Mask}\big(\boldsymbol{X}_t, \boldsymbol{m}_t^{\text{cand}}\big) = \operatorname{Mask}\big(\boldsymbol{X}_t', \boldsymbol{m}_t^{\text{cand}}\big) \quad\Rightarrow\quad \operatorname{Mask}\big(f_\theta\big(\boldsymbol{X}_t, \boldsymbol{m}_t^{\text{cand}}\big), \boldsymbol{m}_t^{\text{cand}}\big) = \operatorname{Mask}\big(f_\theta\big(\boldsymbol{X}_t', \boldsymbol{m}_t^{\text{cand}}\big), \boldsymbol{m}_t^{\text{cand}}\big)$$

The scorer reads the materialization mask only to sanitize padding. It never reads the hard action mask; training owns label and bootstrap admission, and online inference owns the final selectable set. Thus changing the hard mask cannot change raw conditional values or feasibility logits. Duplicate-row and valid-count tests further prevent accidental set normalization from redefining the value of an unchanged physical candidate.

Local-frame encoding removes arbitrary global origin conventions without claiming exact SE(3) equivariance. Complete root/current-relative candidate poses and the candidate-from-target pose retain translation and rotation relationships represented by the shared PoseTW encoder [Bro+21]. Provenance tests additionally exclude target gains, meshes, associations, and current candidate renders from the actor graph. Previously selected privileged depth belongs only to an explicit non-deployable state protocol.

Finally, $h = 1$ has no bootstrap, $h > b_t$ is rejected rather than clamped, and an $h > 1$ target may depend only on a factual successor queried at $h - 1$. No monotonicity across horizons is assumed because admissible actions may have negative immediate gain.

4.5 Target-Conditioned Finite-Horizon Value

Implementation: partial **Evidence: pending**

The selected method is A1–H0–S0 with a scalar requested horizon, direct continuous Huber regression, fitted-Q recursion, and hard-masked online selection. The implementation path is tested, but learned exact-Q2 accuracy and policy evidence remain absent.

Literature: [De +20, EGW05, HGS15, Sch+15]

Promotion gate: populate a frozen held-out exact-Q2 receipt; establish oracle headroom; evaluate endpoint recovery on untouched scenes

Development source: aria_nbv/aria_nbv/vin/models/target_finite_horizon.py; aria_nbv/aria_nbv/lightning/qh_module.py; aria_nbv/aria_nbv/lightning/qh_q2_certification.py; aria_nbv/aria_nbv/oracle/pipelines/online_qh.py; aria_nbv/tests/lightning/test_qh_q2_certification.py

4.5.1 Conditional scorer

The actor input contains root semidense and supported EFM3D summaries, target geometry admitted by the declared source protocol, relative candidate geometry, the factual selected-pose prefix, remaining budget, and materialization support. Oracle gains, target crops, meshes, associations, and audit lineage are excluded. The current `root_moments_v1` scene summary and H0 pose history make this an executable `S0-pose` baseline, not a task-sufficient reconstruction state.

Before the hard action mask is applied, the scorer emits

$$\left(Q^{\mathrm{cond}}_{h,\theta,e,i}, \ell^{\mathrm{feas}}_{t,i}\right) = f_\theta\big(s_t, e, q_{t,i}, h\big), \quad 1 \le h \le b_t \le H_{\max}, \quad h = b_t \text{ when omitted}$$

for every materialized row. `conditional_q` alone carries return units and enters regression, Bellman backup, and online ranking. Feasibility logits are a separate auxiliary prediction; multiplying them by value would mix meanings. The training adapter owns hard-valid label and bootstrap admission, while online inference owns the final masked selection.

The selected A1 interaction lets each candidate query shared scene, target, history, budget, and horizon tokens. A0 supplies the matched independent-row control. Both use the same geometric query and value decoder. Their comparison therefore asks whether query-dependent state reading helps under a fixed state and objective; it does not isolate attention independently of parameter count or runtime.

4.5.2 Scalar requested horizon

Using the finite-horizon return and conditional value defined in Section 2.4, the return for target e over exactly the requested residual horizon h is

$$G_{t,e}^{(h)} = \sum_{k=0}^{h-1} \gamma^k r_{t+k}^e, \quad 1 \leq h \leq b_t \leq H_{\max}$$

and the learned value is

$$Q_{h,e}^{\star}(s_t, i) = \sup_{\pi \in \Pi^{\text{act}}} \mathbb{E}_{\pi}\left[G_{t,e}^{(h)} \mid s_t, a_t = i\right], \quad i \in \mathcal{A}_t, \quad 1 \leq h \leq b_t \leq H_{\max}, \quad Q_{0,e}^{\star}(s, i) = 0$$

One model call follows the frozen interface

$$f_{\theta\left(s_t^{\text{S0-pose}}, \boldsymbol{T}_{r\leftarrow e}, \boldsymbol{a}_e, \{q_{t,i}\}_{i=1}^{N_q}, h\right)} \to \left(\left\{Q_{h,\theta,e,i}^{\text{cond}}\right\}_{i=1}^{N_q}, \left\{\ell_{t,i}^{\text{feas}}\right\}_{i=1}^{N_q}\right)$$

The notation Q_H names the bounded family rather than a collection of separately configured models. Remaining budget b_t describes the factual state; h selects one return from the triangular domain $\{(b_t, h) : 1 \leq h \leq b_t \leq H_{\max}\}$. Omitting h requests the full factual residual budget. The mathematical boundary $Q_0 = 0$ closes recursion, whereas zero in executable tensors denotes padding and is never a learned query.

Dense labels train $h = 1$ for every hard-valid candidate. For $h > 1$, only the factual selected action has a successor observation, so recursive supervision is necessarily narrower. The current recursive construction and exact-Q_2 surface populate the factual diagonal $h = b_t$, with exact Q_2 executable at $(b_t, h) = (2, 2)$ but its held-out receipt still pending. Dense Q_1 additionally supports $(b_t, 1)$ across realized budgets. Current bundles record trained horizons rather than realized pairs, while deployed online inference requests the diagonal and applies that horizon gate. Before any off-diagonal $h > 1$ query is exposed, the bundle and runtime must add pair-bound training and evaluation evidence and reject unsupported pairs. A wide syntactic interface is not evidence of wide learned capability.

4.5.3 Direct continuous objective

The decoder maps each candidate representation directly to the continuous, root-normalized finite-horizon return. This regression target preserves metric order and additive return units. The implemented Huber penalty is quadratic for small residuals and linear beyond its fixed threshold. Losses are first averaged within realized state and then within horizon. Consequently, a state with more valid candidates does not receive more weight merely because it contributes more rows, and abundant one-step states do not silently dominate a sparser recursive horizon. Reported calibration and support remain stratified by horizon.

At $h = 1$, evaluation can use dense candidate labels to report continuous error, within-state pairwise ranking, and greedy regret. Factual $h > 1$ transitions do not supply counterfactual values for unselected rows, so longer-horizon ranking and regret remain undefined until exact or independently oracle-rescored candidate tables exist. Recording zero support is more faithful than inventing a metric from the selected transition alone.

4.5.4 Fitted-Q recursion

Batch fitted Q iteration turns the fixed replay collection into successive supervised problems [EGW05]. The immediate reward is

$$r_t^e = \frac{\Delta_t^e - \Delta_{t+1}^e}{\max(\Delta_0^e, \varepsilon)}$$

and for $h > 1$ the selected method uses the lower-horizon target

$$y_t^{(h,e)} = r_t^e + \gamma B_t^{(h,e)}$$

where the continuation value is detached, frozen, or supplied by a delayed copy. The structural rule is $Q_h \leftarrow Q_{h-1}$: a horizon never bootstraps from itself. The successor maximum intersects the factual horizon support with the hard action set,

$$\mathcal{A}_{t+1}^{(Q,h-1)} = \left\{ i : m_{t+1,i}^{\text{act}} = 1 \wedge m_{t+1,i}^{Q,h-1} = 1 \right\}$$

and Double Q separates row selection from delayed evaluation [HGS15]. This estimator may limit max bias, but it cannot repair unsupported actions, aliased state, or missing selected observations.

The recursion estimates greedy continuation over the generated, hard-valid finite support under the named state and target-source protocols. It is not the Monte Carlo return of the behavior policy that produced the retained chain, unless that behavior policy selects the same continuation; nor is it an optimum over ungenerated continuous camera poses. This distinction fixes the meaning of a successful fit before any policy comparison is attempted.

4.5.5 Why exact horizon two is decisive

When the selected action has a successor table with dense one-step labels, its finite-support two-step target is computable exactly:

$$y_t^{(2,\text{exact})} = r_t^e + \gamma_t \max_{j:m_{t+1,j}^{\text{train}}=1} r_{t+1,j}^e$$

This makes horizon two the first non-myopic falsification test. A unit test can inject exact Q_1 values and prove that the tensor path implements the equation. A frozen population test instead measures learned recursion error,

$$\varepsilon_t^{(2)} = \left| y_t^{(2,\text{recursive})} - y_t^{(2,\text{exact})} \right|, \quad \varepsilon_t^{(2)} \le \tau_{\text{abs}} + \tau_{\text{rel}} \left| y_t^{(2,\text{exact})} \right|$$

on held-out supported rows. The latter tests the learned Q_1 approximation, state encoding, masks, successor linkage, and recursive target construction together. It is therefore model evidence rather than another implementation test.

The held-out evaluation begins with the complete eligible chain population rather than only rows on which an exact target is available. Chains that terminate or lose support before $h = 2$ remain in the denominator with zero exact rows. Scene- and target-stratified coverage, error, uncertainty, numeric tolerance, candidate width, generator, recipe, and behavior policy are frozen before inspection. Low error on a small supported subset cannot justify a longer horizon when minimum support or coverage fails.

Exact Q_2 is necessary but not sufficient: policy claims additionally require positive equal-budget oracle headroom and held-out endpoint recovery.

The resulting evidence boundary is explicit. The current implementation supports the `S0-pose` scorer, direct objective, hard-masked recursion, and exact-

Q_2 evaluation path. The scientific target additionally requires an evaluated actor-visible `v1_observed` corpus, a causal observation-updated state, a frozen held-out exact-Q_2 receipt, positive oracle headroom, and endpoint recovery. The `v1_observed` admission and writer–reader path is implemented, but that intermediate has not passed these evidence gates. Until they pass, this chapter establishes an executable method and its falsification tests, not a successful non-myopic actor.

5 Experimental Design

The experiments separate feasibility evidence from policy evidence. Feasibility requires a frozen scene-level population, reproducible target and candidate support, valid replay linkage, and measured oracle throughput. Policy inference additionally requires a held-out scene split, matched acquisition budgets, independent endpoint oracle evaluation, and an implemented actor-visible decision rule. The current train-only bandwidth pilots address only the first category: incomplete runs and resource failures characterize the generation system, but they cannot support comparisons between policies.

5.1 Study Population and Evidence Gates

> **Validation TODO:** Preregister the eligible population, exclusions, primary estimands, aggregation unit, number of independent runs, uncertainty interval, minimum meaningful effect, and multiplicity policy before inspecting confirmatory results.
>
> Source: experiment manifest and analysis specification
>
> Gate: immutable analysis plan plus matched held-out policy table

5.1.1 Candidate-Generation Realism Analysis Contract

RQ4 treats the scene as the independent unit. Candidate-support quality-control observations are computed per state, averaged across states within scene, and finally macro-averaged across the scene cohort; no candidate row is treated as an independent scene replicate. These within-profile diagnostics remain descriptive. Any comparative claim instead uses a paired per-scene difference over eligible scene/target/root tasks observed under both profiles, equal candidate budgets, and the frozen candidate-generation manifest. Its report identity is `paired_scene_mean_difference`, with the paired scene count as its denominator and an interval computed over the paired scene differences.

Missing or failed states are retained in the audit tables. Support counts and fractions use the preregistered failure values stated in Table 4; an undefined target-relative direction prevents that paired scene from entering a geometric contrast rather than creating an angle by imputation. The excluded-scene count and reason remain reported. Thus the descriptive state–scene macro summaries and the paired comparative estimand have distinct populations and cannot be substituted for one another.

Minimum operational support is reported as a worst-case diagnostic; the fifth percentile of per-state valid support is the more stable lower-tail estimand, accompanied by actor-valid fraction, configured-family zero rate, target-side balance, and circular target-relative orbit span. Failed roots and zero-valid configured family/state pairs remain part of the denominator. The projection fraction, oracle opportunity, and jitter compliance are secondary diagnostics: projection is calibrated framing rather than visibility, oracle opportunity is finite-support headroom rather than policy performance, and jitter compliance must preserve the nonzero production seminar-jitter invariant.

The source population comprises ASE/ATEK snippet windows from scenes for which the configured GT mesh and object-box table resolve. A frozen manifest assigns entire scenes to train, validation, or test before model selection; no scene may cross these boundaries through another snippet. Each reported run records the manifest hash and the exact counts of scenes, snippets, admitted target tasks, rollout chains, transitions, and retained candidate rows. A capped train-only pilot is therefore a throughput and support probe, not a sample from which held-out policy performance can be estimated.

The present data generator defines oracle target tasks by seeded sampling from geometry-valid GT OBB rows. Its task-coverage report therefore describes the available GT pool, sampled tasks, classes, scenes, and later oracle-evaluation failures. It does not measure proposal matching, IoU ambiguity, projected visibility, or actor-observation support. Those quantities belong to a future observed-target selector required for deployable-input claims. Until that selector exists, GT target geometry may define labels and bounded oracle references but cannot be presented as actor-visible input. The privileged-supervision boundary in Figure 1 applies equally to render-derived evidence: dense GT candidate depth may produce labels, returns, or explicitly named privileged ablations, but it is not a legal actor input. A separate teacher policy, distillation path, or current-belief renderer remains a hypothesis until implemented and evaluated, so none is promoted to a main-text process figure.

The first policy gate is an actor-visible myopic scorer over the same finite candidate table intended for Q_H. The scorer must expose one value per candidate, respect the hard action mask, and be assessed by candidate ranking, calibration, and oracle-rescored selected actions. The existing scene-level VIN scorer is historical substrate; it is not a target-conditioned control until the observed target descriptor is wired into the model and evaluated on a frozen held-out split.

The second policy gate estimates whether bounded oracle lookahead has headroom over one-step oracle greedy:

$$\Delta_{\text{look}} = J_e^{(H)}(\pi_{\text{oracle-look}}) - J_e^{(H)}(\pi_{\text{oracle-1}})$$

Only if the preregistered analysis classifies this headroom as meaningful is Q_H evaluated for closure of the separate actor-visible learned-myopic-to-oracle-lookahead endpoint gap:

$$\eta_Q = \frac{J_e^{(H)}(\pi_Q) - J_e^{(H)}(\pi_{\text{learned-1}})}{J_e^{(H)}(\pi_{\text{oracle-look}}) - J_e^{(H)}(\pi_{\text{learned-1}}) + \varepsilon}$$

Success is measured by matched endpoint oracle evaluation, not predicted values or training loss. If lookahead has no meaningful headroom, the result is scoped to the frozen split, target protocol, candidate generator, horizon, branch factor, and validity regime. If headroom exists but the learned model does not close the prescribed actor-visible-myopic-to-oracle-lookahead gap, target observability, action support, replay coverage, reward construction, and model capacity remain separate candidate explanations.

Claim	Primary evidence	Decision rule
Population	scene-split manifest and coverage bundle	Inference is restricted to the frozen held-out scene population.
Task protocol	GT pool, sampled tasks, classes, and oracle failures	Current evidence is oracle-task coverage, not observed-target matching.
Myopic control	ranking, calibration, and oracle-rescored selections	Actor-visible target conditioning must be implemented before comparison.
Planning headroom	Δ_{look} and learned-control gap-closure ratio η_Q	Q_H gap closure is evaluated only after a meaningful oracle-headroom gate.

Table 9: Objective-to-evidence matrix.

5.2 Replay Eligibility and Finite-Horizon Learning Gate

The factual rollout tables determine which records may supervise each learning problem. The action-validity mask defines actions selectable under the admitted oracle geometry contract. The stricter Q-training eligibility additionally requires a valid target/ground-truth label state and finite target-root-gain and diagnostic target-RRI labels. Padding, actor validity, one-step training eligibility, transition eligibility, modality presence, source role, and horizon availability remain separate masks. Masked rows remain available for support and failure analysis but cannot enter action selection, supervised loss, or bootstrap maximization.

All eligible candidate rows can support dense one-step supervision. Exact H=2 supervision is narrower: the factual first action must have a stored reward and either an explicit terminal outcome, whose continuation is exactly zero, or a valid successor step with finite one-step root-gain labels for every hard-valid successor candidate. This completeness condition makes the masked successor maximum exact over the stored action set; one finite label is insufficient because an unlabeled hard-valid action could have the true maximum. General recursive supervision is narrower again: it requires a factual selected action, reward, terminal flag, discount, a defined training horizon, and—when nonterminal—a reproducible successor state, hard mask, and lower-horizon factual support. The derived `q_h/` arrays align these fields on a padded state–candidate view; they do not create labels for unobserved transitions, make selected GT depth actor-visible, or turn sparse long-horizon action support into dense support.

Learning surface	Purpose	Minimum factual evidence
dense $h = 1$	immediate candidate value	actor-selectable row with finite one-step root-gain label
exact $h = 2$	base-case finite-support value	selected reward plus either terminal outcome or complete finite one-step labels over the hard-valid successor action set
recursive $h > 1$	variable-horizon fitted value	selected transition, terminal and discount; nonterminal rows also require successor actor state, hard mask, and lower-horizon target support
behavior return	policy-conditioned Monte-Carlo control	complete retained reward prefix and behavior-policy identity

Table 10: Required replay evidence by learning target. Dense immediate labels, exact H=2 targets, recursive optimal-continuation targets, and behavior-policy returns are distinct supervision surfaces.

Gate	Available evidence	Required before inference
Oracle data	target labels, masks, lineage, replay validation, and selected-depth persistence	held-out coverage, source-role audit, and matched endpoint evaluation
Myopic control	scene-level VIN substrate	actor-visible target-conditioned Q_1 scorer and frozen checkpoint
Finite-horizon scorer	feature-matched A0/A1–S0-pose/root-moments scalar-horizon controls and fitted-Q seam	exact-Q2 certification, parameter/runtime report, compatible checkpoint, frozen state protocol, and oracle-rescored policy
Requested horizons	fail-closed scalar query and exact $h \leftarrow h-1$ recursion tests	supported targets through $H_{\max}$, positive headroom, and per-horizon validation
Dynamic Q_H	selected-observation persistence and planned state update	typed dynamic-state reader, deterministic fusion, source masks, and held-out policy evaluation
Policy claim	train-only feasibility pilots	completed held-out paired comparison under equal budget

Table 11: Learning-readiness gates. A runnable scorer establishes executable readiness, not a scientific policy result.

Implementation: partial **Evidence: pending**

The scalar requested-horizon A0/A1 controls, dense-Q1 plus selected-recursion Double-Q learner, feasibility auxiliary, manifest-bound trained-horizon support, and bounded stratified exact-Q2 certification surface are implemented. A1 remains the default; a qualifying population receipt, task-sufficient dynamic state, and comparative policy evidence remain unavailable.

Literature: [De +20, EGW05, FMP19, HGS15, Kum+20]

Promotion gate: populated held-out exact-Q2 receipt, supported H>2 targets, compatible checkpoint, frozen state protocol, and independent held-out oracle re-evaluation

Development source: aria_nbv/aria_nbv/data_handling/qh_data/batching.py; aria_nbv/aria_nbv/lightning/qh_datamodule.py; aria_nbv/aria_nbv/lightning/qh_module.py; aria_nbv/aria_nbv/lightning/qh_q2_certification.py; aria_nbv/aria_nbv/rollouts/qh_reader.py

The finite-candidate value model decodes actions only over valid candidate rows:

$$\boldsymbol{u}_{t,i} = \mathrm{Enc}_\theta\big(\mathcal{I}_{t,e}, q_{t,i}\big)$$

$$a_t^\theta = \operatorname*{argmax}_{i:m^{\mathrm{act}}_{t,i}=1} Q_{h,\theta,e}(s_t, i)$$

The masked argmax is already the discrete decision rule. A separate actor network and online data collection are not required to train or execute this finite-candidate policy. Batch fitted Q iteration explicitly learns a greedy Q function from a fixed collection of transitions by repeatedly solving supervised regression problems [EGW05].

5.2.1 Training unit, causal batching, and prediction support

A retained rollout chain supplies several realized decision states rather than one monolithic recurrent sample. The root is one state; after each selected action, its factual successor is another state with an updated current pose, selected prefix, remaining budget, and candidate table. The `q_h/` reader collates these states on padded batch and state axes. At state t, every actor carrier is restricted to the causal prefix available before action a_t: a later state may be present elsewhere in the same tensor batch, but its selected poses, observations, and candidate table are not inputs to the prediction at t.

The scorer evaluates the realized states and their candidate rows in parallel. For one scalar horizon query per state it emits one conditional value for every materialized candidate row. This parallel tensor layout is not recurrent inference, autoregressive rollout generation, or temporal attention across the rollout. The current scorer does not advance the environment, feed one prediction back into its next state, or iteratively refine one candidate value. During policy execution it is invoked again only after the selected action has produced a genuinely new causal state and candidate table. Causal masking across the state axis would become relevant only for a future joint temporal state encoder; it is not required when each state already carries its independently materialized causal prefix.

Prediction support and loss support are therefore different. The scorer may emit finite values for every materialized candidate. Dense one-step loss uses every hard-valid candidate with a finite immediate label. For $h > 1$, recursive loss is limited to the factual selected transition. Nonterminal targets additionally require a supported factual successor; terminal targets have zero continuation. The padded candidate dimension must not be read as dense multi-step supervision, and an unselected candidate receives no invented successor or return.

5.2.2 Scalar requested-horizon targets

The scorer represents $Q_{\theta(s_t,e,i,h)}$ for each residual horizon h admitted by $1 \leq h \leq b_t \leq H_{\max}$. Remaining budget b_t is a factual field of the represented state; requested horizon h selects how many rewards the current value is meant to summarize. Consequently, a shorter query $h < b_t$ changes the value estimand without changing the state or pretending that less acquisition budget was available. Truncating the return of an eight-step suffix to five steps is conceptually valid only when that five-step target is explicitly constructed and factually supported; changing b_t itself is not ordinary data augmentation. The boundary target is

$$r_t^e = \frac{\Delta_t^e - \Delta_{t+1}^e}{\max(\Delta_0^e, \varepsilon)}$$

and the recursive target is

$$y_t^{(h,e)} = r_t^e + \gamma B_t^{(h,e)}$$

Here **recursion** means Bellman target recursion across factual successor states. For a nonterminal $h > 1$ target, the current reward is combined with a shorter-horizon successor value queried at $h-1$; a terminal selected transition has zero continuation and requires no successor state. This construction combines the Bellman decomposition of action value into immediate reward and successor value with temporal-difference bootstrapping from an estimate of that successor [SB18, Secs. 3.5–3.6 and 6.1, pp. 58–67 and 119–124]. The lower-horizon prediction is treated as a fixed regression target by stop-gradient, a frozen stage checkpoint, or a delayed target copy. The defining relation is $Q_h \leftarrow Q_{h-1}$ rather than $Q_h \leftarrow Q_h$, and the executable learner rejects a nonterminal successor whose factual horizon is not exactly $h-1$. This terminology does not denote an RNN hidden-state loop, repeated invocation of a refinement block, or backpropagation through an unrolled future rollout. Fixed-horizon TD was introduced precisely for predictions over a bounded number of future rewards and avoids same-horizon self-bootstrapping; its horizon functions may use shared parameters and parallel updates [De +20].

The executable joint objective uses one shared horizon-conditioned network. Its public scorer interface still admits one scalar h per state. Privately, Lightning duplicates the complete actor batch along the leading batch axis and concatenates two query domains into one scorer transaction:

1. query $h = 1$ for every realized state and fit every finite hard-valid candidate reward;
2. query the factual residual horizon $h = b_t$ for the same realized states, retaining recursive loss only for selected transitions with $h > 1$;
3. select the successor action with the online scorer at factual $h-1$ and evaluate it with the delayed target scorer;
4. average candidate losses within each Q1 state, state losses within each horizon, and non-empty horizon means uniformly;
5. persist the realized per-horizon state and candidate counts, report state-normalized loss and absolute calibration error for every supported horizon,

report pairwise ranking and greedy selected-action regret only where a dense counterfactual candidate table exists, and reject online queries absent from manifest-bound training support.

This private batching avoids a second model transaction while preserving the scalar-horizon estimand. It does not add a public vectorized horizon axis and it does not enumerate every off-diagonal query $1 < h < b_t$. Sampling or enumerating additional supported state–horizon pairs would be a separate, versioned learning-contract change with its own weighting and support rules. The present objective preserves one inference interface and shared encoders while making target lineage and weighting explicit. Staged backward induction, fixed-H networks, separate per-horizon heads, and broader off-diagonal horizon enumeration remain matched alternatives rather than implicit runtime behavior.

For remaining horizon two, the store supplies an exact target whenever the successor table has dense one-step labels:

$$y_t^{(2,\text{exact})} = r_t^e + \gamma_t \max_{j:m_{t+1,j}^{\text{train}}=1} r_{t+1,j}^e$$

This target uses no learned successor value or target network. The executable base-case evaluation has two layers: exact-table injection tests the recursion implementation, while the frozen-bundle certification compares the learned recursive target against the factual control using

$$\varepsilon_t^{(2)} = \left|y_t^{(2,\text{recursive})} - y_t^{(2,\text{exact})}\right|, \quad \varepsilon_t^{(2)} \leq \tau_{\text{abs}} + \tau_{\text{rel}}\left|y_t^{(2,\text{exact})}\right|$$

The latter is evaluated on a deterministic bounded sample from a metadata-only census. Its versioned stratum contains scene, target row, configured horizon, candidate-width bin, candidate- and rollout-configuration hashes, and behavior policy. Within each stratum wave, the selector prefers previously unseen scenes before taking another chain from an already represented scene. This rule improves scene coverage without pretending that several rows from one scene are independent replications.

The `qh-exact-q2-certification-receipt-v2` receipt binds the scorer and module configuration, actor-state and learning contracts, ordered test-store manifests and paths, test provenance, selection seed and bounds, and absolute and relative tolerances. Its independent unit is the pair of ordered-store-manifest digest and scene identity. Thesis-core promotion requires at least five selected independent units, at least one factual selected-action exact Q2 row in every selected unit, and the same error gate to pass in every unit. Pooled row-level error remains diagnostic and cannot compensate for a failing scene. The receipt exposes a denominator ladder from eligible census chains through materialized successors and complete hard-valid successor labels to factual selected-action exact Q2 rows. It also reports per-stratum support and provenance-bound fixed-support CORAL saturation separately from recursion error. Missing support is a failed gate rather than zero error. Development-schema `qh-exact-q2-certification-receipt-v1` receipts remain readable historical evidence but cannot promote a claim. Agreement with the exact control remains necessary but insufficient for interpreting longer horizons: positive oracle-lookahead headroom must come from independent held-out endpoint evaluation. A persisted terminal-step role contrast is retained as a diagnostic proxy and is manifest-bound, but it cannot promote an $h > 2$ claim.

5.2.3 Double-Q and behavior-return controls

Double Q changes how a noisy learned successor maximum is estimated; it does not define the scalar horizon interface. The online path selects

$$B_t^{(h,e)} = \begin{cases} Q_{h-1,\theta^-,e}\Big(s_{t+1}, \operatorname{argmax}_{i \in \mathcal{A}_{t+1}^{(Q,h-1)}} Q_{h-1,\theta,e}(s_{t+1}, i)\Big) & \text{if } \; h > 1 \wedge d_t = 0 \wedge \mathcal{A}_{t+1}^{(Q,h-1)} \neq \emptyset \\ 0 & \text{otherwise} \end{cases}$$

and the delayed path evaluates $Q_{|(\theta)}(s_{t+1}, e, j^\star, h-1)$. This selector/evaluator split can reduce overestimation caused by maximizing noisy action values [HGS15]. It remains an ablation against the simpler frozen lower-horizon maximum. It is relevant in an offline setting only because a learned maximum is present, not because online learning is planned.

As distinguished from the greedy value objective in Section 4.5.4, a retained chain also yields the truncated Monte-Carlo target

$$G_{t,e}^{(h)} = \sum_{k=0}^{h-1} \gamma^k r_{t+k}^e, \quad 1 \le h \le b_t \le H_{\max}$$

for its behavior policy μ. Regression to this fixed target is a useful policy-conditioned control, but it estimates Q^μ, not the greedy finite-support value $Q^\star$, unless μ is explicitly the target continuation policy. Behavior returns from random-valid, greedy, softmax, and oracle-lookahead chains must therefore remain identified rather than pooled as if they represented one optimal value function.

Double Q addresses one maximization bias. It does not create missing successor transitions, make unsupported actions reliable, or repair state aliasing. CQL and BCQ motivate explicit offline-support diagnostics because a greedy learned policy can select actions whose multi-step consequences are weakly represented in the behavior data [FMP19, Kum+20]. Conservative regularization is introduced only if those diagnostics reveal systematic unsupported-action overestimation.

5.2.4 Horizon-balanced replay and evaluation

Dense one-step rows vastly outnumber selected-action targets at longer horizons. An unweighted row mean can therefore optimize the myopic task while obscuring longer-horizon failure. The implemented learning contract freezes candidate-within-state, state-within-horizon, equal-horizon aggregation as `uniform-horizon-state-balanced-dense-q1-selected-recursion-v2`. It binds this rule with the maximum horizon, replay candidate/rollout configuration identities, behavior-policy vocabulary, reward/return semantics, and discount, and rejects cross-stage mismatches. The bundle additionally records realized state and candidate counts for every trained horizon; syntactically valid horizons without positive training support fail closed at online inference. Checkpoint admission binds the actor-state identity separately. Alternative sampling or weighting schemes require new versioned learning identities. Candidate alternatives are:

- a horizon-stratified sampler;

- per-horizon loss weights w_h;
- or a fixed number of admitted targets per horizon and scene.

Every run reports, separately for each h:

- admitted states, selected actions, behavior policies, candidate families, scenes, and targets;
- value loss and signed target error;
- candidate ranking and top-action regret where oracle comparison is available;
- bootstrap, terminal, and no-valid-successor fractions;
- online/target disagreement for Double-Q runs;
- marginal selected target RRI and the selected-chain cumulative target RRI diagnostic;
- cumulative target root gain, kept distinct from cumulative RRI;
- endpoint performance of the masked policy requested at the remaining budget.

A single scalar validation loss is insufficient for model selection unless its horizon aggregation is frozen in advance. Cross-stage corpus admission enforces the same maximum horizon, reward and return semantics, discount, state/source protocol, candidate/reason vocabulary, replay-support identity, and horizon-weighting rule.

The optimality claim remains bounded under either interface. A learned finite-horizon value can approximate the best continuation only within the sampled finite candidate generator, hard-validity regime, represented actor state, and offline transition support. The same checkpoint cannot silently mix a pose-only state with privileged selected-depth, sensor-like, or actor-visible dynamic states. Longer horizons increase—not decrease—the need for selected-observation geometry and a sufficiently Markov scene state.

The current scorer-independent learner establishes infrastructure readiness only. Its horizon metrics are clustered by factual state: candidate errors are first averaged within state, pairwise ranking accuracies are first formed within state, and greedy regret contributes one value per supported state. Only dense Q1 currently has ranking/regret support; selected-transition recursion at longer horizons reports zero counterfactual support. Confirmatory interpretation requires the source-owner scorer decision, a canonical oracle-task corpus, dense-Q1 and exact-Q2 controls, supported targets for every claimed horizon, cross-stage learning-contract equality, checkpoint compatibility, a frozen actor-state protocol, horizon-specific support diagnostics, and matched held-out endpoint oracle re-evaluation. A dynamic-state claim additionally requires selected-observation fusion and no-future-observation leakage tests.

5.3 Policy Comparison and Statistical Protocol

Implementation: partial **Evidence: pending**

The aggregation and artifact-driven reporting seam exists, but confirmatory paired policy evidence does not.

Promotion gate: frozen held-out manifest, completed paired endpoints, and validated report bundle

Development source: aria_nbv/aria_nbv/rollouts/reporting.py; docs/typst/thesis/experiment_data.typ

The experimental unit is the scene. Every comparison is paired by scene, target task, root state, candidate-generation distribution, hard validity regime, and acquisition horizon. Planner depth may differ because it defines the decision rule, but every policy receives the same acquisition budget. Repeated snippets, targets, and seeds are aggregated within scene before the primary comparison so that densely sampled scenes do not dominate the estimand.

The primary estimand is the paired per-scene difference in fixed-budget endpoint target-quality gain, using $J_e^{(H)}$. Its $\Delta_0^e + \varepsilon$ denominator is distinct from the $\max(\Delta_0^e, \varepsilon)$ denominator of cumulative target-root gain, while state-relative RRI has a state-varying denominator; both remain secondary diagnostics rather than alternate names for the endpoint. The centralized analysis artifact freezes the scene aggregation function, uncertainty procedure, interval level, comparison set, and minimum meaningful effect before final aggregation. Invalid-action rate, runtime, path length, and task/candidate coverage likewise diagnose mechanism and feasibility but do not replace the endpoint estimand. The current replay store alone cannot estimate this endpoint comparison because it does not persist an independent post-budget reconstruction for every policy; confirmatory analysis therefore requires matched endpoint oracle re-evaluation or an equivalent frozen endpoint record.

Support and failure strata are fixed before policy inspection. The report distinguishes source rows without an admitted target task, admitted tasks whose oracle evaluation fails, roots without a valid action, trajectories that terminate before the shared budget, and completed paired trajectories. A policy that cannot continue remains in the comparison with no further gain; it is not silently dropped. These strata delimit the analysis population and expose whether a policy difference is instead a support or systems failure.

Policy	Decision input	Acquisition budget	Scientific role
π_{rand}	actor-visible	H	valid-action lower reference
$\pi_{\text{learned-1}}$	actor-visible	H	learned myopic control
$\pi_{\text{oracle-1}}$	oracle immediate reward	H	one-step oracle-greedy comparator
$\pi_{\text{oracle-look}}$	bounded oracle lookahead	H	conditional finite-support upper reference
π_Q	actor-visible	H	planned learned finite-horizon gap closure

Table 12: Matched policy comparison. All rows acquire the same number of views; oracle access and planner depth define the decision rule rather than the acquisition budget.

The first inferential comparison is bounded oracle lookahead against one-step oracle greedy. Its evaluation contract freezes one target mesh and crop, ASE-depth source, renderer and backprojection, fusion and point cap, and point–mesh metric configuration for every policy. Repeated evaluation must reproduce the same scores within the declared numerical tolerance; it verifies deterministic execution under this contract, not invariance to a different mesh, sampling process, or metric. The analysis artifact also freezes the meaningful-headroom rule before learned-policy inspection. Learned-control gap-closure ratios are reported only after oracle headroom passes that gate and only when their distinct actor-visible-myopic-to-oracle-lookahead denominator is admissible. The learned comparison then contrasts Q_H with the actor-visible myopic control under the same endpoint evaluation and acquisition budget; because neither learned target-conditioned control is presently complete, this comparison remains prospective.

Validation TODO: Populate the evidence chain in order: evaluation-contract identity and deterministic repeatability, candidate and target support, oracle-lookahead headroom, myopic-control calibration, finite-horizon gap closure, then representation and architecture ablations. Missing upstream evidence blocks downstream claims rather than becoming a zero result.

Source: thesis objective-to-evidence contract

Gate: artifact-backed Results bundle

Validation TODO: Run row-shuffle, mask-isolation, duplicate-row, valid-count, frame-transform, target-source-dropout, and horizon-boundary tests for every model admitted to the policy comparison.

Source: geometric acceptance contract

Gate: architecture validity report

Open decision: Freeze the scene aggregation, interval procedure, interval level, comparison family, and meaningful-headroom threshold in the resolved analysis manifest rather than in thesis prose.

Source: artifact-driven reporting plan

Gate: confirmatory analysis freeze

5.4 Outcome Logic

Meaningful, stable lookahead headroom establishes only that the frozen finite-candidate setup contains exploitable non-myopic structure. No meaningful headroom is a setup-specific negative result, not evidence that target-aware planning is universally myopic. A non-reproducible or mismatched evaluation contract, or insufficient paired support, blocks downstream policy claims. Conclusions remain conditional on the frozen metric contract and are not claims of representation independence. Likewise, train-only bandwidth pilots and failed generation attempts, including memory-exhaustion failures, inform compute configuration and throughput planning only; incomplete artifacts cannot establish headroom or policy superiority.

6 Results

Validation TODO: Replace fixture- and readiness-oriented output with confirmatory results that answer each research question using matched endpoint evaluation, denominators, exclusions, aggregation units, independent-run uncertainty, and immutable artifact provenance.

Source: confirmatory report bundle and analysis manifest

Gate: submission evidence bundle passes all report assertions and every stated result resolves to raw and derived artifacts

The thesis report bundle is loaded through the strict schema checked in `experiment_data.typ`. Its declared evidence class is *pilot*. Schema validity establishes only that provenance, missingness, and table contracts are readable; it does not turn a development fixture or an incomplete pilot into scientific evidence.

Result	Required evidence	Status
Population and targets	population facts with denominators in every validated store	not available
Candidate-support QC	five descriptive diagnostics with the frozen state–scene macro identities and exact scene denominators	not available
Paired policy effect	paired per-scene effect, interval, exact paired-scene count, and headroom decision in every validated store	not available
Resource feasibility	wall time, peak GPU memory, and storage in every validated store	not available

Table 13: Evidence availability in the loaded report bundle. A status of “not available” means that no thesis result is inferred from fixture values, train-only attempts, or an incomplete store.

6.1 Candidate and Store Feasibility

The frozen 100-scene Phase-A proposal audit attempted $6,000$ candidates and admitted $3,146$ into the compact valid shells. It nevertheless failed its support gate: 44 applicable state–family cells had no selected row, 24 states missed the aggregate non-forward target-aware-family floor, and 8 states missed the root-support threshold. All 100 reviewed source rows, scenes, and target states were represented without exclusions. Because this no-render audit contains no reward labels, flat-gain status is unavailable with label and eligible-state denominators both zero. The result diagnoses proposal support; it provides no evidence about RRI, candidate quality, rollout throughput, or policy performance and does not admit broad rollout generation.

Separately, training-source rollout attempts reached mesh rendering, target-specific oracle scoring, and selected-action replay. An out-of-memory failure during unbatched candidate rendering and subsequent memory-bounded attempts identify rendering as a scale gate; they do not establish population throughput or storage feasibility.

Figure 7: Candidate-family survival for one deterministic scene from each of the ten persisted Phase-A audit strata. Each cell reports compact-valid-shell membership divided by attempted rows for one factual state and proposal family; the complete 100-state matrix remains in the hash-bound evidence bundle. The underlying artifact geometry uses correction revision `phase-a-target-aligned-z-up-v1`; the stored shells, support counts, verdict, and execution identity are unchanged. The artifact is bound to clean execution revision `2baf7cf6b276b81c50d01d45b152016d7cf68033` and SHA-256 `6d33e9e3d68737c8a6a5589ae5117c1e4d7fcaa89056fcfcaec1d315e4509c83`.

6.2 Target-Task Coverage

The implemented pipeline samples geometry-valid GT target tasks and records target and validity fields in the rollout schema. The loaded bundle does not contain a validated population and exclusion table for the study. Scene coverage, eligible-target coverage, invalid-reason frequencies, and their denominators are consequently unavailable as results.

6.3 Policy Comparison

The primary estimand is not estimable from the current evidence because no validated held-out bundle contains the matched policy outcomes and aggregation inputs. Oracle repeatability, oracle-lookahead headroom, learned one-step performance, finite-horizon recovery, uncertainty intervals, and sensitivity analyses are therefore not reported.

6.4 Runtime and Storage Gate

The runtime and storage claim requires completed validated stores whose resolved manifests, failure records, and resource summaries refer to the same run. Until that evidence exists for both rollout profiles, the large-scale data-generation gate remains pending. The current attempts justify memory-bounded rendering and retained failure provenance, but no extrapolation to cluster throughput or dataset volume.

7 Discussion

Remove or rewrite before submission: Rewrite the discussion after confirmatory results exist. The present implementation-readiness narrative and architecture bridge registry are development material; the final chapter must interpret measured answers, alternative explanations, scope, limitations, and failure modes.

Source: Chapter 6 and the frozen analysis plan

Gate: every interpretation points to a reported result or is explicitly bounded as a limitation

The current evidence supports a narrow implementation conclusion. ARIA-NBV has an executable finite-candidate path that separates actor-visible policy inputs from oracle-only target geometry, applies invalidity as a hard decision constraint, evaluates target-specific reconstruction change offline, and exposes the resulting store through one typed reporting seam. This establishes that the proposed experiment can be represented and audited; it does not establish that one candidate family, rollout policy, representation, or learned model performs better than another.

The rollout attempts also expose a systems limitation. Counterfactual scoring repeatedly renders a large mesh for a candidate set, so renderer memory and latency constrain feasible branch factor and rollout volume before statistical efficiency becomes relevant. An out-of-memory observation is evidence for batching and resource measurement, not for or against the candidate distribution or planning objective. Completed validated stores are required before throughput, storage, or failure rates are generalized.

The scientific limitations are more restrictive. Current attempts use training sources, no held-out paired policy table supports the endpoint estimand, metric repeatability and meaningful oracle-lookahead headroom remain unestablished, and the target-conditioned finite-horizon comparison has not yet been demonstrated. The present evidence therefore cannot distinguish negligible non-myopic structure from inadequate candidate support, target observability, replay coverage, representation support, or model failure.

7.1 Evidence-conditioned architecture bridges

Implementation: exploratory **Evidence: pending**

Alternative representations and policies remain part of the thesis design registry, but they are promoted only as responses to diagnosed bottlenecks.

Promotion gate: a measured limitation that the proposed bridge directly tests

Development source: Table 7; Table 8

If the local EFM3D field is the limiting variable, broad semidense memory, sparse occupied/free/unknown state, target-centred re-lifting, or logged appearance features provide progressively stronger representation tests. If compact descriptors lose relevant spatial structure, point, sparse, or equivariant encoders become justified. If independent candidate queries leave errors correlated with the valid candidate set, DeepSets or masked candidate interaction becomes relevant. If selected-view history fails specifically by approach direction, signed first- plus second-moment memory, followed only if needed by spherical harmonics, becomes the targeted ablation. A renderable 3DGS state is appropriate only when candidate rendering or soft instance membership is itself the measured missing capability.

Continuous and hierarchical policies change the action contract rather than merely increasing model capacity. A target-then-pose policy could first select or mask a target token and then condition pose queries on target, horizon, and step; multiple targets can share scene encodings and be evaluated in parallel. This remains post-core work because continuous pose generation requires separate feasibility, safety, support, and simulator-realism evidence. The finite-candidate model is retained as the calibrated control even if a continuous policy is later introduced.

Sequence or recurrent models are similarly conditional. They become scientifically motivated when errors persist after explicit history, time, horizon, and scene-memory conditioning, not simply because the dataset contains trajectories. Looping or recurrent refinement must preserve causal masks and candidate-row equivariance and must be compared against the same fixed-budget endpoint oracle evaluation.

Research TODO: After the primary evidence chain is complete, promote only bridges tied to a measured failure: EVL support, target observability, directional history, valid-set interaction, long-horizon credit, or action-support mismatch.

Source: method design-space registry

Gate: artifact-backed failure analysis

The next evidential step remains procedural: complete validated stores under resolved manifests, freeze the eligible population and exclusions, establish oracle stability and headroom, train the matched controls, and only then interpret architectural differences. The registry above preserves concrete follow-up hypotheses without presenting them as validated conclusions.

8 Conclusion

Validation TODO: Rewrite the conclusion as direct, evidence-calibrated answers to RQ1–RQ4. The current conditional outcome tree is useful development guidance but is not a final scientific conclusion.
Source: Chapter 6; Chapter 7
Gate: confirmatory results and discussion support one concise answer per research question

This thesis defines a leakage-auditable experiment for target-conditioned finite-candidate next-best-view planning. Its present contribution is the separation of actor-visible state from oracle supervision, the target-specific reconstruction objective, hard validity and replay contracts, and an artifact-driven reporting seam that keeps provenance and missingness attached to later results.

The available evidence does not answer whether bounded oracle lookahead improves fixed-budget target reconstruction over oracle-greedy or whether a learned finite-horizon policy closes the separate endpoint gap from an actor-visible learned-myopic control to oracle lookahead. The current training-source rollout attempts establish pipeline reachability and reveal a renderer resource gate; the development report fixture establishes the data contract only. Neither supports a held-out policy claim, a population-level effect, or a scale estimate.

The final scientific conclusion is therefore conditional on evidence that is not yet available. A stable oracle metric and positive paired lookahead effect would define measurable headroom for the evaluated finite support. Negligible headroom would be a setup-specific negative result. Unstable oracle evaluation would block planning claims, and stable headroom without the prescribed learned-control gap closure would remain a learned-control failure with several unresolved mechanisms. These outcomes delimit the thesis without extending it to continuous control, online reinforcement learning, or real-device deployment.

Development roadmap

This page is the compact strategic view of thesis development. It is omitted from submission output. Scientific claims remain owned by the active thesis; executable behavior and measurements remain owned by source, tests, configuration, and immutable evidence artifacts.

Implementation: partial **Evidence: pending**

Reviewed 31 Aug 2026 refresh due 14 Sep 2026. Target: reproducible full-draft freeze by 30 November 2026, followed by a safety buffer until submission on 7 January 2027.

Promotion gate: Close P1 before confirmatory policy claims.

Development source: `docs/typst/thesis/development/roadmap.toml`; active thesis, package, test, configuration, and evidence owners

TL;DR

Done **Pilot infrastructure**

Candidate-benchmark smoke evidence, a bounded two-scene orbit MVP, and a pilot rollout report are committed. These prove the pipeline can produce reviewable artifacts, not that population-scale support or policy gains are established.

Now **Evidence substrate**

Freeze the multi-scene store, scene-level split, camera/depth and target-RRI fidelity, candidate-support admission, and throughput receipts while the thesis narrative and conceptual figures remain under review.

Blocked **Confirmatory claims**

Lookahead-headroom, learned-value, and population-coverage claims remain blocked until paired multi-scene evidence and an immutable experiment registry exist. The current rollout bundle is explicitly pilot evidence.

Next **Paired evidence to freeze**

Run equal-budget baselines, train and evaluate the finite-horizon value model under actor-visible inputs, synthesize results and limitations, and reach the reproducible full-draft freeze recorded below.

Critical path

The strategic dependency is evidence-first: close the substrate, establish paired baseline headroom, evaluate the learned value model, synthesize the claims, and freeze the full draft. The December-to-January interval is a safety buffer, not planned scientific work.

P1 Freeze experiment substrate → **P2** Paired policy baselines → **P3** Learned value evaluation → **P4** Results and discussion synthesis → **P5** Reproducibility and full-draft freeze

Figure 8: Critical development gates from experiment substrate to the 30 November 2026 full-draft freeze.

Milestones

ID	Phase	Dates	State	Exit gate
P0	Pilot infrastructure	7 Jul–30 Aug 2026	**done**	Committed smoke and MVP artifacts compile and remain explicitly bounded as pilot evidence.
P1	Freeze experiment substrate	31 Aug–30 Sep 2026	**in progress**	Multi-scene immutable store, scene-level split, camera/depth and target-RRI fidelity, candidate-support admission, and throughput receipts.
P2	Paired policy baselines	1 Oct–20 Oct 2026	**planned**	Equal-budget random-valid, greedy, and bounded-lookahead comparisons report uncertainty, support, and oracle receipts.
P3	Learned value evaluation	21 Oct–10 Nov 2026	**planned**	Held-out ranking, calibration, and paired policy value are reported under actor-visible inputs.
P4	Results and discussion synthesis	11 Nov–22 Nov 2026	**planned**	Final tables, failure cases, coverage limits, and claim dispositions resolve to the immutable evidence registry.
P5	Reproducibility and full-draft freeze	23 Nov–30 Nov 2026	**planned**	Full draft, reproducible configurations, final figures, and release checks pass with no unresolved placeholders.
P6	Safety buffer and submission	1 Dec–7 Jan 2027	safety buffer	Advisor feedback, formal checks, final upload, and the authenticated submission receipt are closed without reopening the scientific core.

Table 14: Evidence-gated schedule from the completed pilot infrastructure to submission. Dependencies are checked against the adjacent roadmap data.

Blockers

- **B1 — Multi-scene experiment substrate** (affects P1): The committed artifacts do not yet establish a fresh multi-scene store, scene-level split, camera/depth fidelity, target-RRI robustness, and representative throughput in one immutable receipt chain.
- **B2 — Candidate-support scale admission** (affects P1, P2): The positive two-scene orbit MVP is explicitly not scale-up admission; family survival, framing, diversity, and proposal regret still require broader evidence.
- **B3 — Confirmatory experiment registry** (affects P2, P3, P4): The available report bundle is a one-scene pilot. Final comparisons need immutable run/config identities, coverage accounting, uncertainty, and claim-to-artifact resolution.

The primary evidence pointers remain beside each record in `roadmap.toml`; the roadmap contract checks their existence, while scientific review decides whether they are sufficient for promotion.

Promotion queue

Promotion queue — candidate: Promote bounded-lookahead and learned-value results only after paired held-out evidence is immutable.
Source: P2-P3 experiment registry and rollout reports target: Results and Discussion gate: positive headroom, actor-visible evaluation, uncertainty, and support coverage

Promotion queue — blocked: Resolve candidate-support and substrate limits before claiming population coverage.
Source: P1 evidence receipts and candidate benchmark manifests target: Experimental Design and Results gate: multi-scene support, family survival, scene-level split, and representative throughput

Promotion queue — deferred: Keep online discrete and continuous target-then-pose extensions outside the core until the offline finite-candidate gates close.
Source: Discussion and roadmap P1-P3 target: Discussion gate: stable offline learned-value evidence and an explicit post-core scope decision

Maintenance contract

`roadmap.toml` is the single owner for the snapshot, dates, states, dependencies, blockers, release checks, and evidence pointers rendered above. Update it at least every 14 days and whenever a gate changes state. `make thesis-roadmap-contract` rejects stale review dates, broken dependencies, missing local evidence pointers, divergence from the thesis submission date, or references to the retired M1 snapshot. Internal trackers and hosted issue state may inform a refresh but are not imported as public thesis truth.

The native table, cards, and critical-path strip are deliberately sufficient: they show exact milestones and the consequential dependency chain without a second diagram data model. A date-dense Gantt can be reconsidered only if this compact projection no longer answers the planning question.

HM/FK07 release gate

These checks are human-owned and apply to the exact final candidate and the author's authenticated records; the repository cannot satisfy them by itself.

- **R1 — Applicable rules:** Confirm the individually assigned programme and SPO version in PRIMUSS; public reading versions establish only the current baseline.
- **R2 — Registration record:** Retain the PRIMUSS confirmation, official start date, individual deadline, assigned examiners, supervisor-specific form requirements, and requested attachments.
- **R3 — Exact upload set:** Before the individual deadline, verify the exact final PDF and every PRIMUSS-marked mandatory document or attachment. A repository build, digest, or upload alone is not submission evidence.
- **R4 — Conditional paper copy:** Provide an identical paper copy only if the examiner or authenticated PRIMUSS record requires one under the current FK07 process.
- **R5 — Final-PDF review:** Check the declaration and AI/tool disclosure in the exact PDF prepared for upload; confirm any mandatory signature or attachment against the authenticated record.
- **R6 — Submission closure:** Finalize with the PRIMUSS action "Abschlussarbeit abgeben", retain the authenticated timestamp, and archive the exact submitted PDF; only that receipt closes the gate.

Official baseline checked 31 August 2026: HM examination regulations, FK07 thesis process, PRIMUSS registration guide, PRIMUSS submission guide. At the full-draft freeze, recheck the official versions and the individually assigned instructions in authenticated PRIMUSS records.

Freeze and submission

By 30 November 2026, the full draft, experiment registry, configurations, figures, and release checks must be reproducible and free of unresolved placeholders. The interval from 1 December to 7 January 2027 is reserved for advisor feedback, formal HM/FK07 checks, final upload, and recovery from unexpected defects. Only the authenticated submission receipt closes the final gate.

9 Development Pilots

9.1 Candidate-Family Support

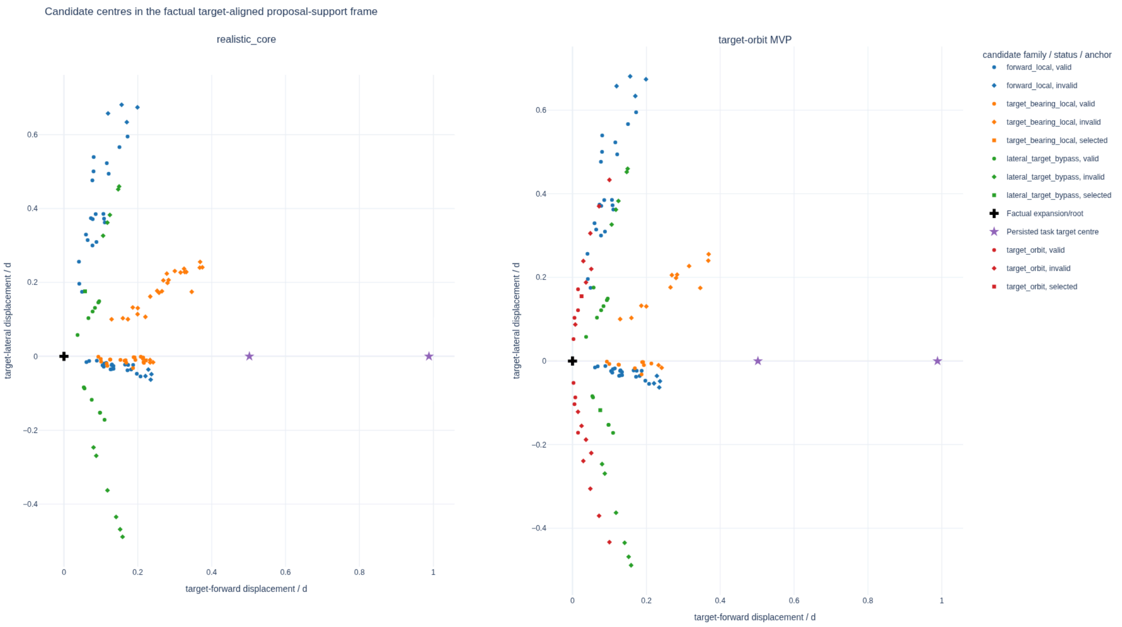


Figure 9: Attempted candidate centres from a two-scene, two-state real-data method audit. Each state is centred on its factual expansion pose, yaw-aligned to the oracle-evaluation target with world Z-up, and divided by its current three-dimensional target distance. The cross marks the expansion/root pose, the stars mark the two oracle task target centres, colour denotes proposal family, and marker shape denotes actor-valid or selected status. The figure diagnoses proposal geometry under the `v0_gt_input` target protocol; it is neither a population estimate nor evidence of actor-visible target discovery or policy quality.

The target-orbit challenger adds attempted support on both target-relative sides in this bounded pilot, while pruning still leaves invalid rows and one configured family/state pair without actor-valid support. The associated immutable evidence manifest records the common-frame reducer, store-manifest hashes, expected state index, committed reduced candidate rows, resolved challenger profile, reproduction commands, metric denominators, and plot hashes. No frame-mixed side-balance or orbit-span values are used. The interactive evidence and Streamlit view can optionally add short arrows for the valid candidates' camera optical axes; the static thesis figure omits them to preserve legibility.

9.2 Target-Frame $\mathcal{S}^2$ Diagnostics

The following frozen figures are generated by the same Python reporting and plotting owners as the stored-rollout application. For the selected real-data shard, the report records these support counts: source samples 1, snippets 1, scenes 1, targets 1, retained rollout chains 1, and selected steps 5. It remains pilot evidence: its purpose is to validate the representation and reporting contract, not to estimate policy quality. The two point-direction views contain 5 factual displacements and 5 factual optical axes. Colour denotes rollout-chain identity, marker shape denotes decision-step identity, and the sphere heat map is computed from the complete equal-solid-angle count grid rather than from the bounded incidence overlay.

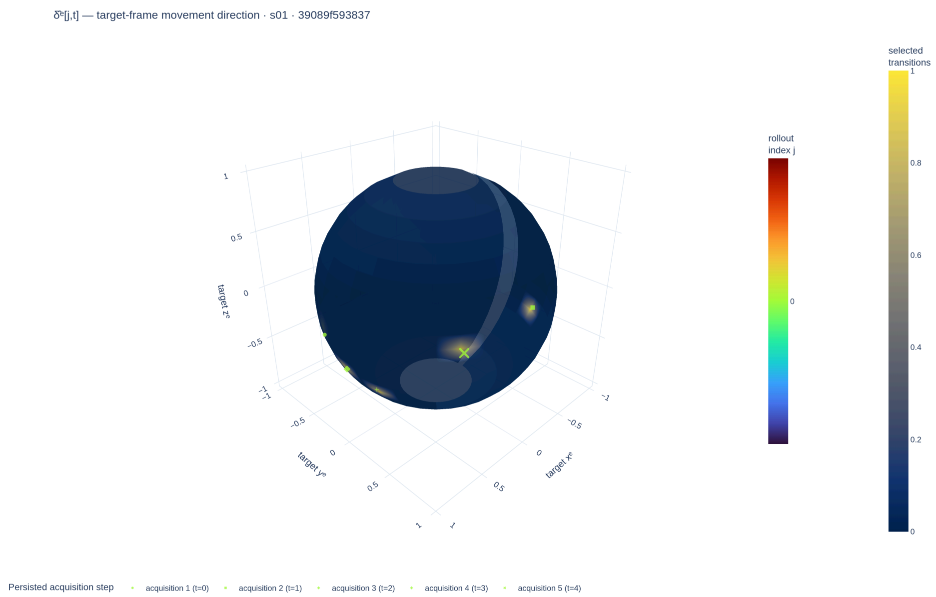


Figure 10: Target-frame $\mathcal{S}^2$ movement directions from the frozen pilot report. This view records how the camera moved; it is not a visibility measurement.

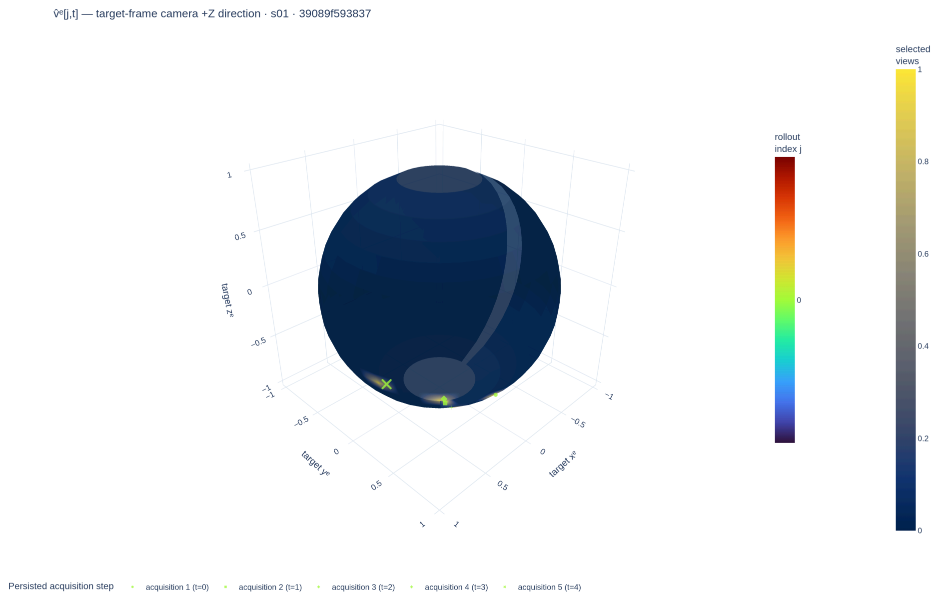


Figure 11: Target-frame $\mathcal{S}^2$ selected-camera optical-axis directions from the same frozen pilot report. This view records where the camera pointed; it is distinct from both camera motion and calibrated surface support.

The calibrated proxy diagnostic integrates 5 selected frusta. Their mean intrinsic field-of-view solid angle is 2.49 sr. On the geometric-mean-scale proxy sphere, one view covers a mean fraction of 0.168, while their union covers 0.312. These fractions express geometric potential support under the proxy and front-facing tests; they exclude true target shape, depth ordering, and scene occlusion.

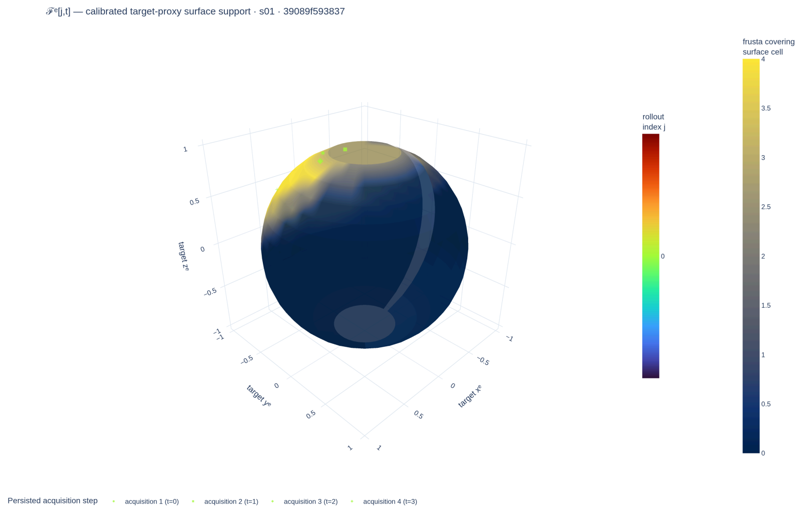


Figure 12: Calibrated selected-frustum support on the target-centred proxy sphere. The surface heat map is the complete equal-solid-angle coverage field; overlaid incidence samples preserve rollout and time heritage. This is not measured target-mesh visibility.

Development Register: Method Alternatives

This register preserves executable but unpromoted controls and future hypotheses outside the thesis-core Method chapter. Entry here means that an idea has a named comparison and promotion gate; it does not mean that the thesis evaluates or recommends it.

Implementation: exploratory **Evidence: pending**

The thesis-core selection remains A1–H0–S0 with direct continuous regression. Every item below is excluded from its central method claim.

Promotion gate: promote only a one-factor comparison with an immutable receipt and untouched scene-disjoint evaluation

Development source: `aria_nbv/aria_nbv/vin/`; `aria_nbv/aria_nbv/lightning/`; `docs/contents/theory/`

State and history alternatives

- **S1 selected surfaces:** the executable privileged control backprojects only factual selected mesh depth, pools the causal point set, and initializes its residual at zero so a fresh S1 scorer matches H0. It lacks free/unknown evidence, is density weighted, and has no confirmatory receipt; it therefore cannot support a geometry-benefit or deployment claim.
- **S2 ray-aware memory:** a planned state would distinguish observed surface, observed free, unknown, support, uncertainty, source, and recency. It requires deterministic fusion and no-future-observation tests before a model comparison.
- **H1 ordered history:** an executable causal Transformer adds relative age to the same selected-pose tokens used by H0. It remains a pose-only capacity ablation until repeated-seed endpoint evidence justifies chronology.
- **Other carriers:** target-centred EFM3D re-lifting, appearance-on-point, sparse TSDF/SDF encoders, object-aware 3D Gaussian splats, and SceneScript context are hypotheses for diagnosed representation failures, not parallel thesis methods.

Interaction alternatives

A2 DeepSets context, A3 masked candidate self-attention, A4 query-local relation bias, A5 recurrent state reading, A6 residual value prediction, and A7+ graph or exact-equivariant layers are ordered escalation points. A level may enter the main method only after A0/A1 are measured under the same state, target protocol, decoder, objective, and evaluation contract and a concrete failure identifies the missing interaction. Candidate-set models must retain row equivariance, invalid-row isolation, duplicate tests, and absolute-value semantics.

Objective and estimator alternatives

- **CORAL decoding** is executable over the same continuous fitted-Q targets, but its ordinal loss and fixed representatives add support-provenance and saturation questions. It is excluded until direct regression supplies the frozen control.
- **Fixed-horizon or separate heads** alter parameter sharing and the public interface. They are ablations against the scalar requested-horizon family, not co-equal definitions of Q_H.
- **Behavior-return Monte Carlo regression** estimates the continuation policy that generated each chain rather than greedy finite-support value unless those policies coincide. It must retain that distinct estimand.
- **One-step plus residual prediction** can test whether calibrated immediate utility is a useful base, but the residual may not be mean-centred across a candidate table because unrelated or duplicate rows would then redefine an action's absolute target.
- **Quantile regression** is justified only after stochastic return variation or state aliasing is measured and a distributional Bellman projection, support, decoding rule, and calibration protocol are frozen.

The promotion sequence is: establish population/action support, measure oracle headroom, learn actor-visible Q_1, pass exact Q_2, and only then introduce the smallest alternative that addresses the observed failure.

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A Reproducibility Record

The report bundle is the single numerical interface between validated rollout artifacts and this manuscript. Its schema version, evidence status, store manifests, typed parameter rows, failure records, and sidecar hashes preserve provenance without duplicating analysis logic in Typst. The document loader rejects schema or status mismatches before any table is rendered. Compact labels and digest prefixes are used below; complete store identifiers, paths, and hashes remain in the machine-readable bundle.

The implementation anchors behind that interface are the immutable `vin_offline/` source store, the task-specific `rollouts.zarr/` replay store, and the derived `q_h/` training cache. Their exact schemas, normalized joins, chunking, codecs, receipt versions, and execution commands remain manifest-owned reproducibility metadata rather than scientific entities in Chapter 3. A reported row is admissible only when the bundle resolves these identities and preserves source role, missingness, and failure records.

Evidence class	Store	Manifest digest	Validated
Development fixture	store S1	not-a-real-m…	yes

Table 15: Loaded development-fixture provenance. The fixture verifies the document interface only and is not scientific evidence.

The bundled development fixture contains synthetic values solely to test typed loading and missingness. Its numerical rows are deliberately not rendered or interpreted.

B Development Diary

Archived source note: This marked diary is a development-only record. It is excluded from submission mode; only evidence-backed definitions, protocols, and results graduate into the thesis body.

Source: thesis mode contract

B.1 Decisions Retained

The thesis keeps the privileged oracle and actor-visible policy as separate contracts, evaluates finite candidate sets under a shared validity mask, and treats target-specific reconstruction improvement as the common comparison signal. Rollout stores retain selected transitions and provenance rather than becoming a second implementation of the oracle or training pipeline.

Open decision: Freeze the scene-disjoint analysis population and the minimum evidence scale only when the resolved manifests and exclusion ledger are available.

Source: experimental-design contract

Gate: confirmatory bundle freeze

B.2 Failures That Changed the Plan

Early rollout attempts reached the mesh-rendering path but exceeded available accelerator memory when candidate views were rendered without a bounded batch. This failure narrowed the immediate claim to implementation readiness and made peak memory, throughput, failure provenance, and completed-store validation explicit scale gates.

Validation TODO: Require completed stores, matching manifests, resource summaries, and recorded failures before extrapolating rollout bandwidth or storage demand.

Source: rollout attempt artifacts

Gate: cluster-readiness evidence

B.3 Rejected Scope

Continuous actions, online simulators, additional scene representations, and alternative reinforcement-learning families are excluded from the core study while the finite-candidate target-conditioned comparison remains unvalidated. They would change both the action contract and the supervision regime, so adding them now would weaken rather than extend the primary experiment.

Remove or rewrite before submission: Delete bridge material that cannot be tied to a concrete limitation or a matched follow-up experiment after the confirmatory analysis is complete.

Source: scope review

Gate: discussion freeze

B.4 Directional-Memory Hypothesis

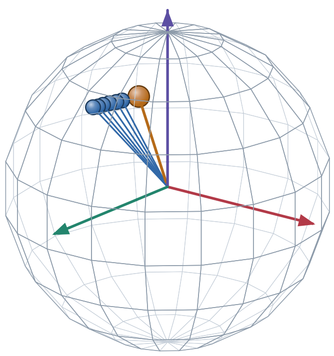
Research TODO: Treat target-centred directional memory as an ablation hypothesis, not an actor feature or contribution. The representation must be implemented, mask-audited, and compared against pose distance and overlap before any predictive claim enters the manuscript.

Source: RQ3 representation hypothesis

Gate: paired held-out ablation

A. Target-centred directions on $\mathbb{S}^2$

actual logged camera centres, expressed in the target-object frame

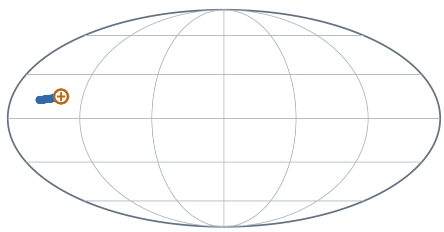


● logged history direction   ⊕ example query

Scene 81286, sample ASE_81286_Atek_000024, target instance 128 frames 0, 2, 4, 6, 8, 10 and query frame 15.

B. Complete-domain view and prospective compression

Mollweide is equal-area; no smooth field or SH lobe is implied



$+x_e$ at map centre · $+z_e$ north · wrap seam at $\pm 180°$

$$M_{\mathrm{dir}(v)} = \sum_{k<t} w_{k(v)} \boldsymbol{d}_{k(v)} \boldsymbol{d}_{k(v)}^{\top}$$

$$\nu_{i(v)} = 1 - \frac{\boldsymbol{d}_i^{\top} \boldsymbol{M}_{\mathrm{dir}} \boldsymbol{d}_i}{\operatorname{tr} \boldsymbol{M}_{\mathrm{dir}} + \varepsilon}$$

This fixture uses uniform weights and a logged future pose only to make the second-moment query inspectable. It does not claim that the learner currently stores $\boldsymbol{M}_{\mathrm{dir}}$ or that this novelty predicts target RRI.

HYPOTHESIS / ABLATION MATERIAL. Keep outside submission claims until the representation exists and paired evaluation tests whether it adds information beyond pose distance and overlap.

Figure 13: Development-only directional-memory fixture. The sphere and Mollweide panels show the same logged ASE camera directions in a target-object frame; the orange query is another logged pose used to inspect one prospective second-moment novelty definition. No smooth field, spherical-harmonic representation, or learned benefit is claimed.

B.5 Next Evidence Gates

The next admissible evidence is a validated report bundle that fixes the study population, records target and candidate exclusions, supplies paired endpoint outcomes with uncertainty, and joins runtime and storage statistics to the same manifests. Only then can the manuscript replace the present availability statements with policy comparisons.

Implementation TODO: Export the confirmatory report bundle, compile both thesis modes against it, and inspect the resulting tables and failure cases before freezing claims.

Source: reporting contract

Gate: claim freeze

Ehrenwoertliche Erklaerung

Ich versichere, dass ich diese Masterarbeit selbstständig verfasst, noch nicht anderweitig für Prüfungszwecke vorgelegt, keine anderen als die angegebenen Quellen oder Hilfsmittel benutzt sowie wörtliche oder sinngemäße Zitate als solche gekennzeichnet habe.

Muenchen, 7. Jan 2027 Jan Duchscherer